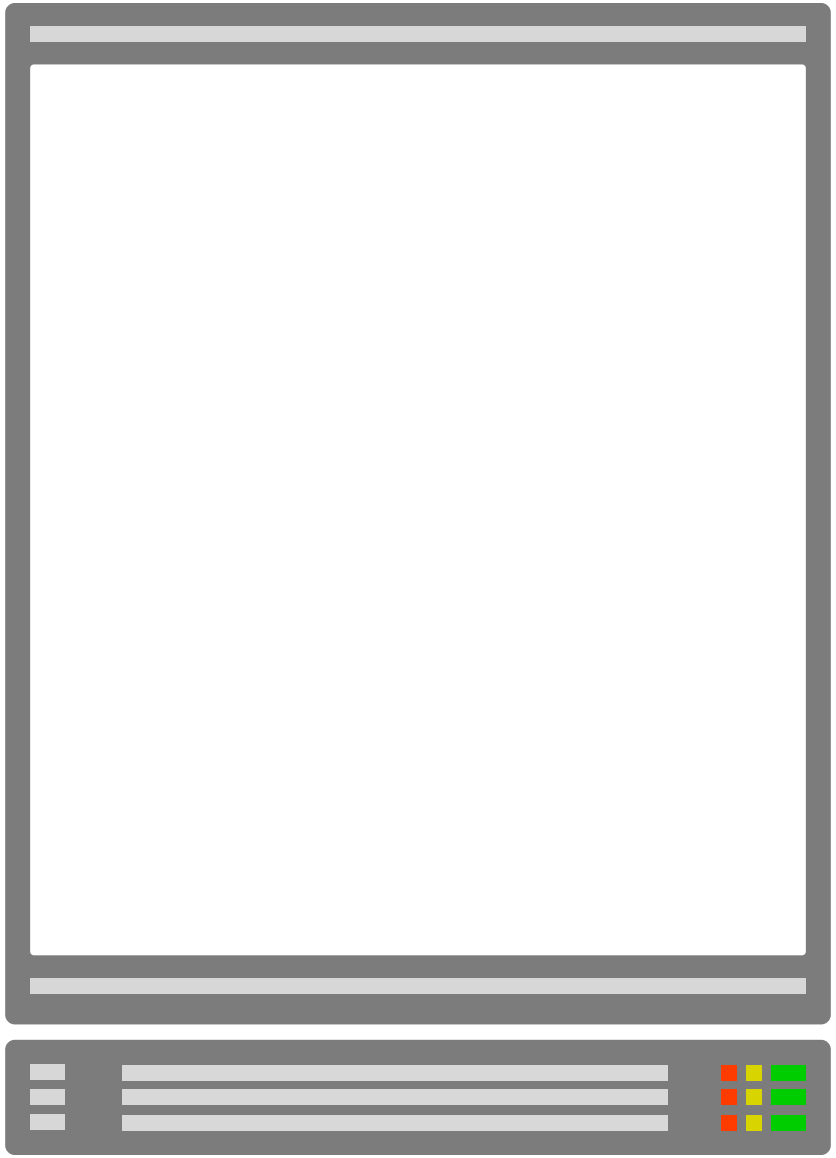
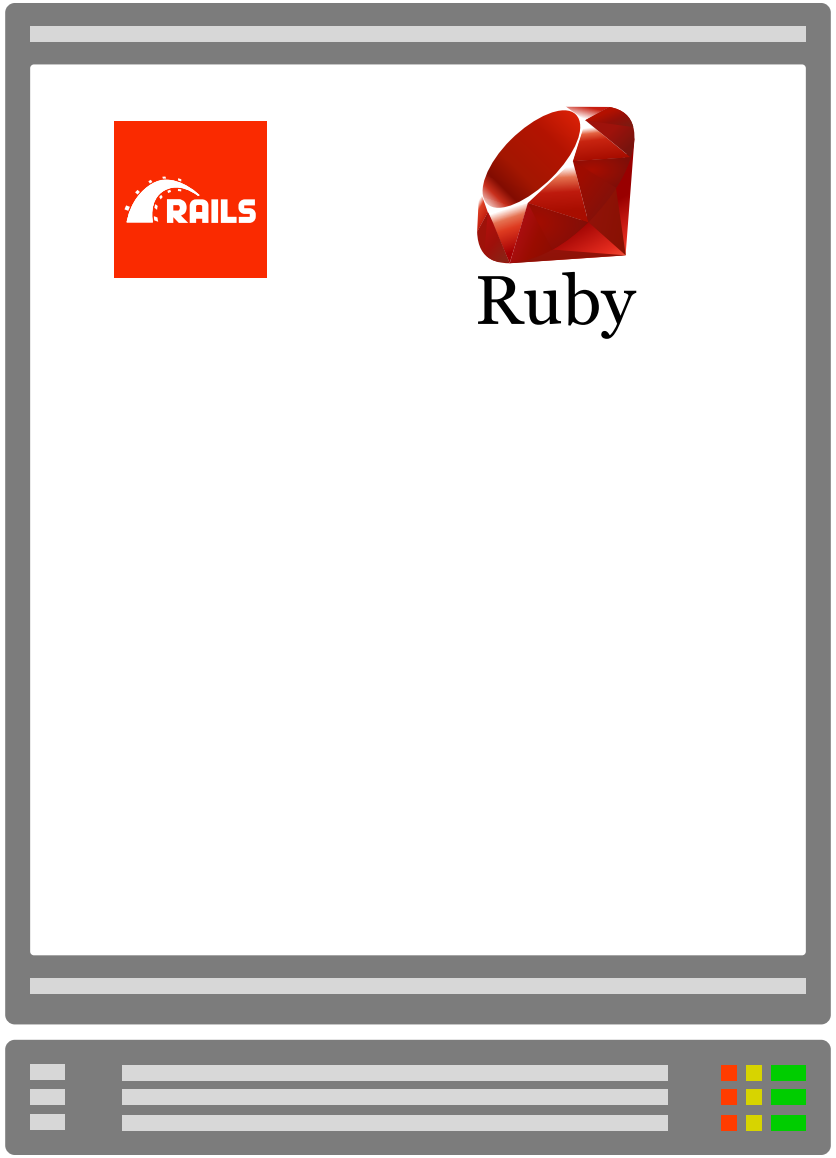
Kubernetes Fundamentals

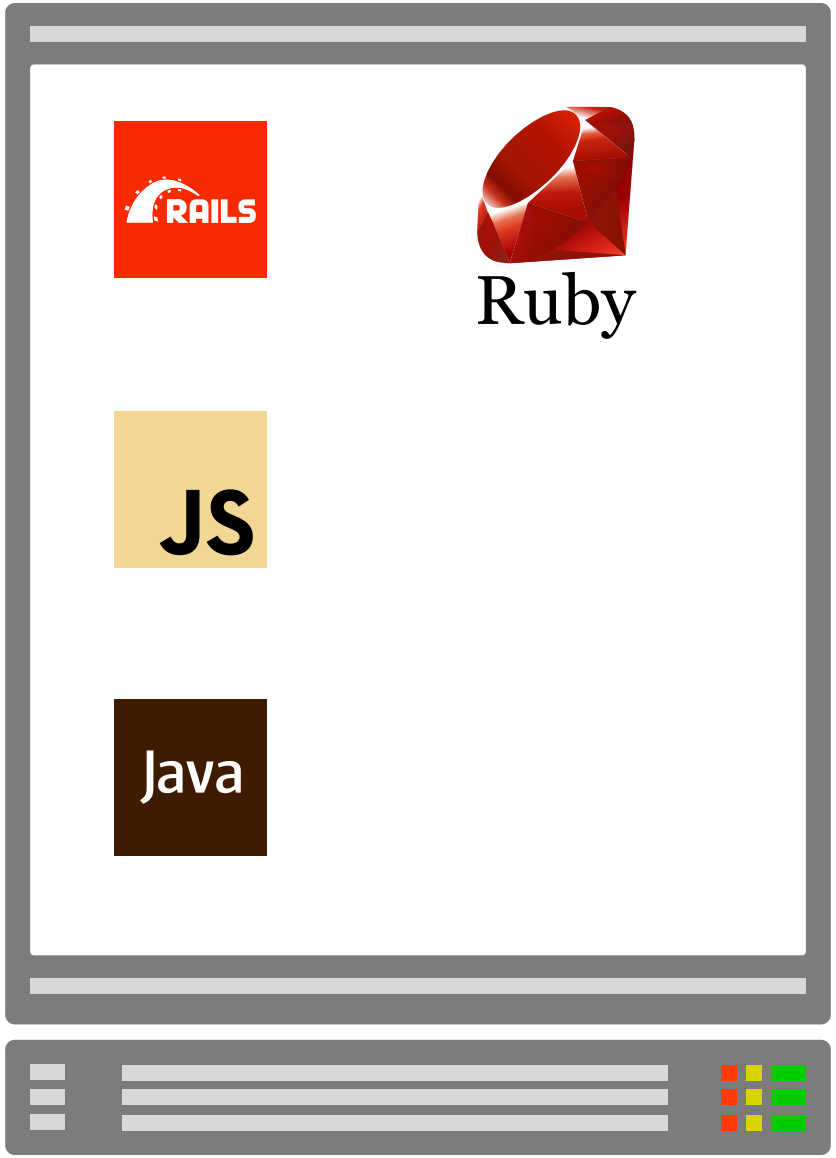
02

Deploying apps in containers

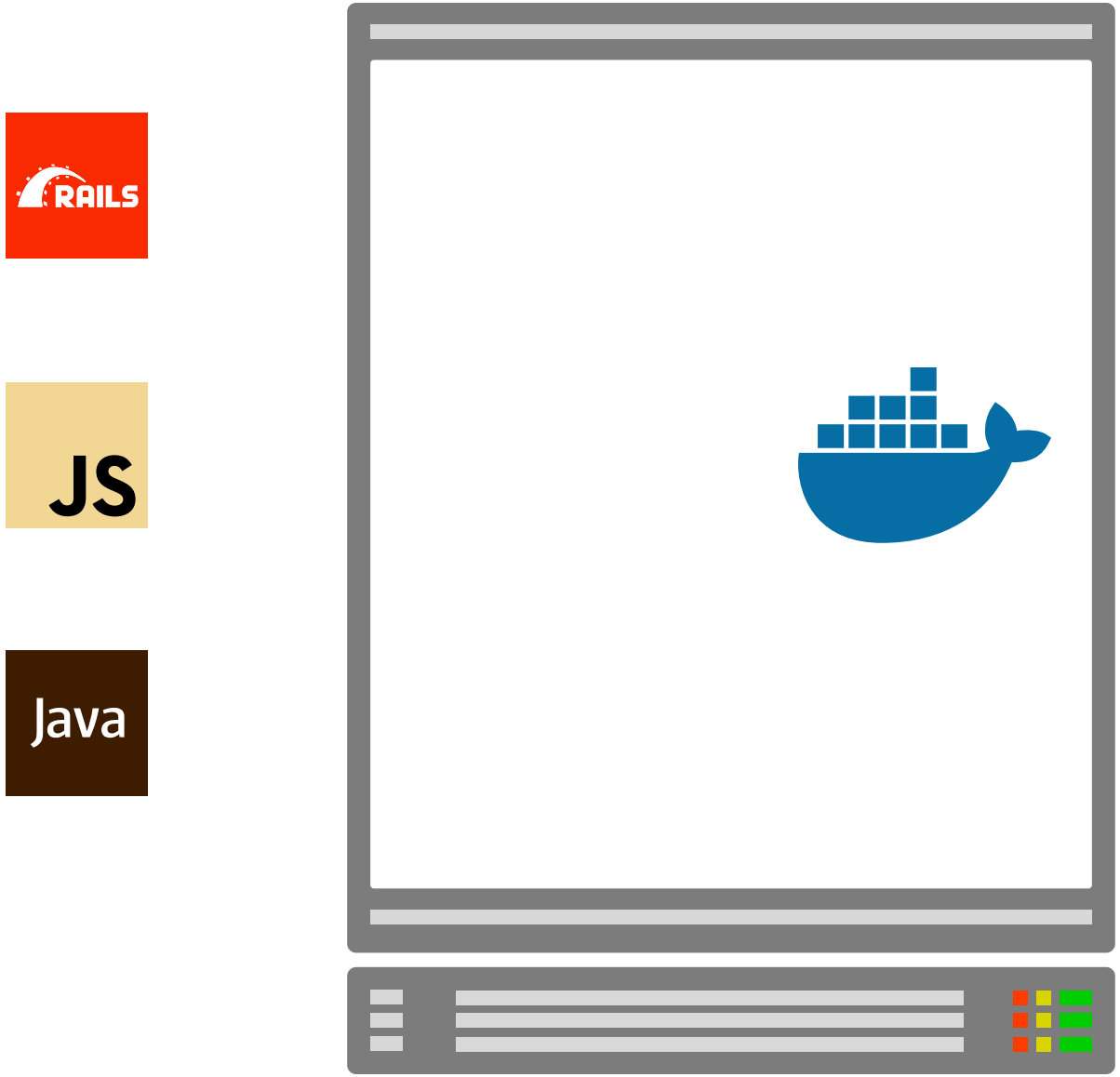

RAILS

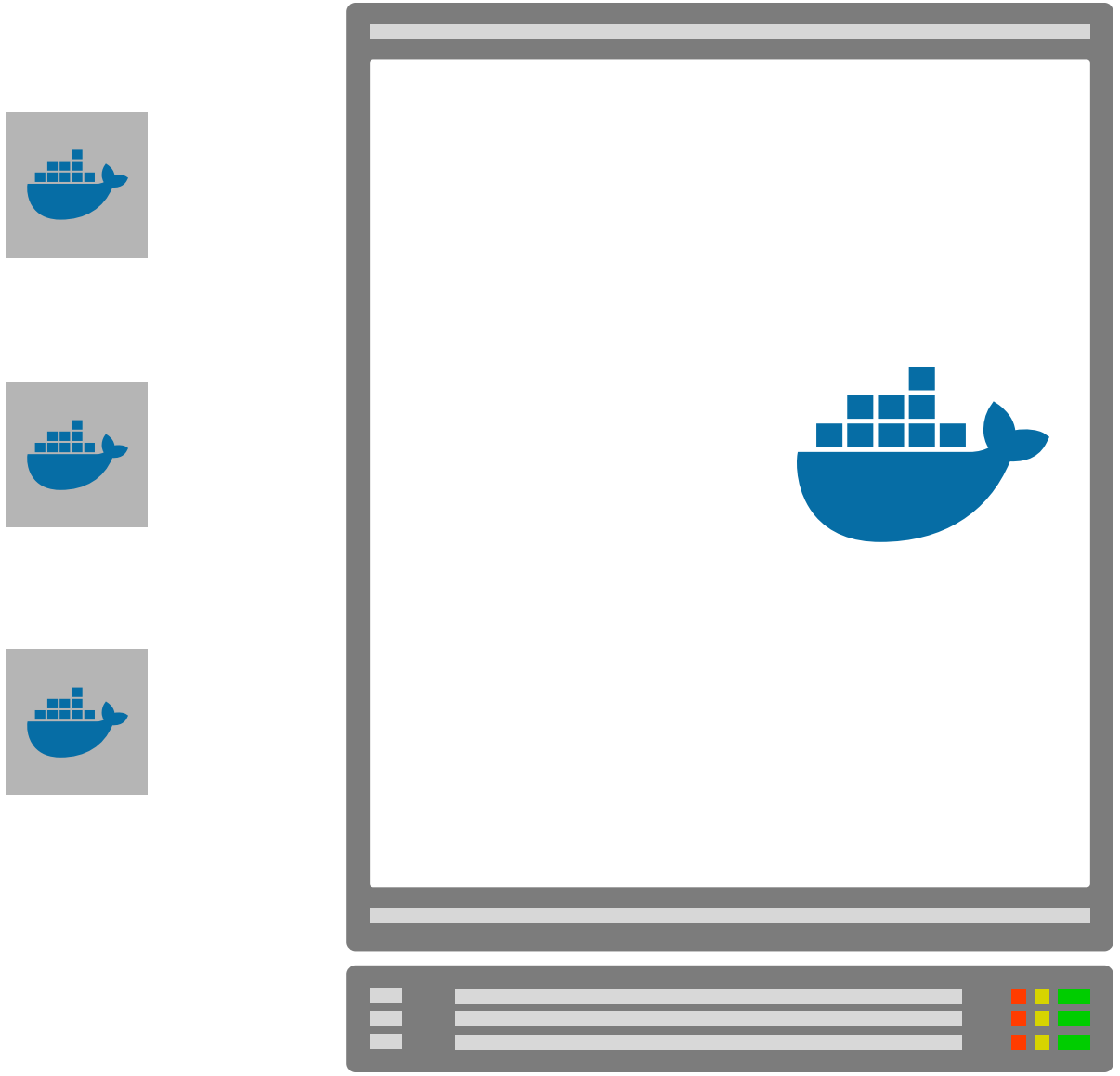


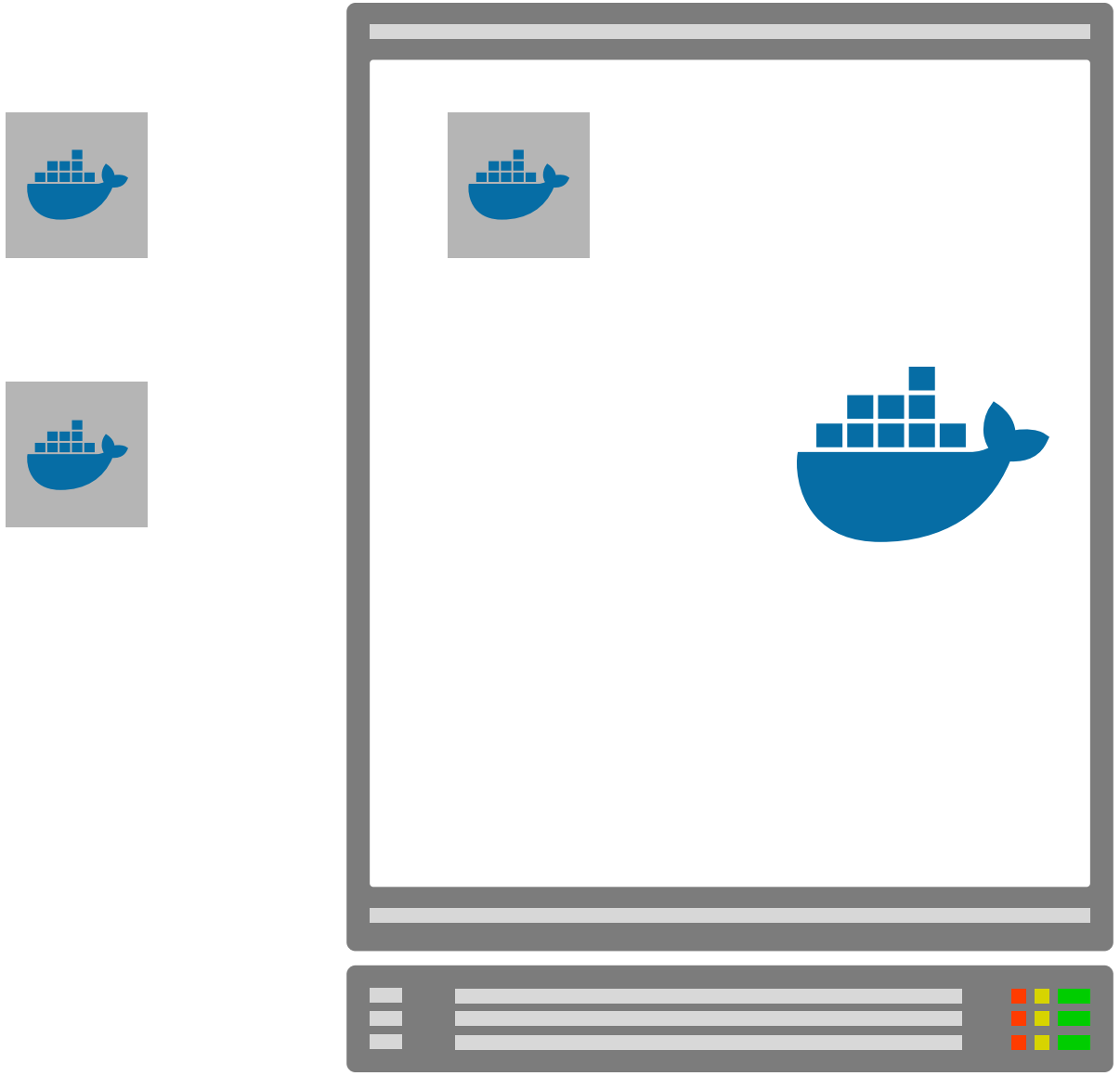

RAILS
Ruby

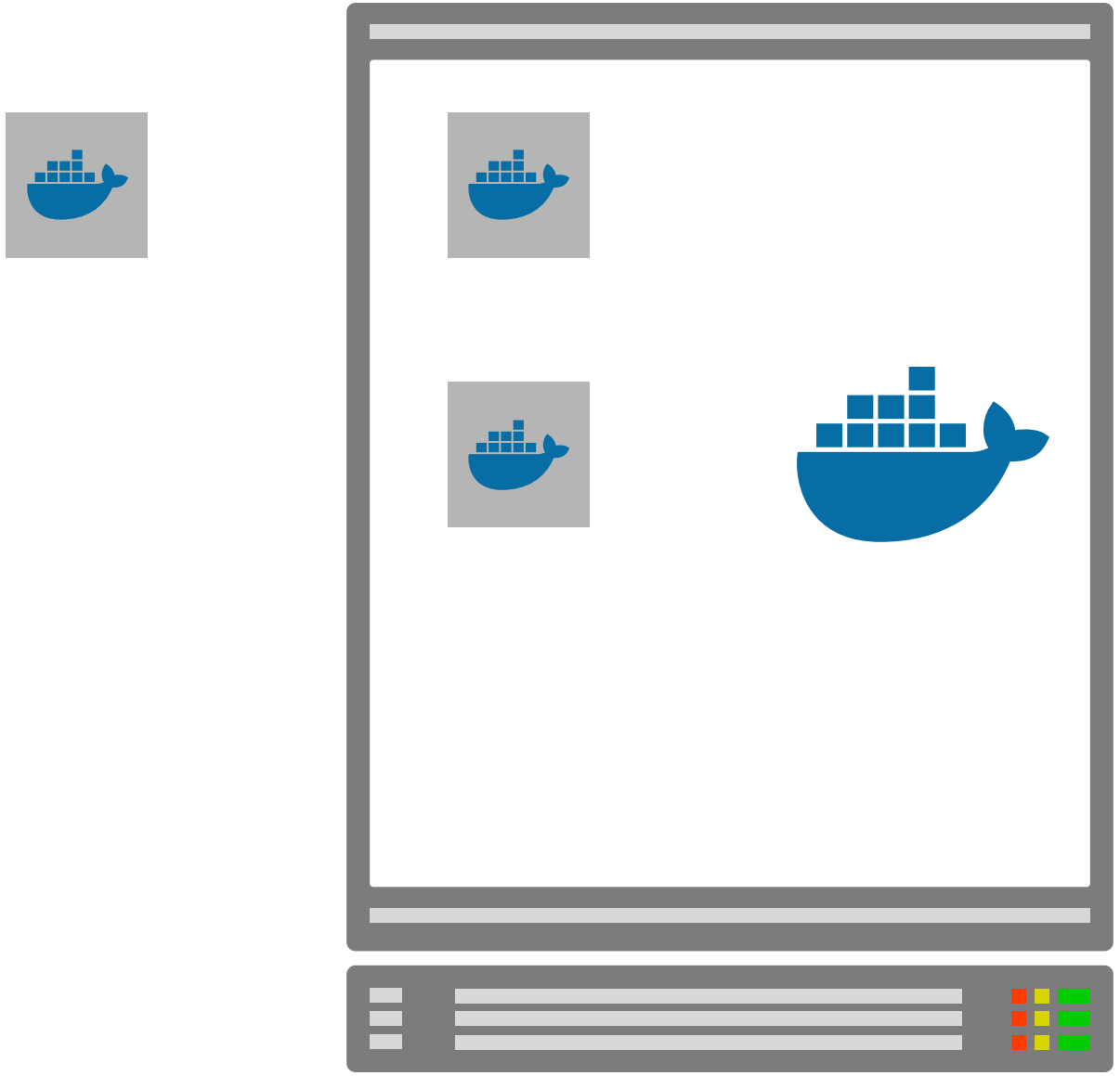

RAILS
Ruby
JS
Java

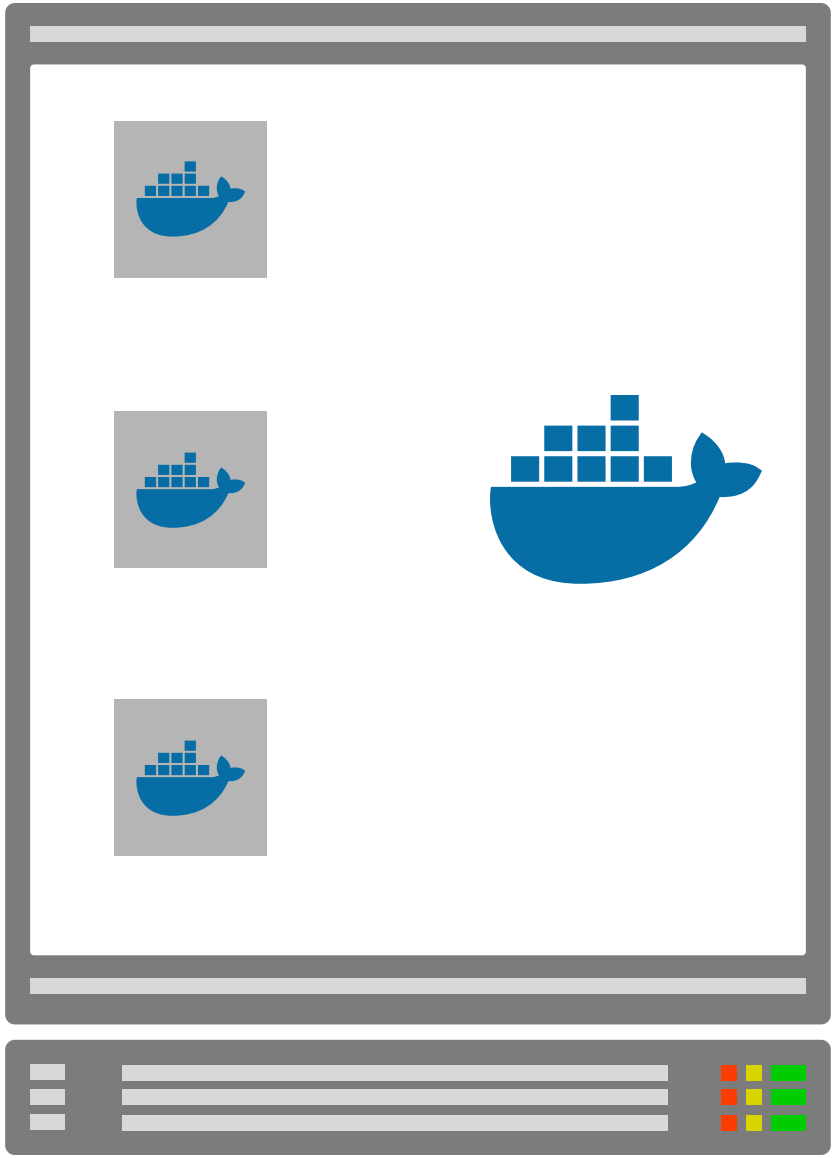

RAILS
Ruby
JS
node
Java
JVM


RAILS
JS
Java



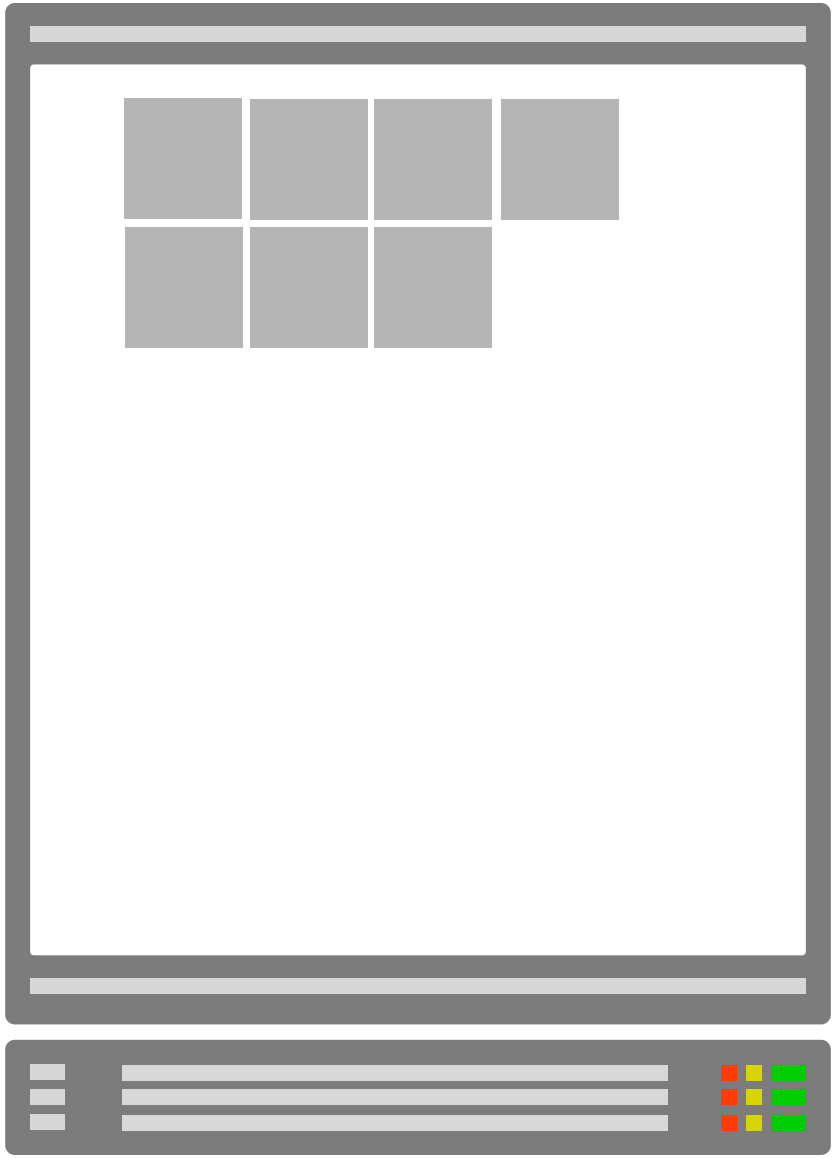


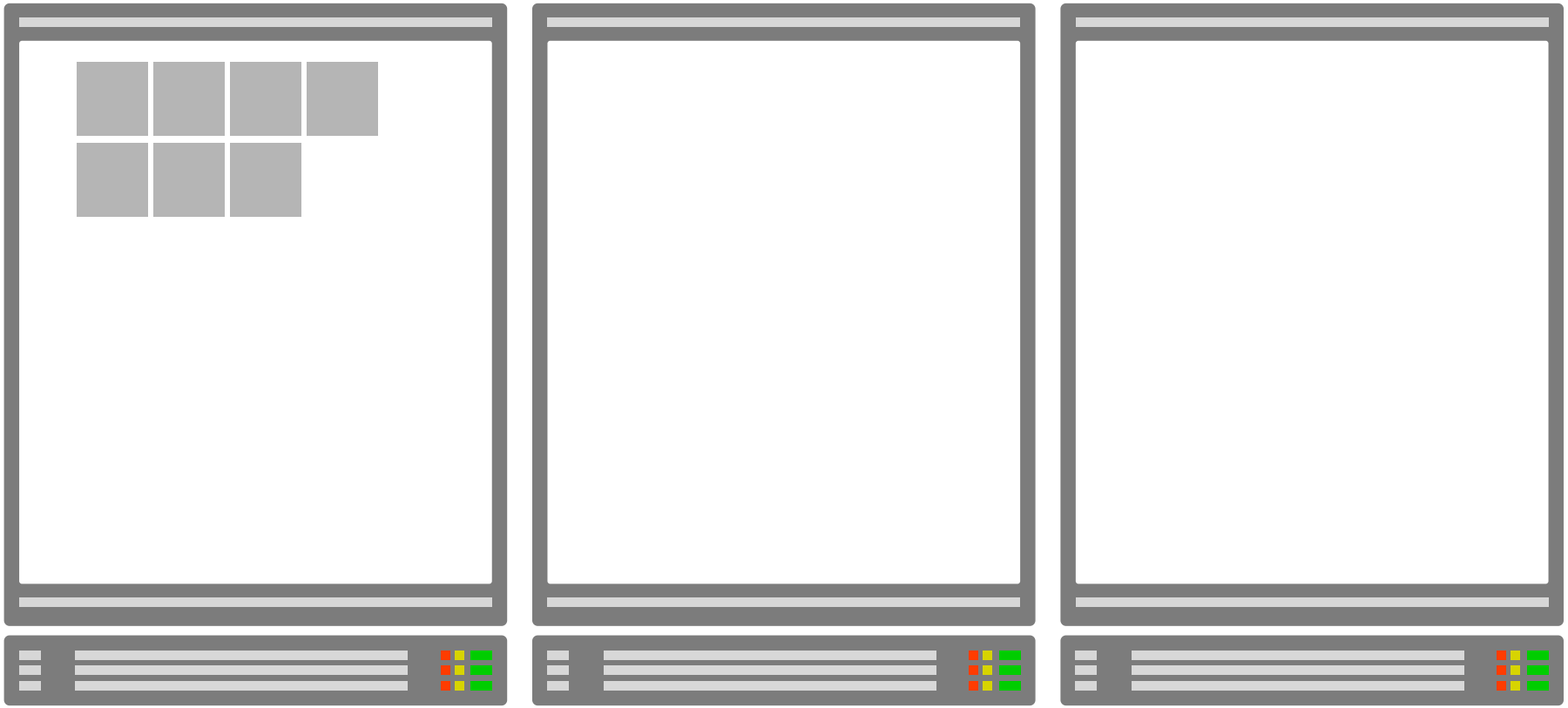


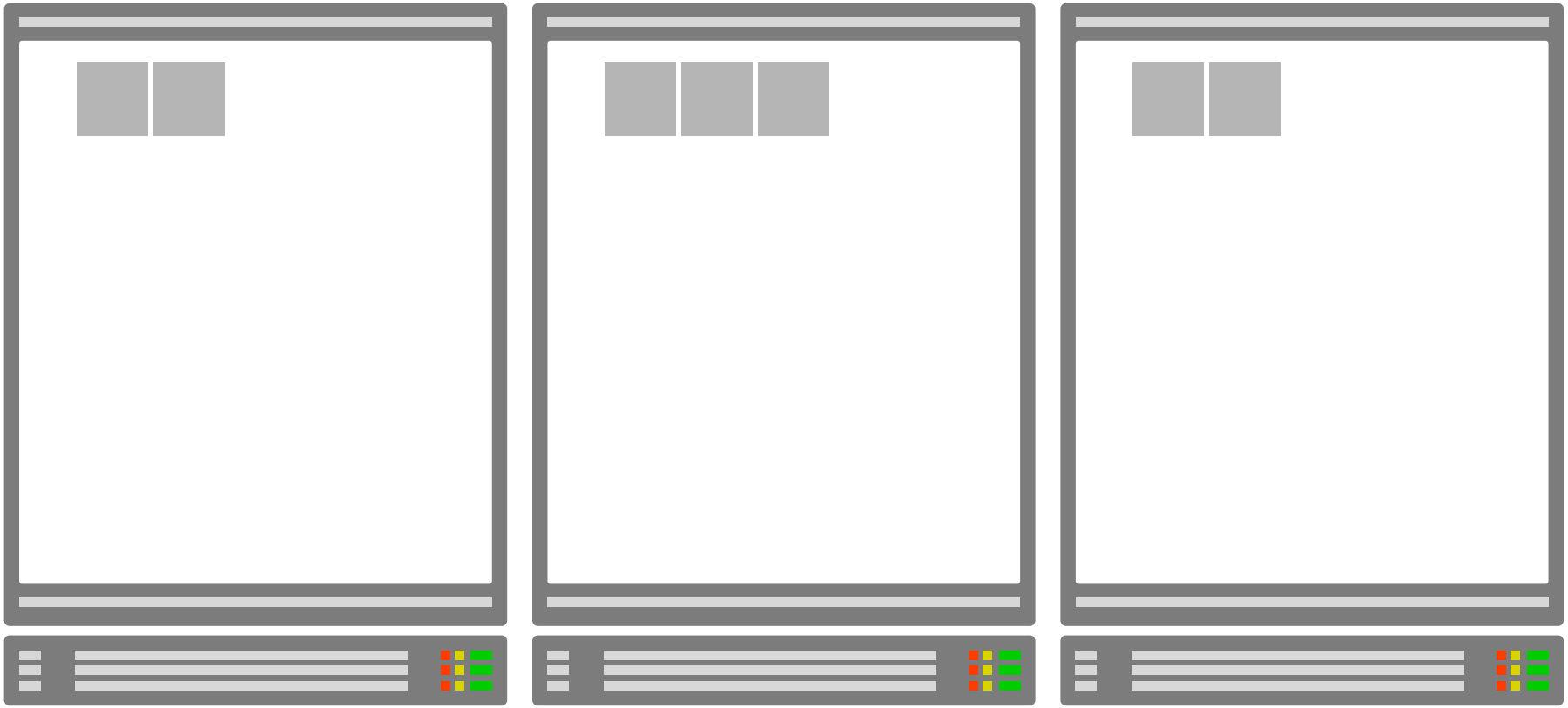


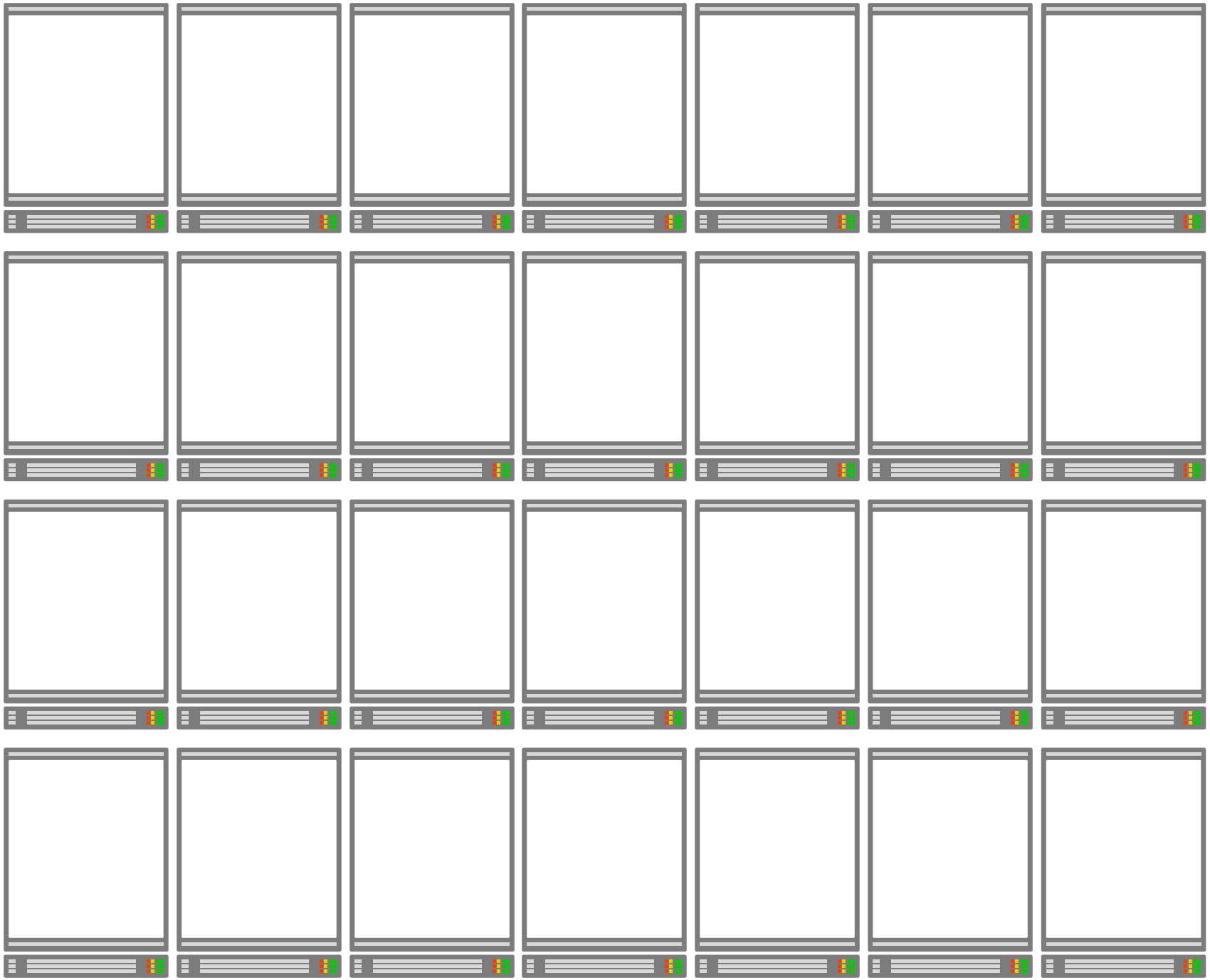
Managing allocations

Challenge 1



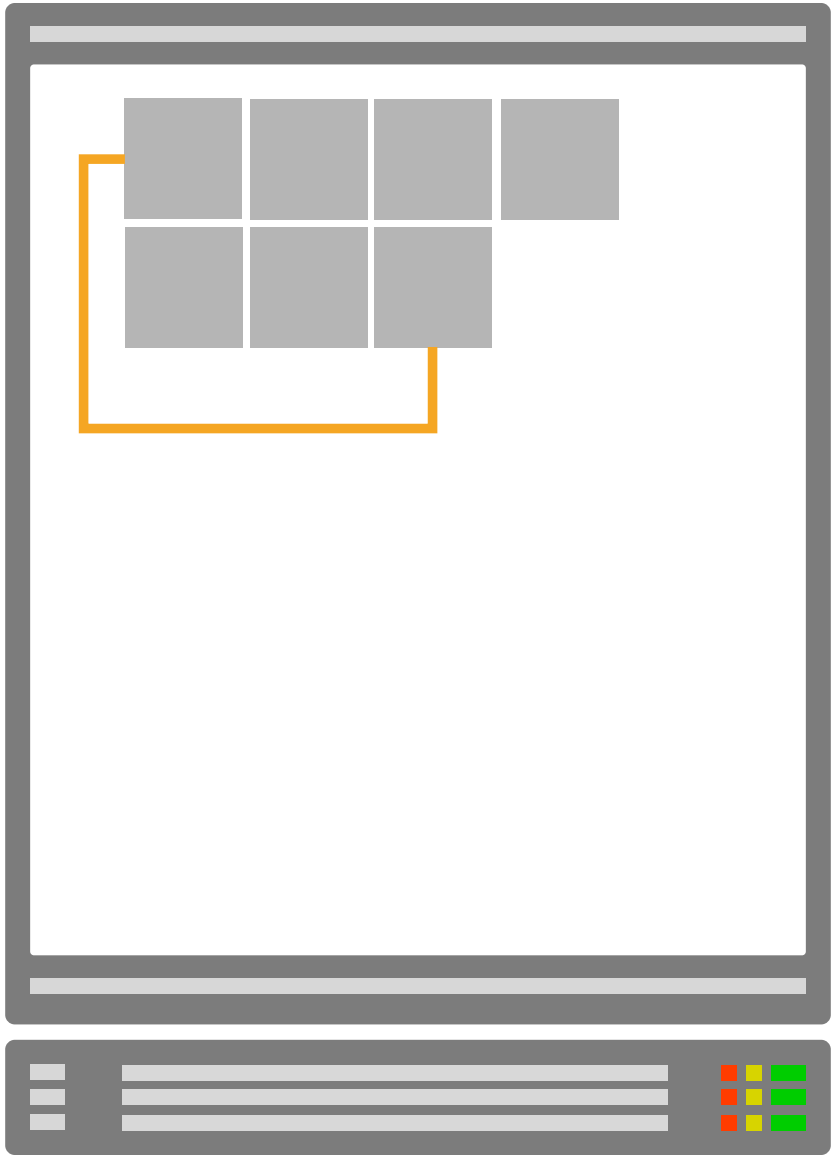


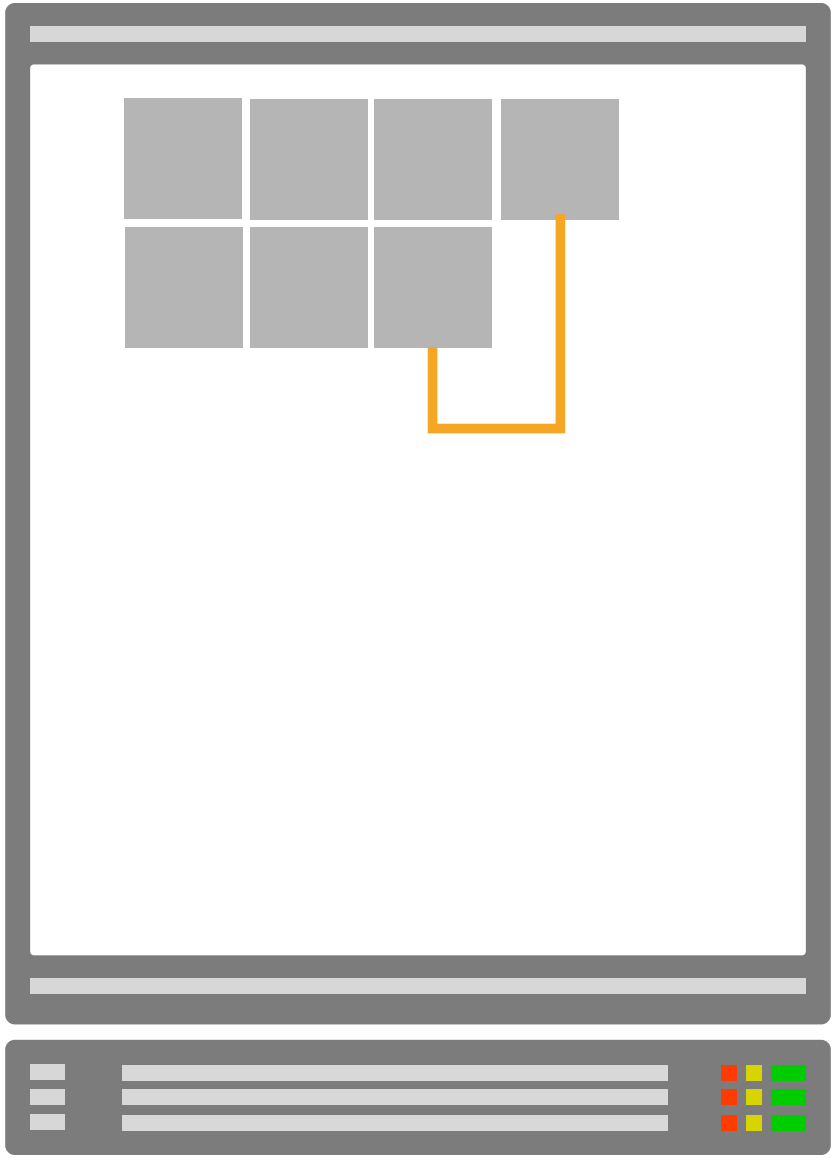


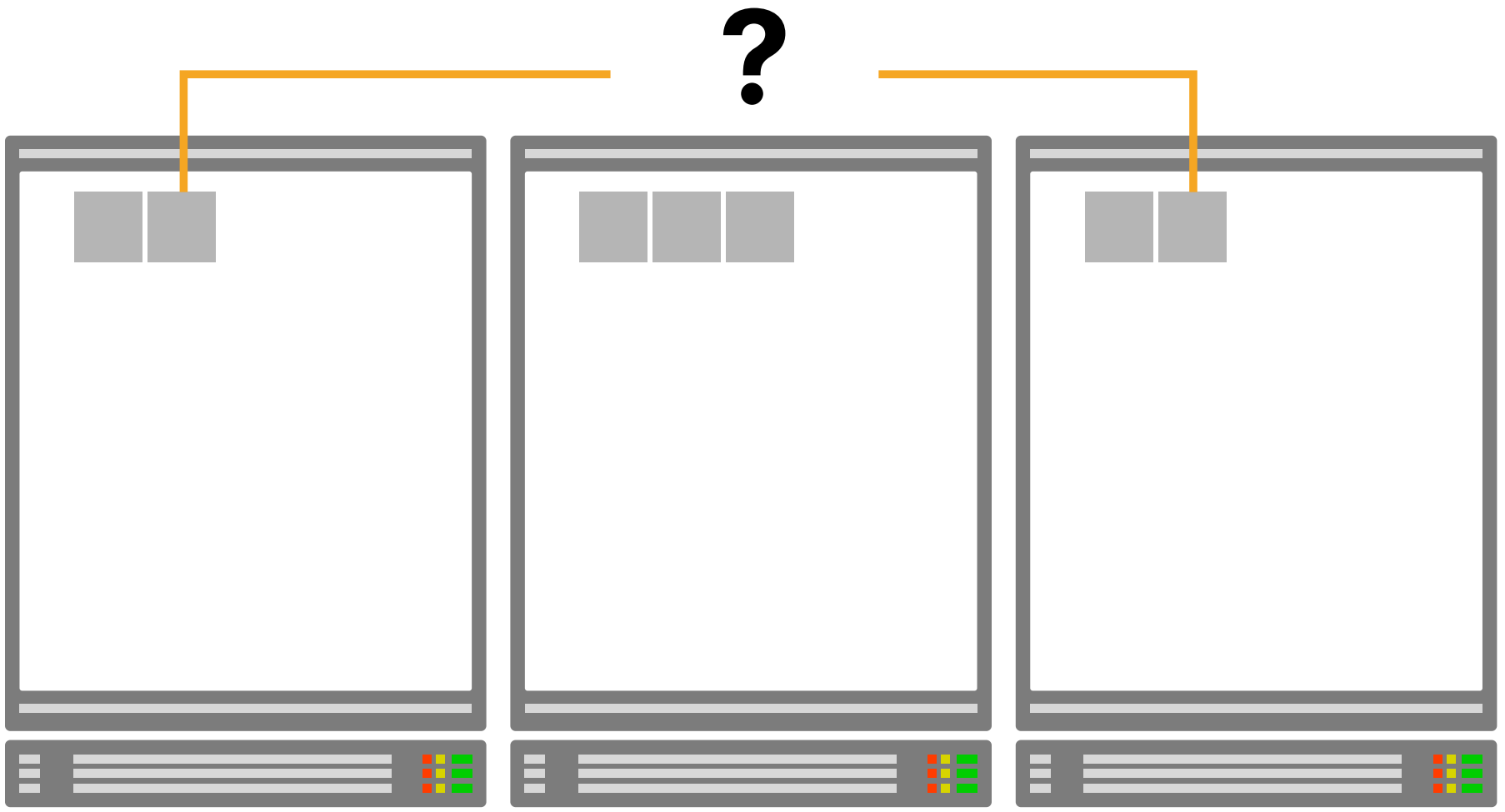


Service discovery and networking

Challenge 2







What if...

Container orchestrators

Data centre as 1 computer

Smart scheduler

Programmable

Container orchestrators

Data centre as 1 computer

Smart scheduler

Programmable

Container orchestrators

Data centre as 1 computer

Smart scheduler

Programmable

Container orchestrators



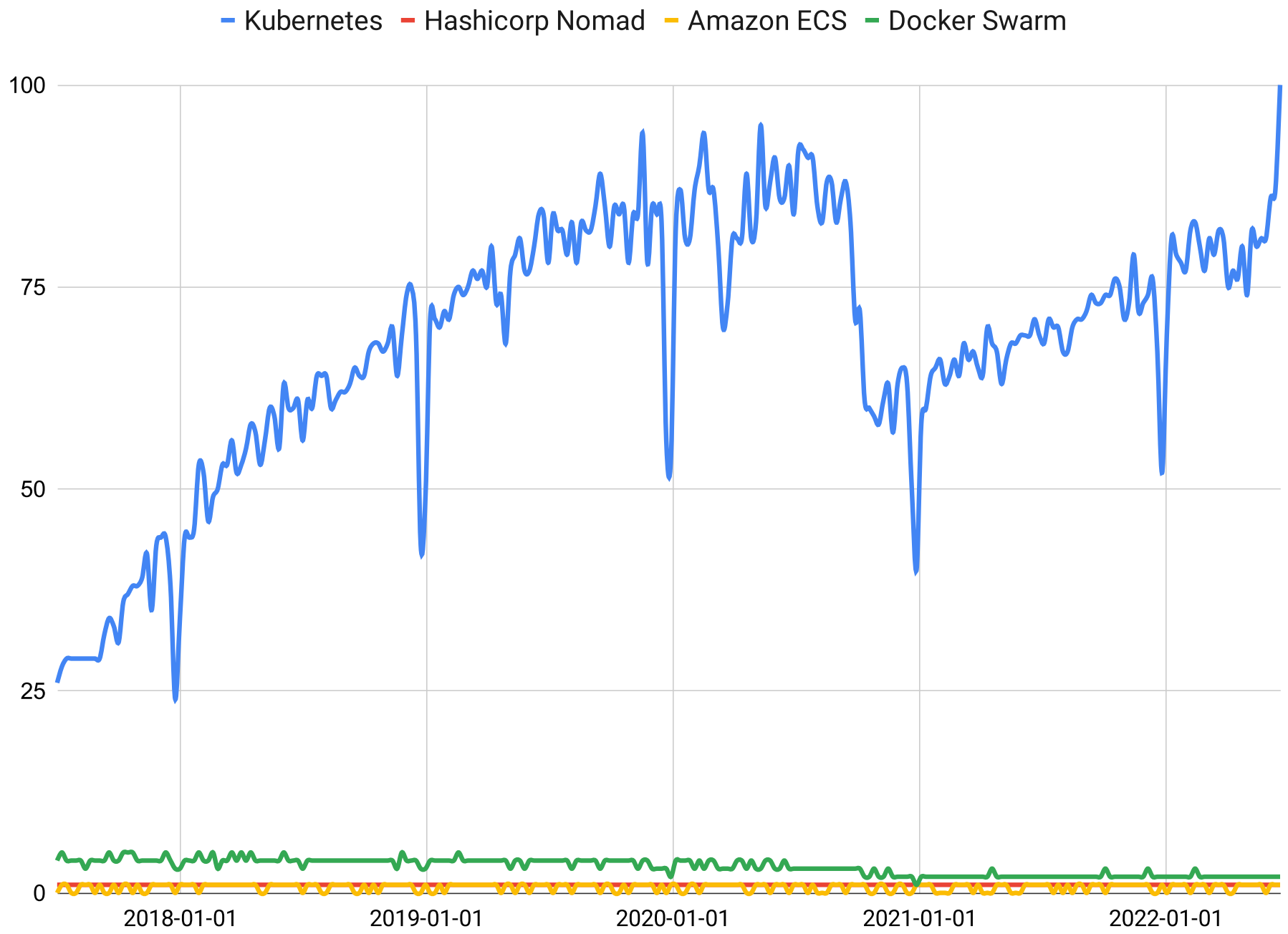
Mesos



HashiCorp
Nomad



kubernetes

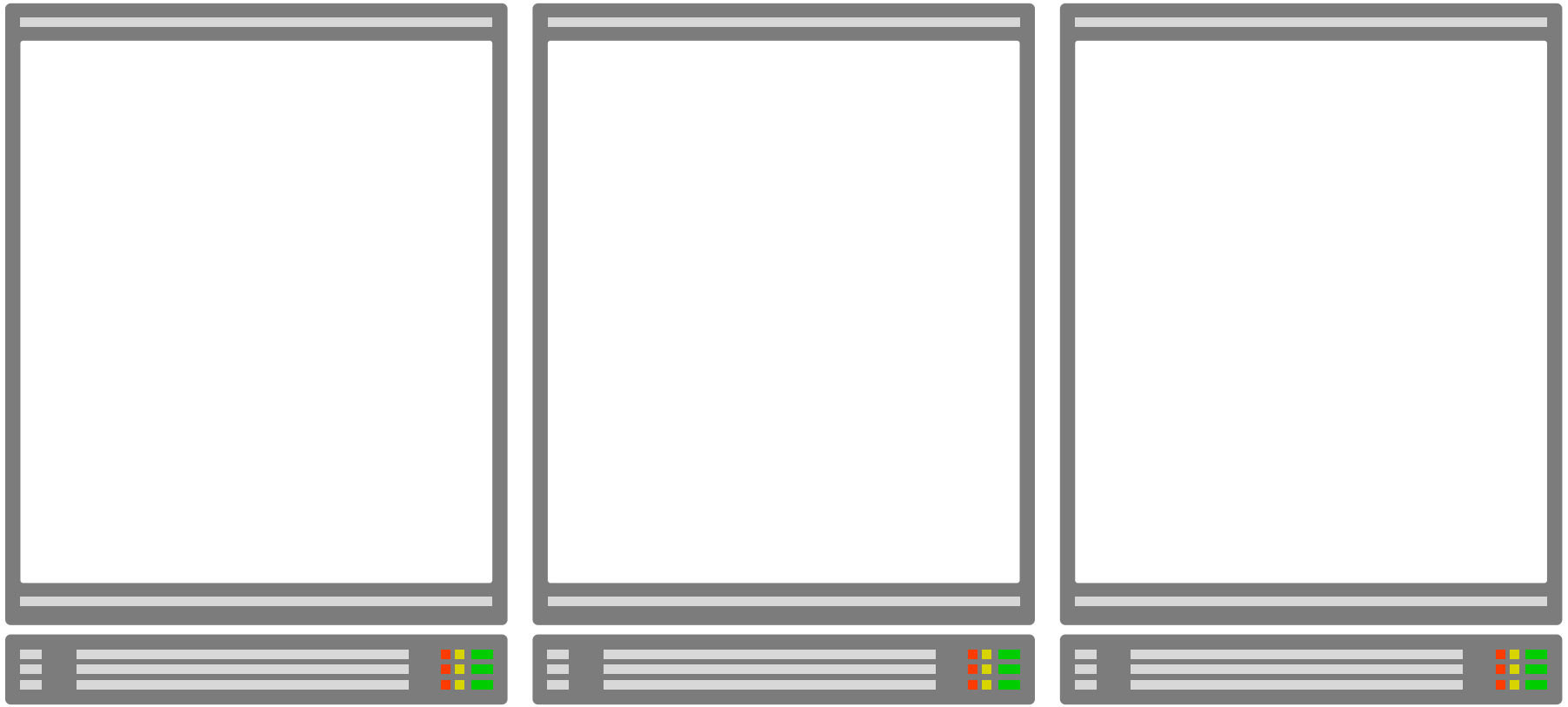

Kubernetes
Hashicorp Nomad
Amazon ECS
Docker Swarm
100
75
50
25
0
2018-01-01
2019-01-01
2020-01-01
2021-01-01
2022-01-01

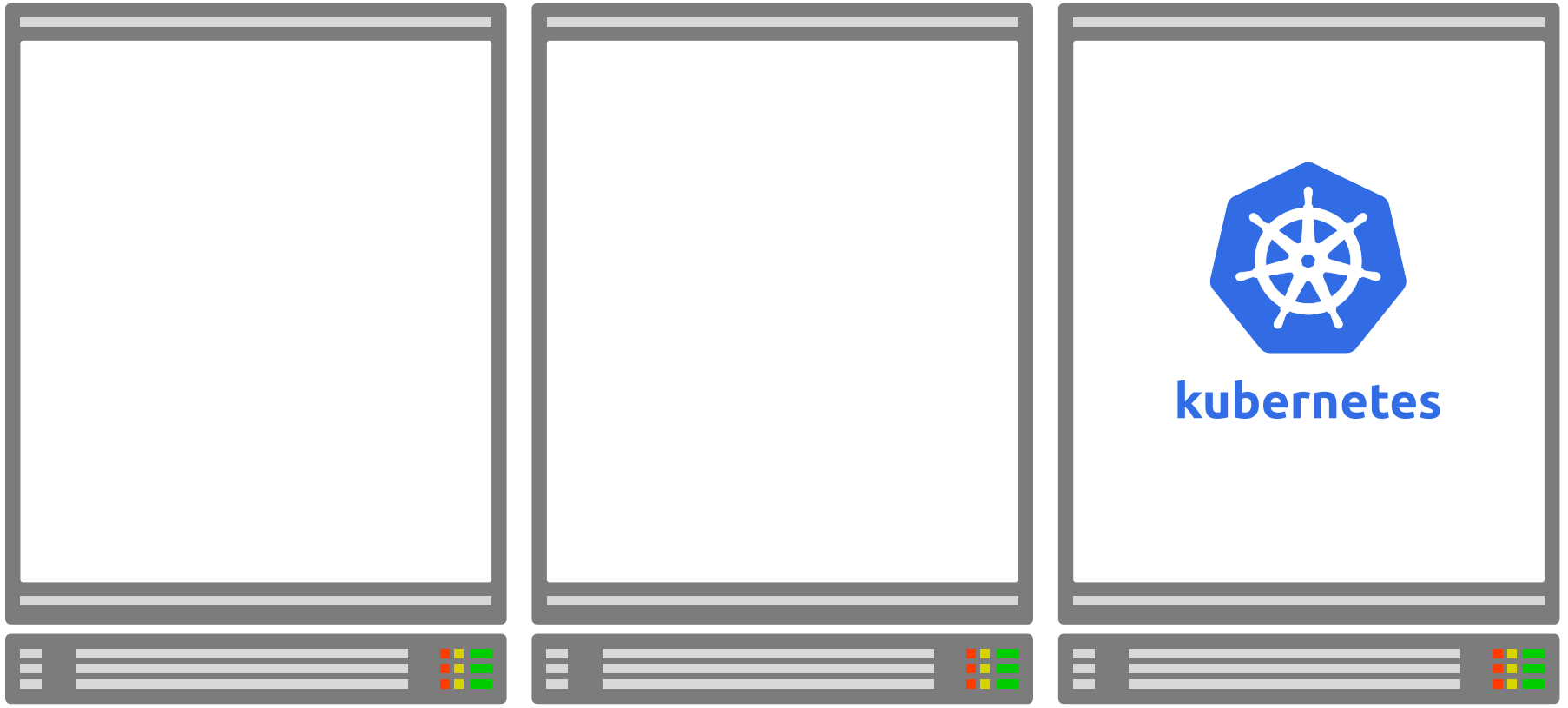


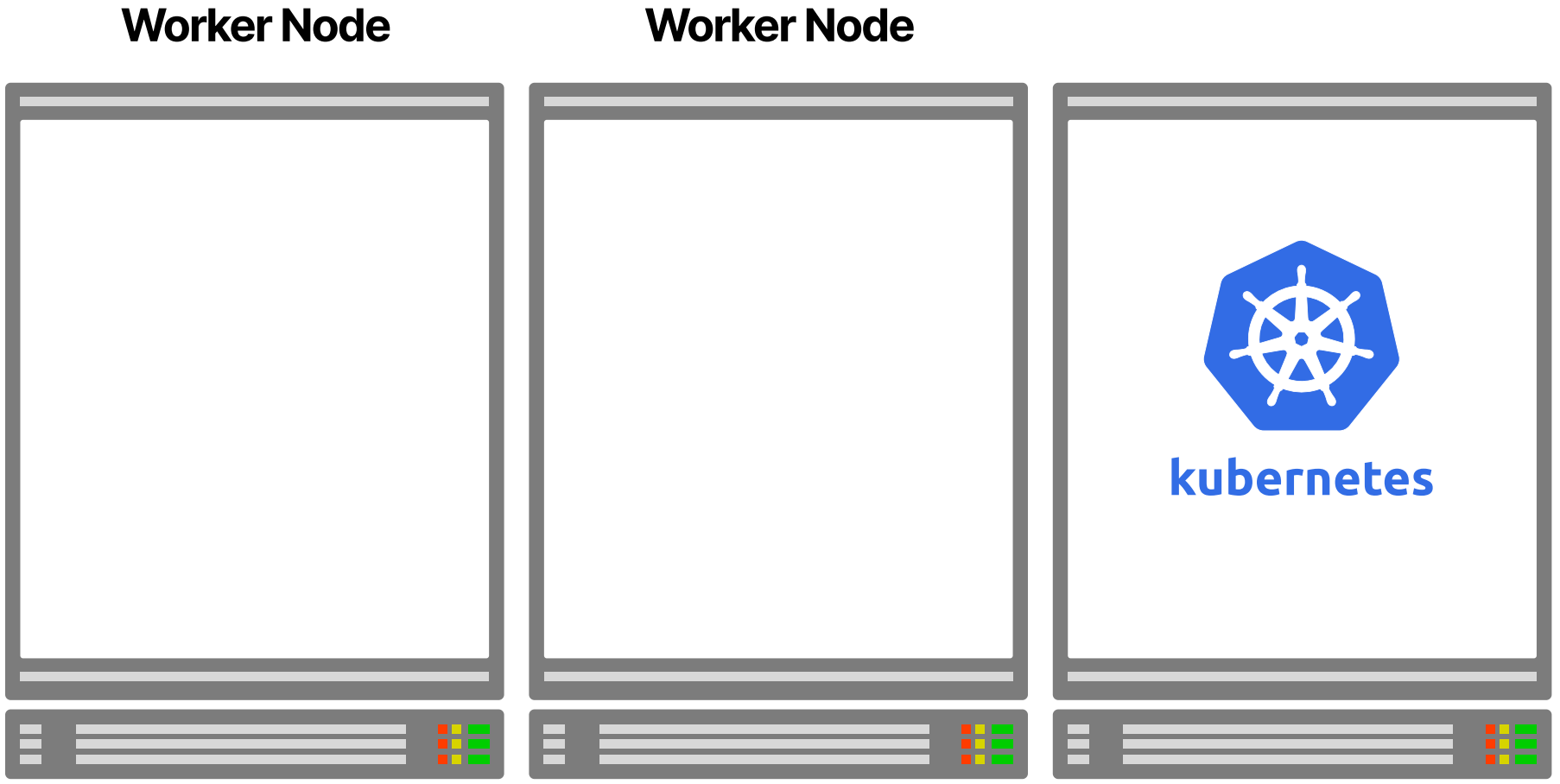
kubernetes

Datacentre as a single computer

01

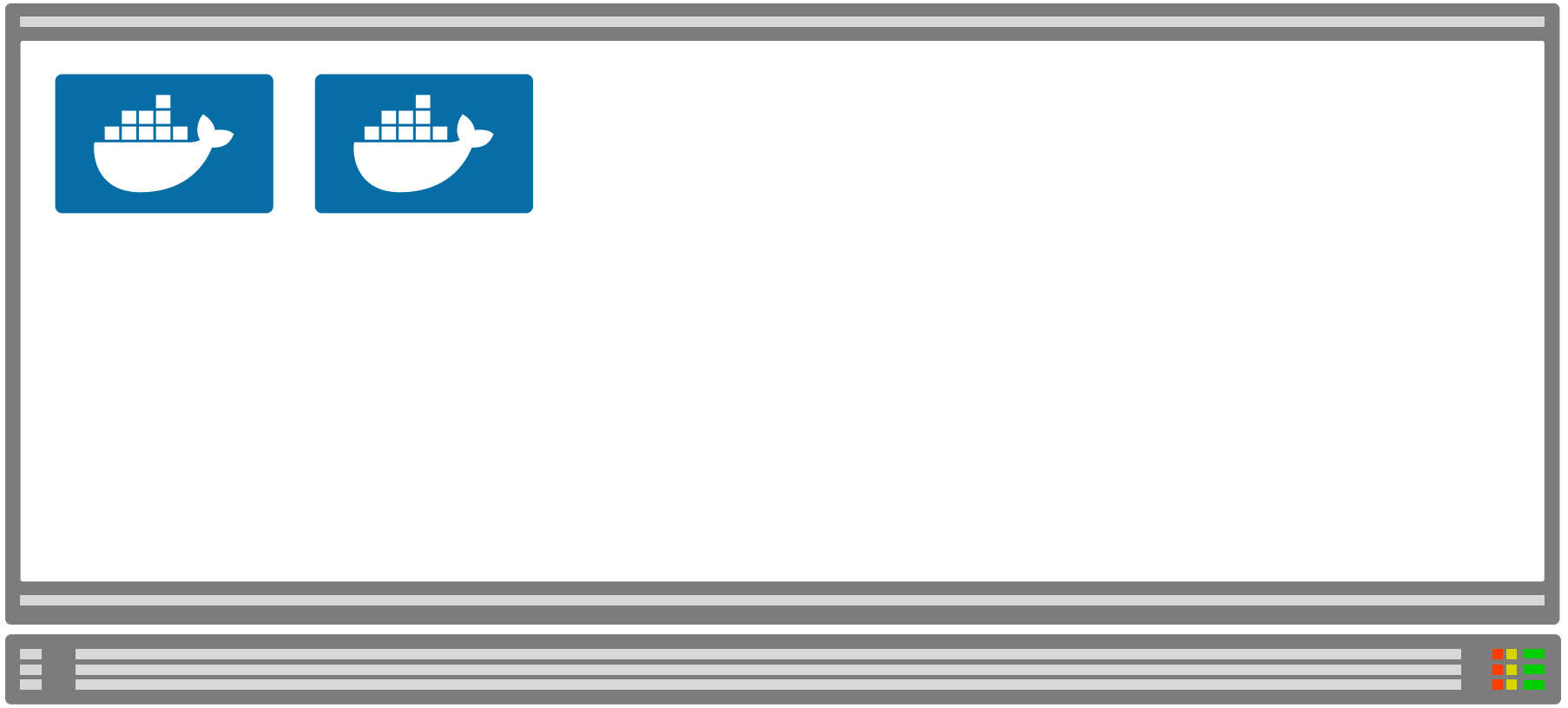


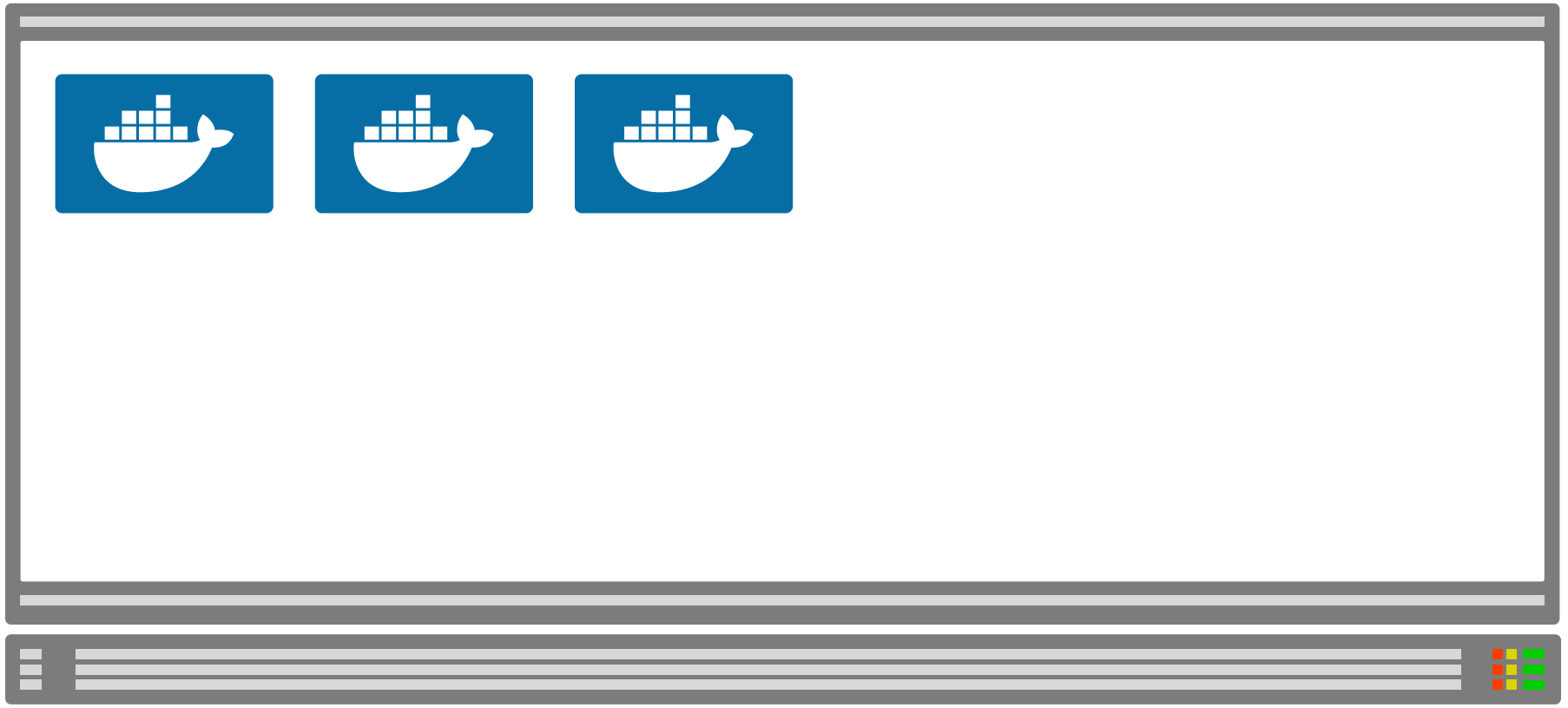

kubernetes

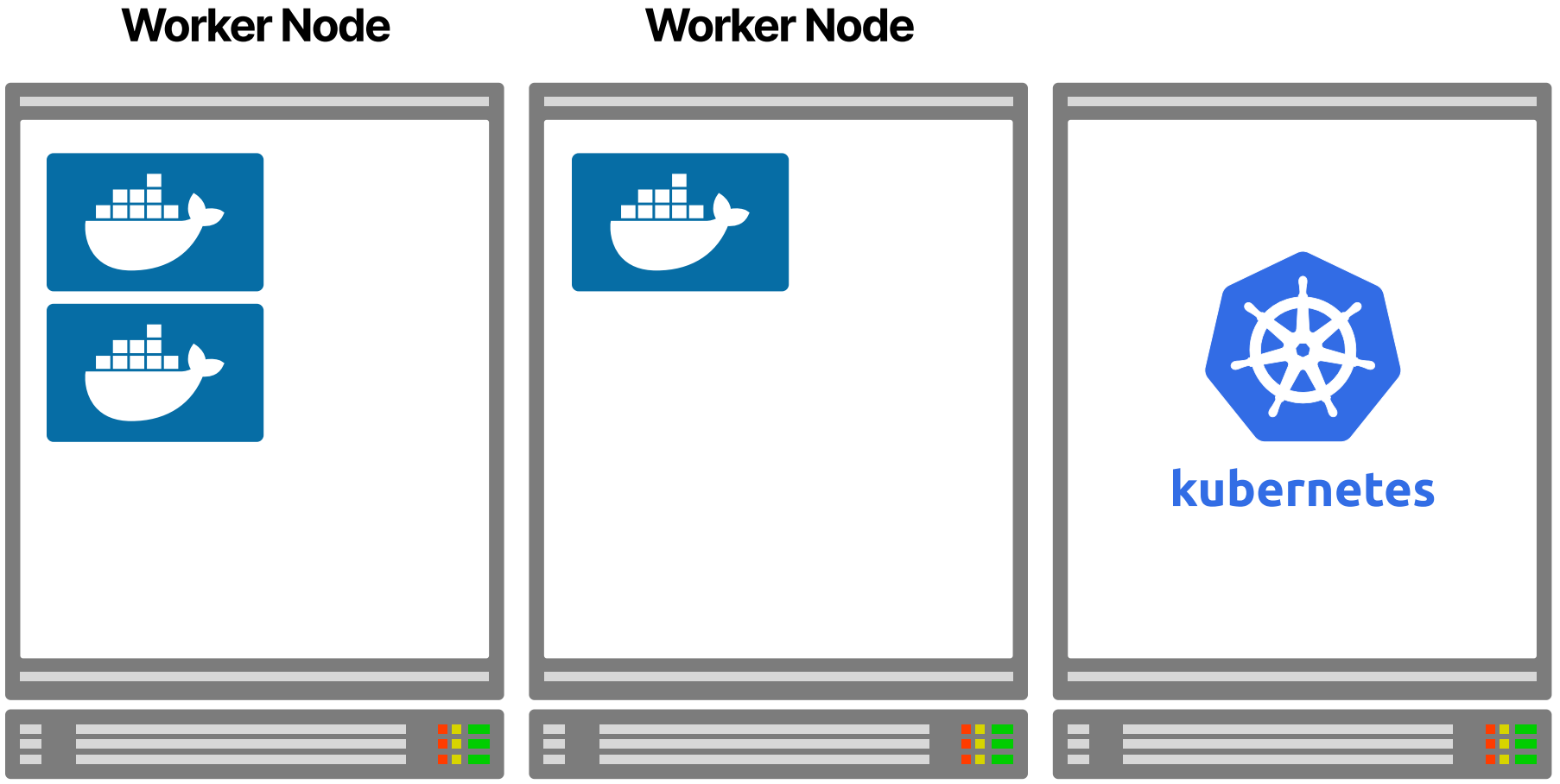

Worker Node
Worker Node
kubernetes



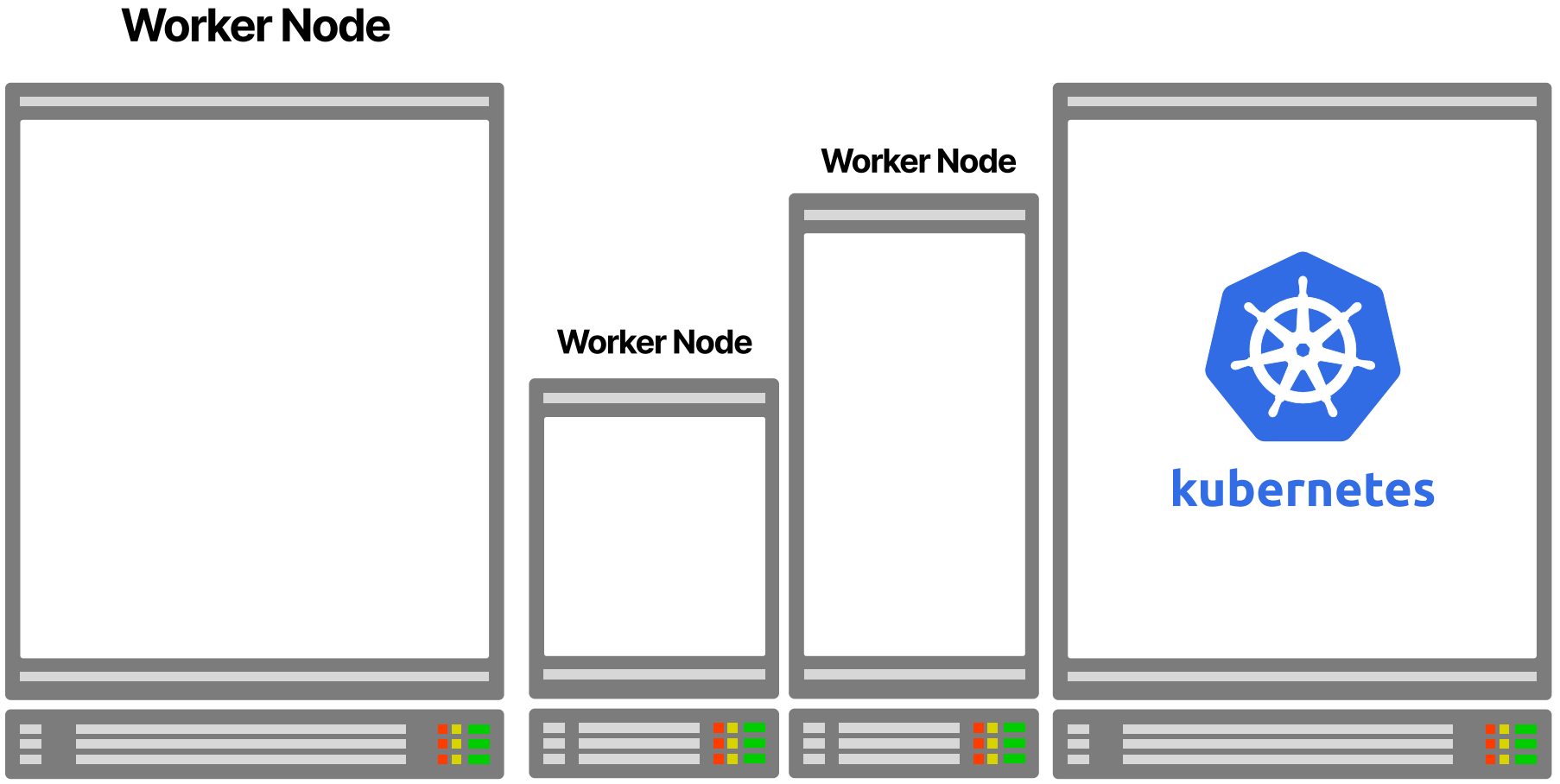


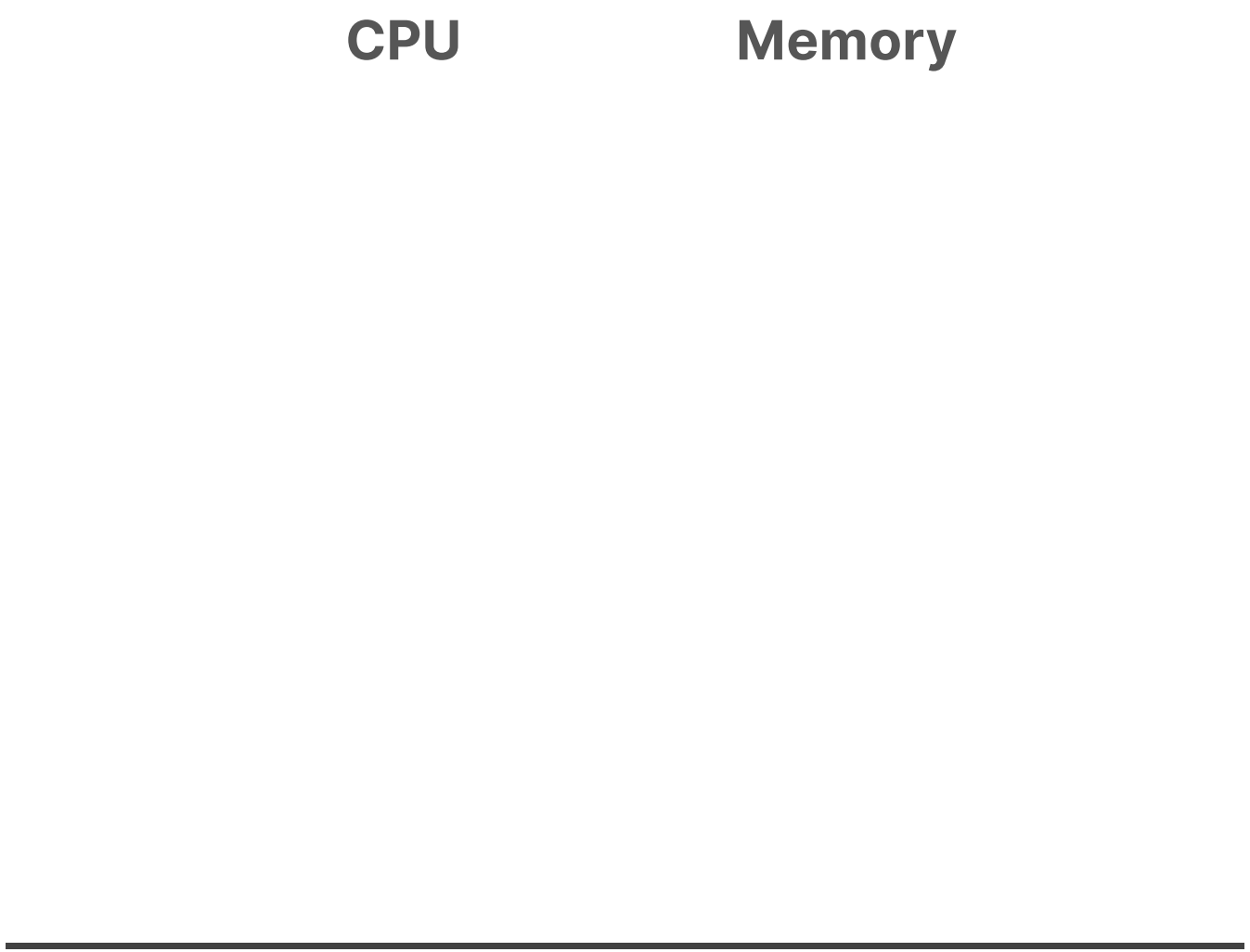


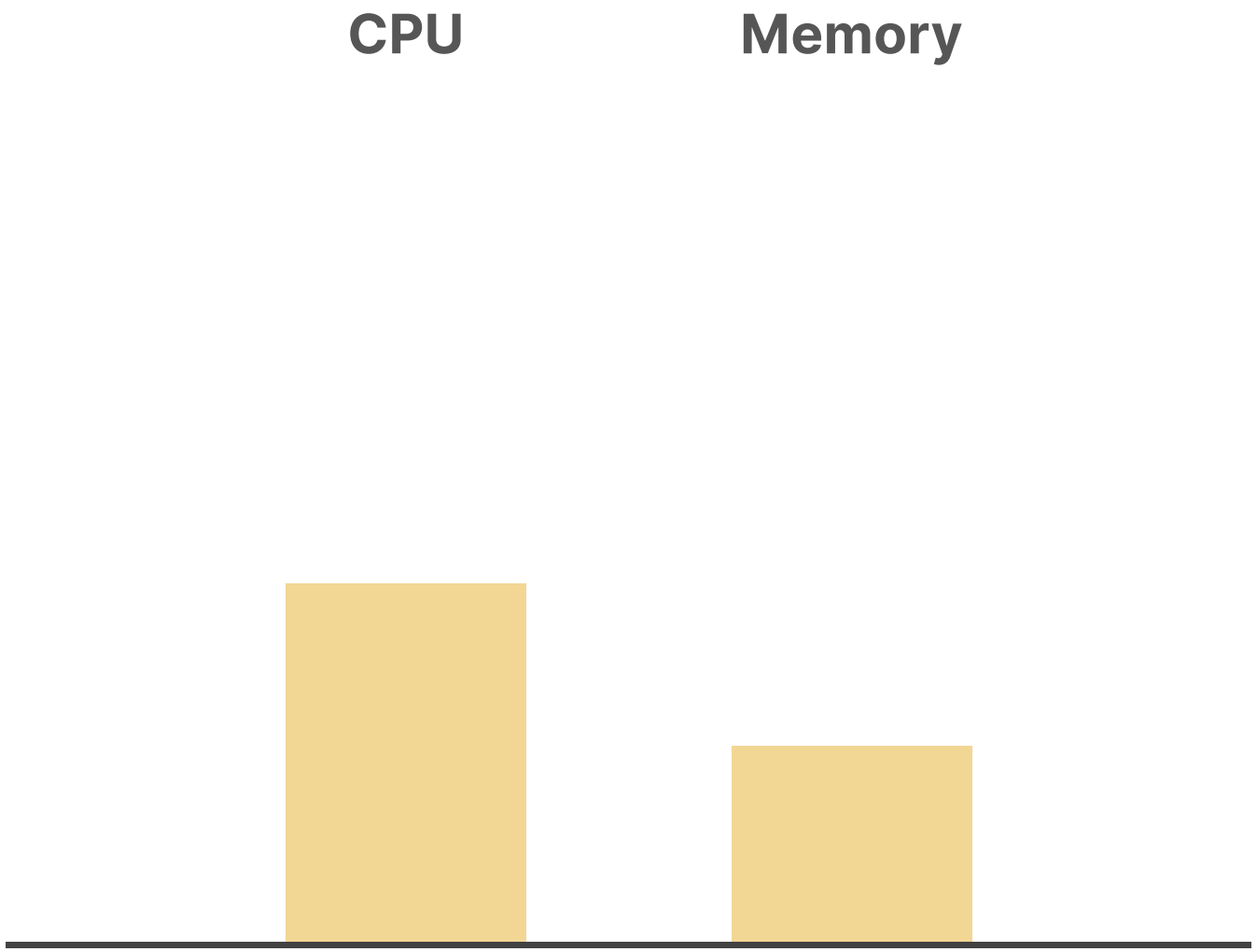


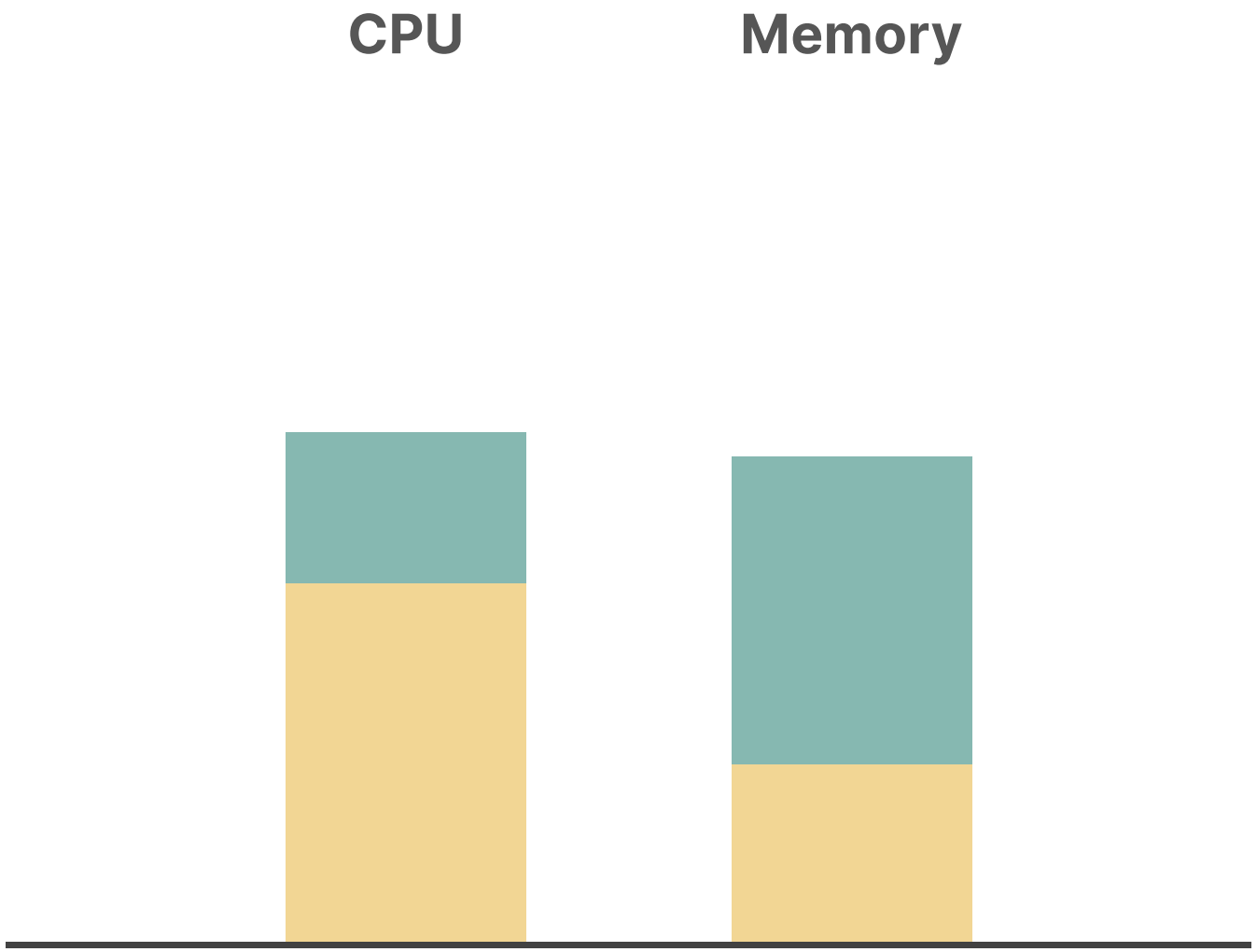

Worker Node
Worker Node
kubernetes

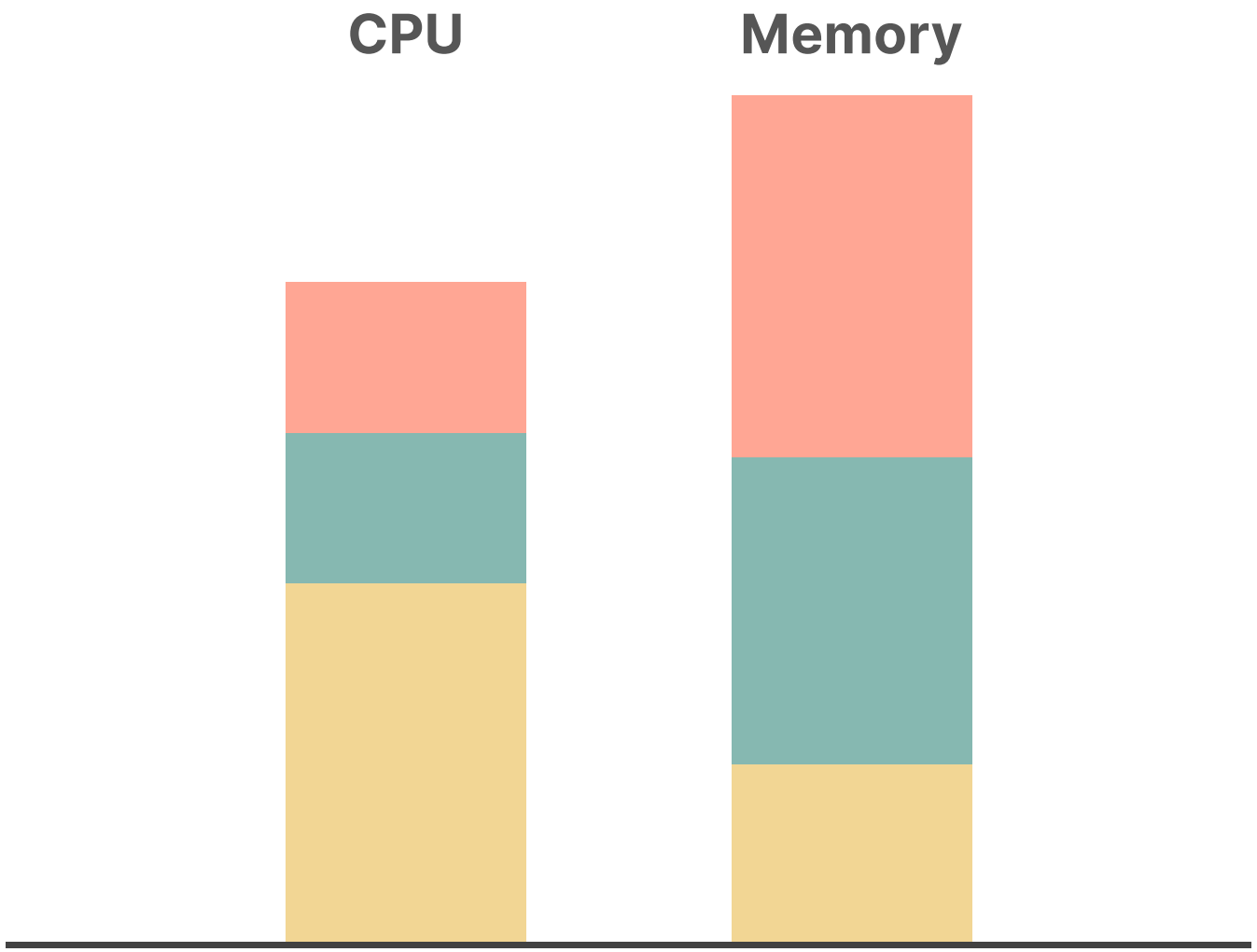
Cluster resources


Worker Node
Worker Node
Worker Node
kubernetes


CPU
Memory

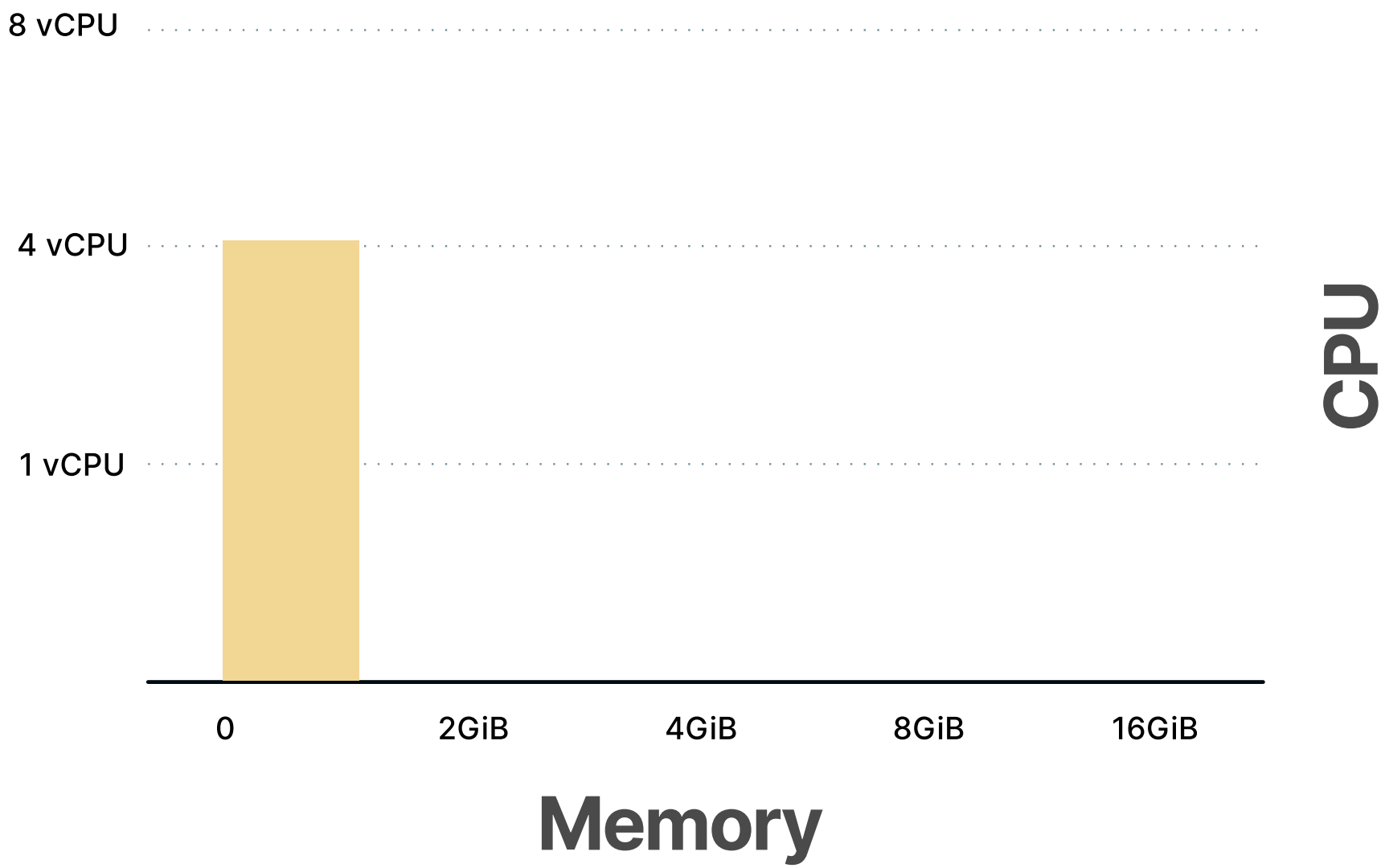

CPU
Memory

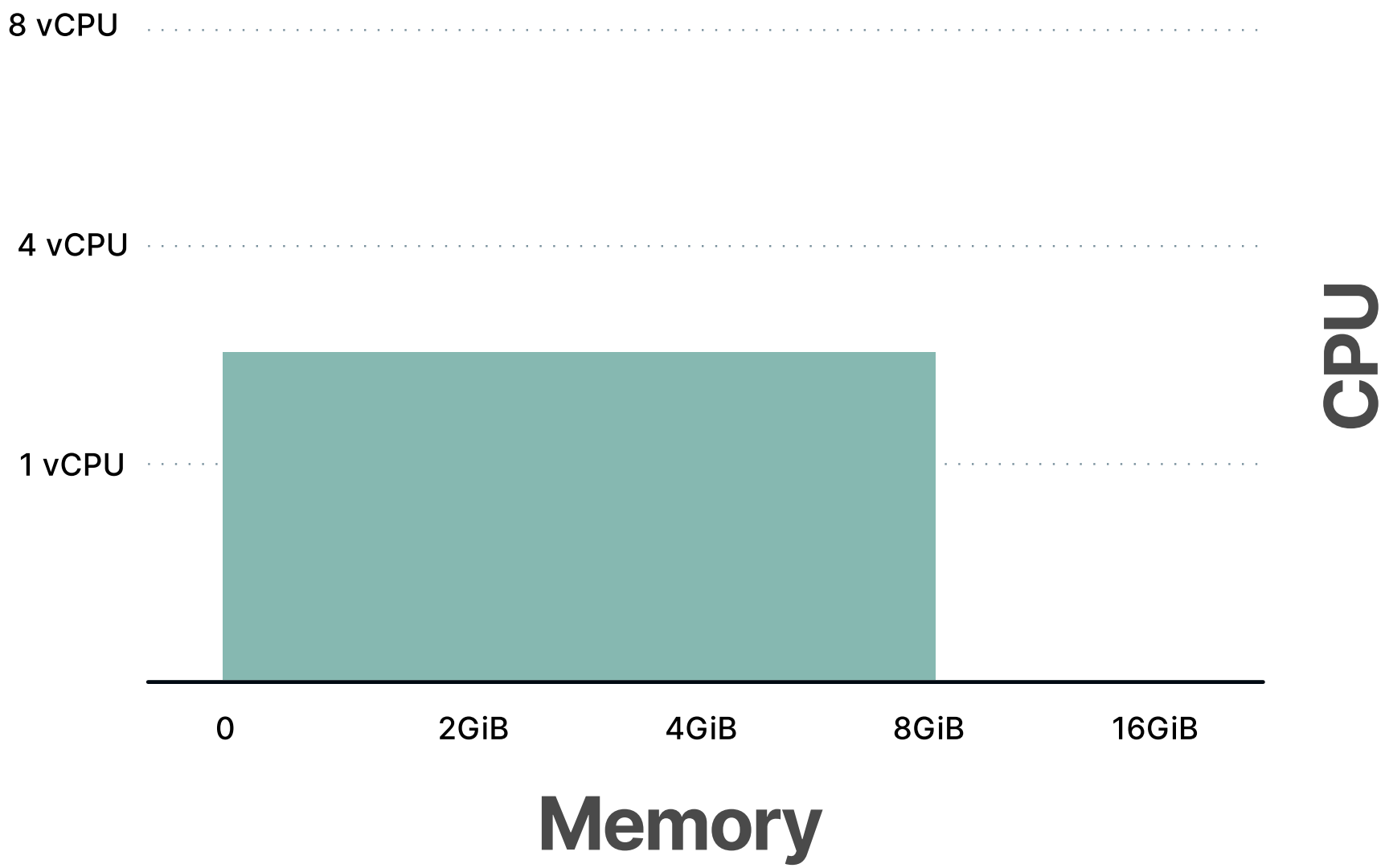

CPU
Memory

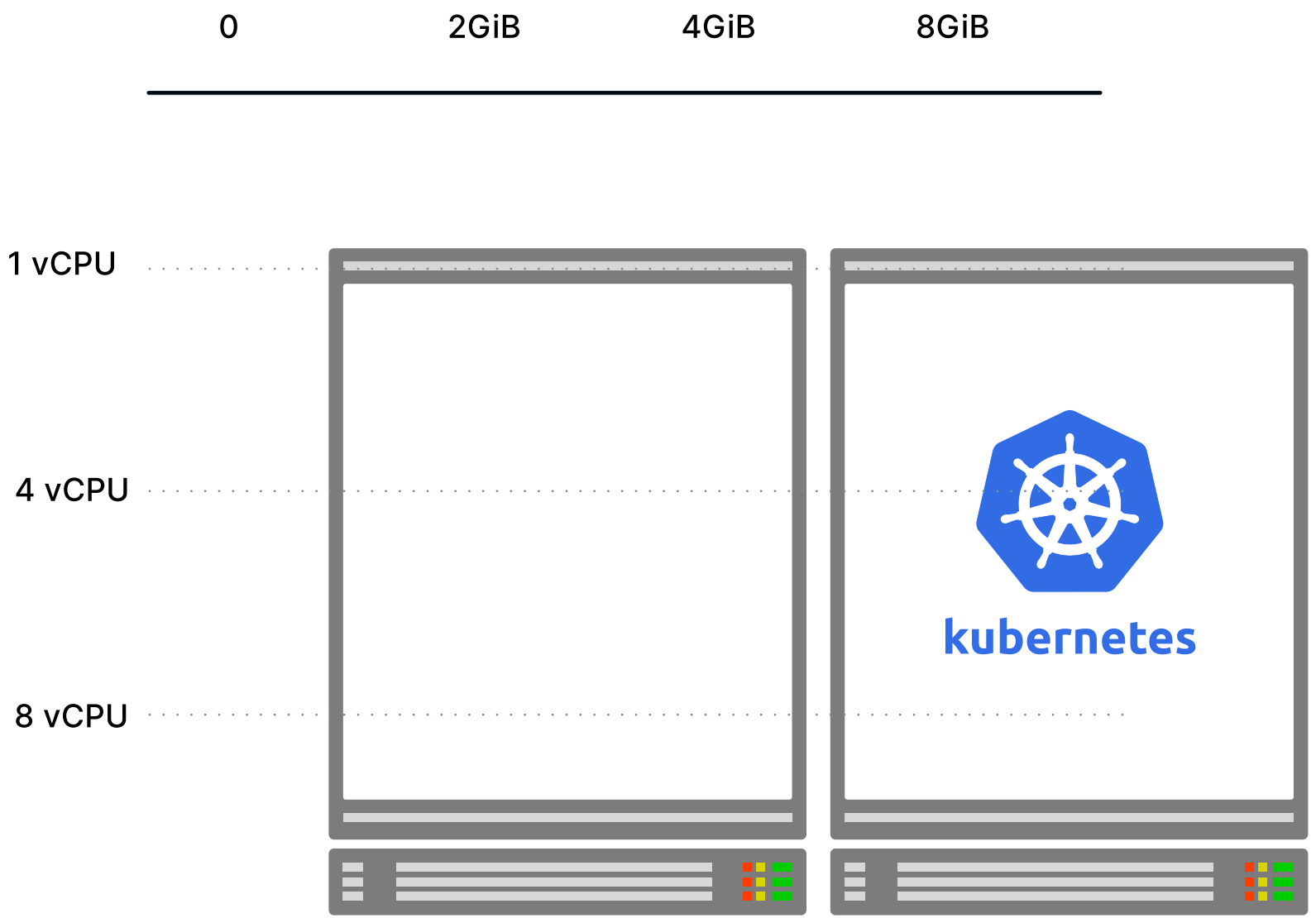

CPU
Memory

Smart scheduler

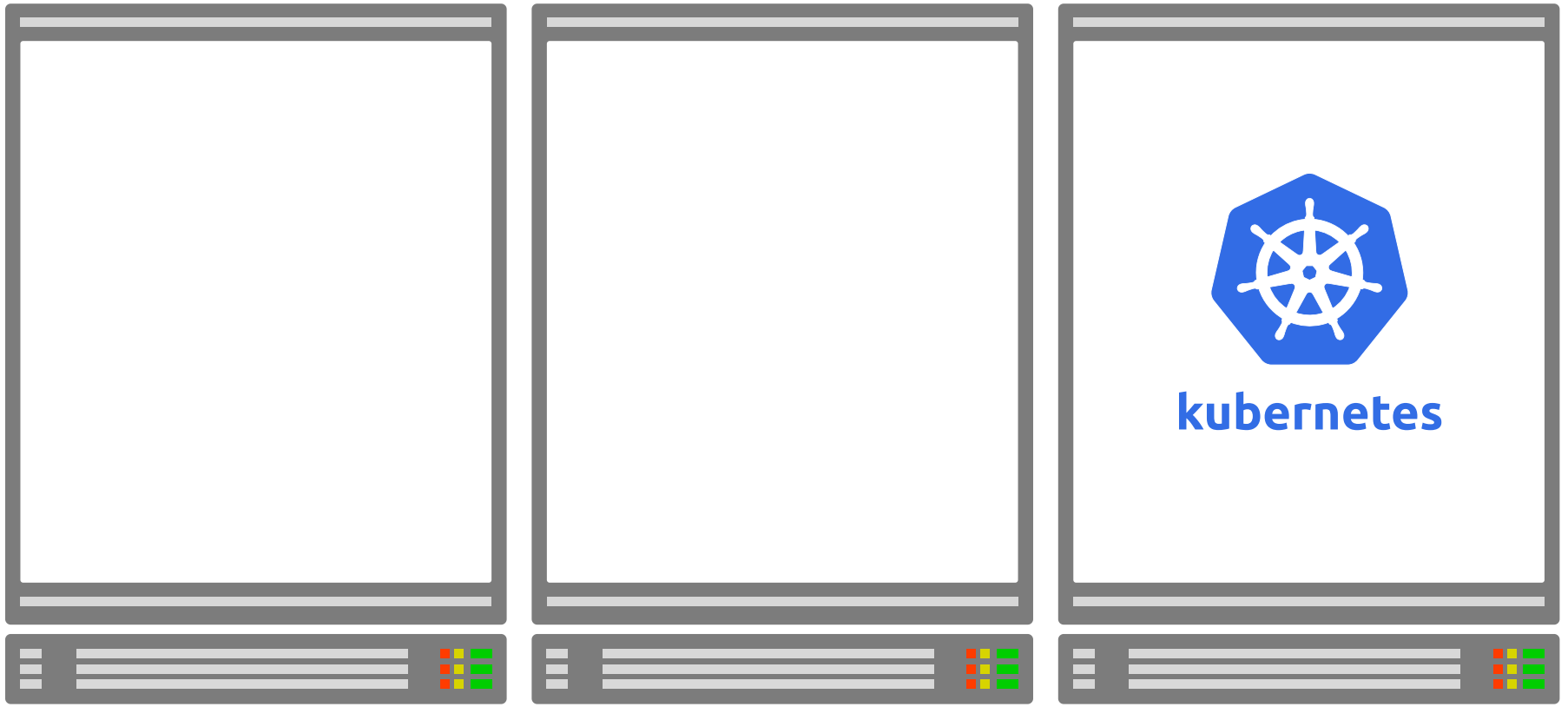
02

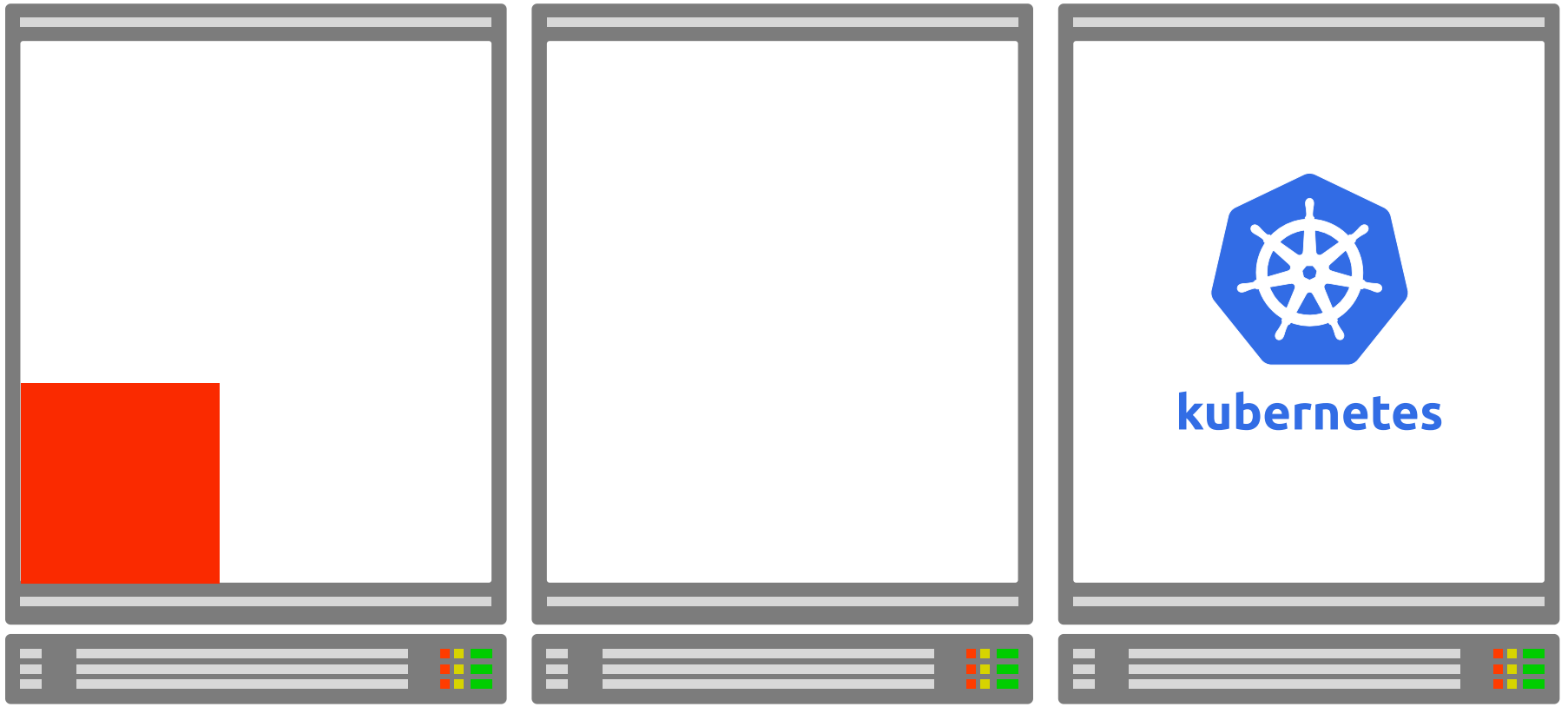

8 vCPU
4 vCPU
1 vCPU
CPU
0
2GiB
4GiB
8GiB
16GiB
Memory


8 vCPU
4 vCPU
1 vCPU
CPU
0
2GiB
4GiB
8GiB
16GiB
Memory

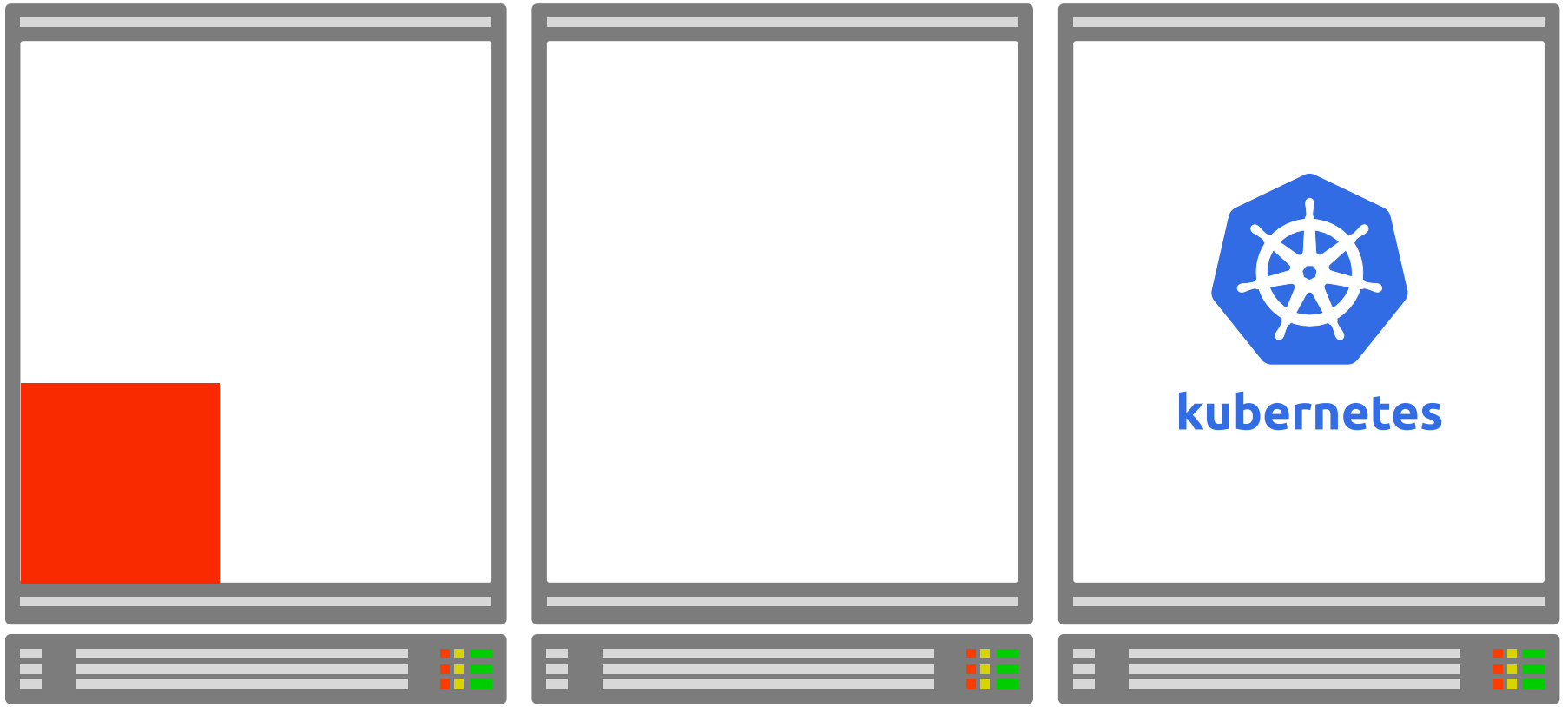

0
2GiB
4GiB
8GiB
1 vCPU
4 vCPU
8 vCPU
kubernetes

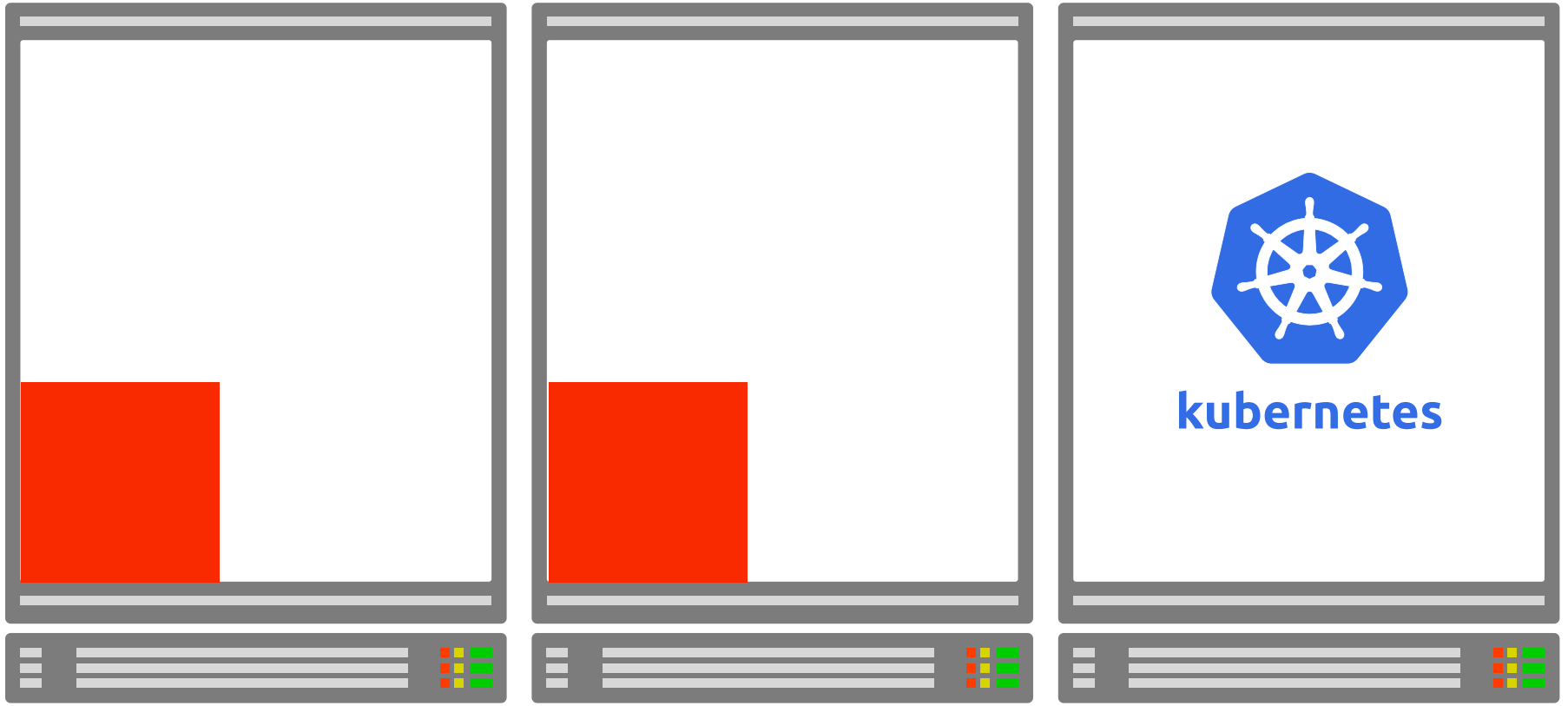
Next container



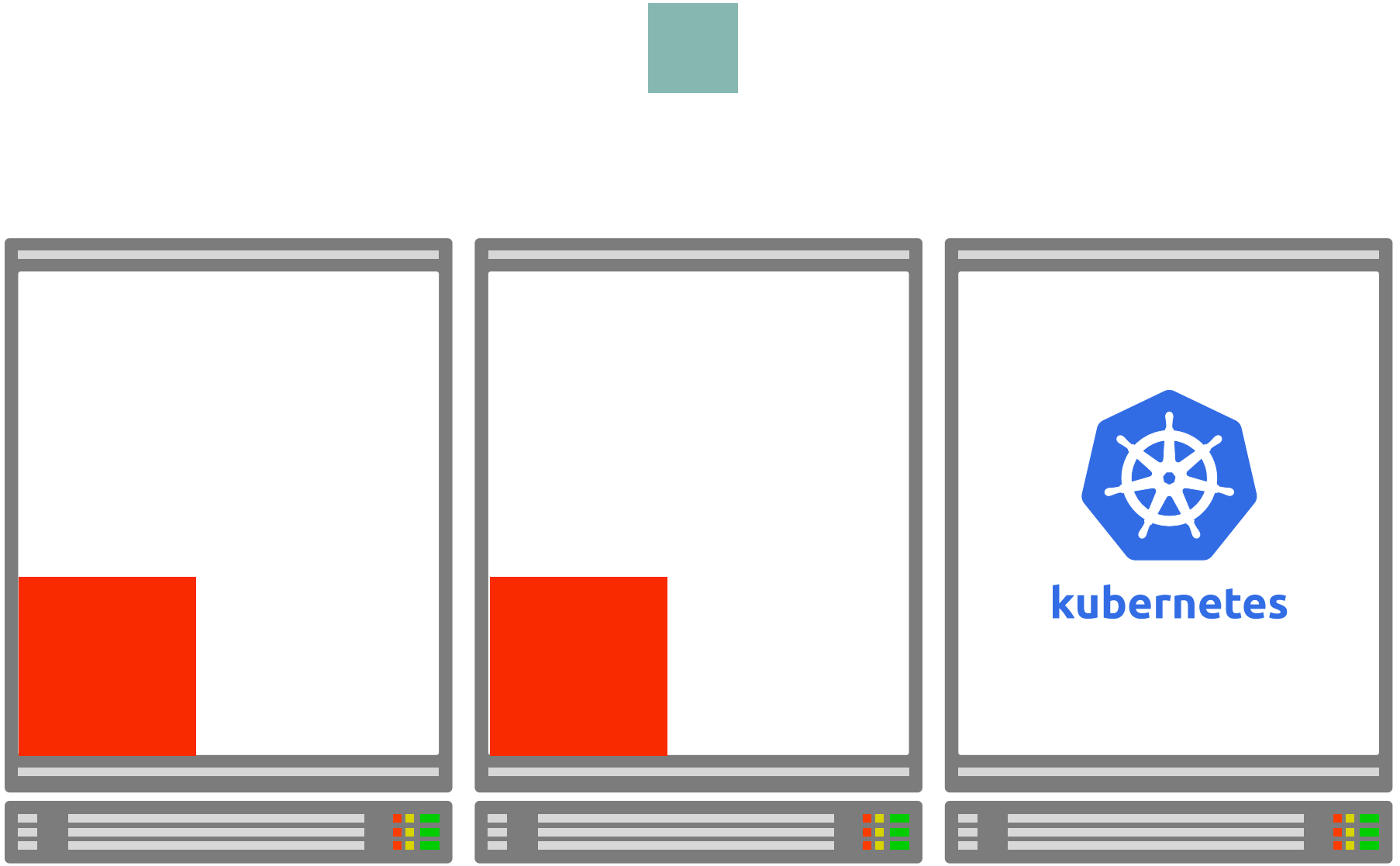

kubernetes

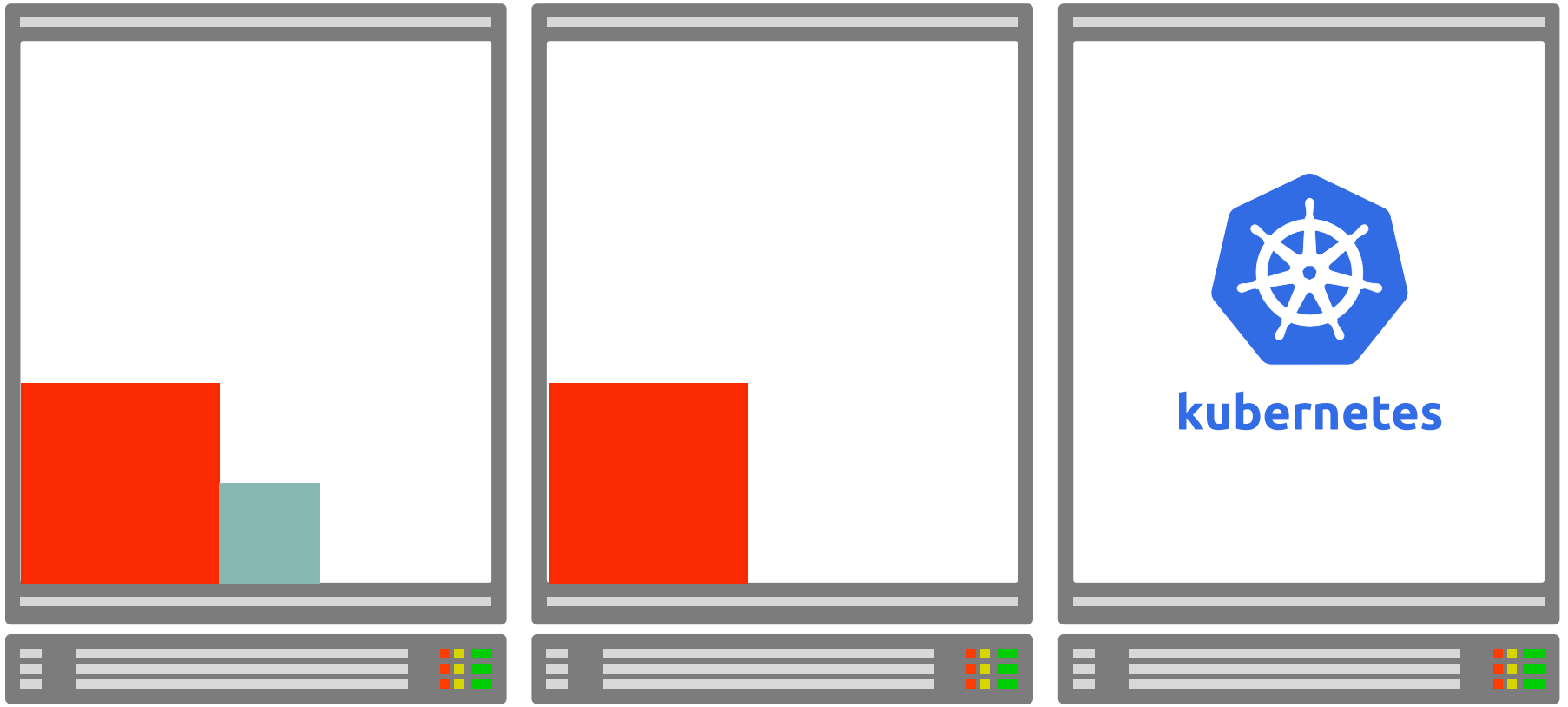
Next container



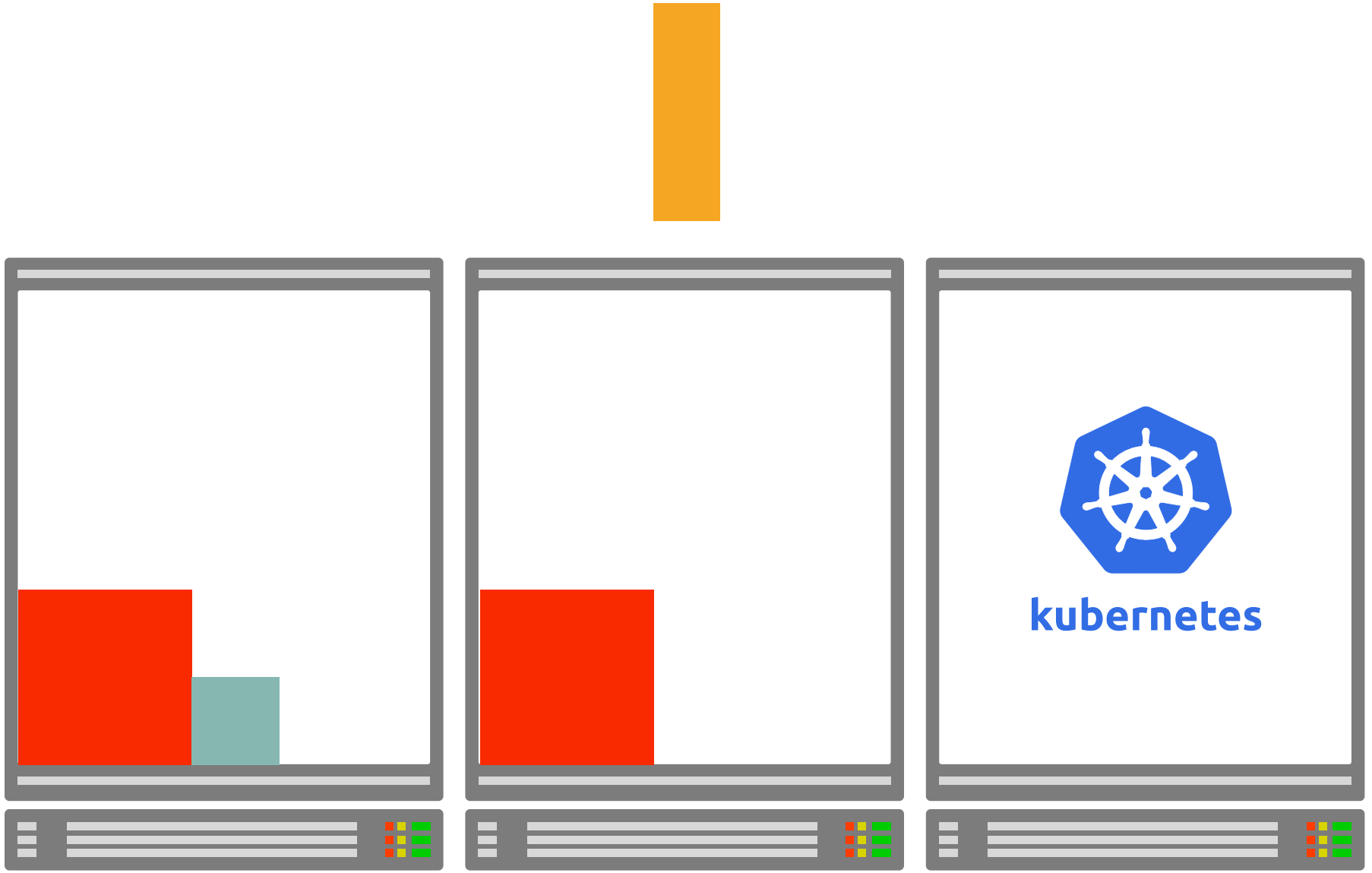

kubernetes

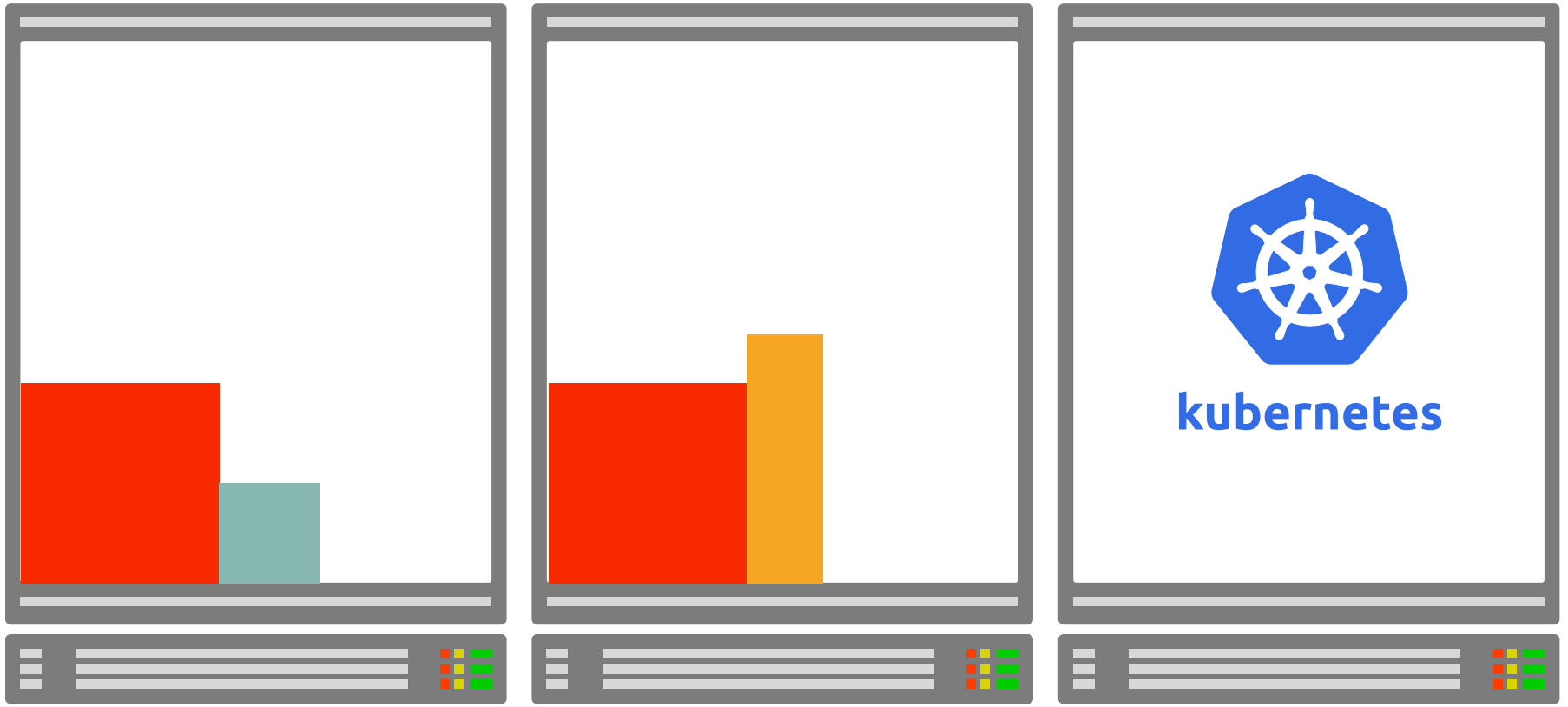
Next container



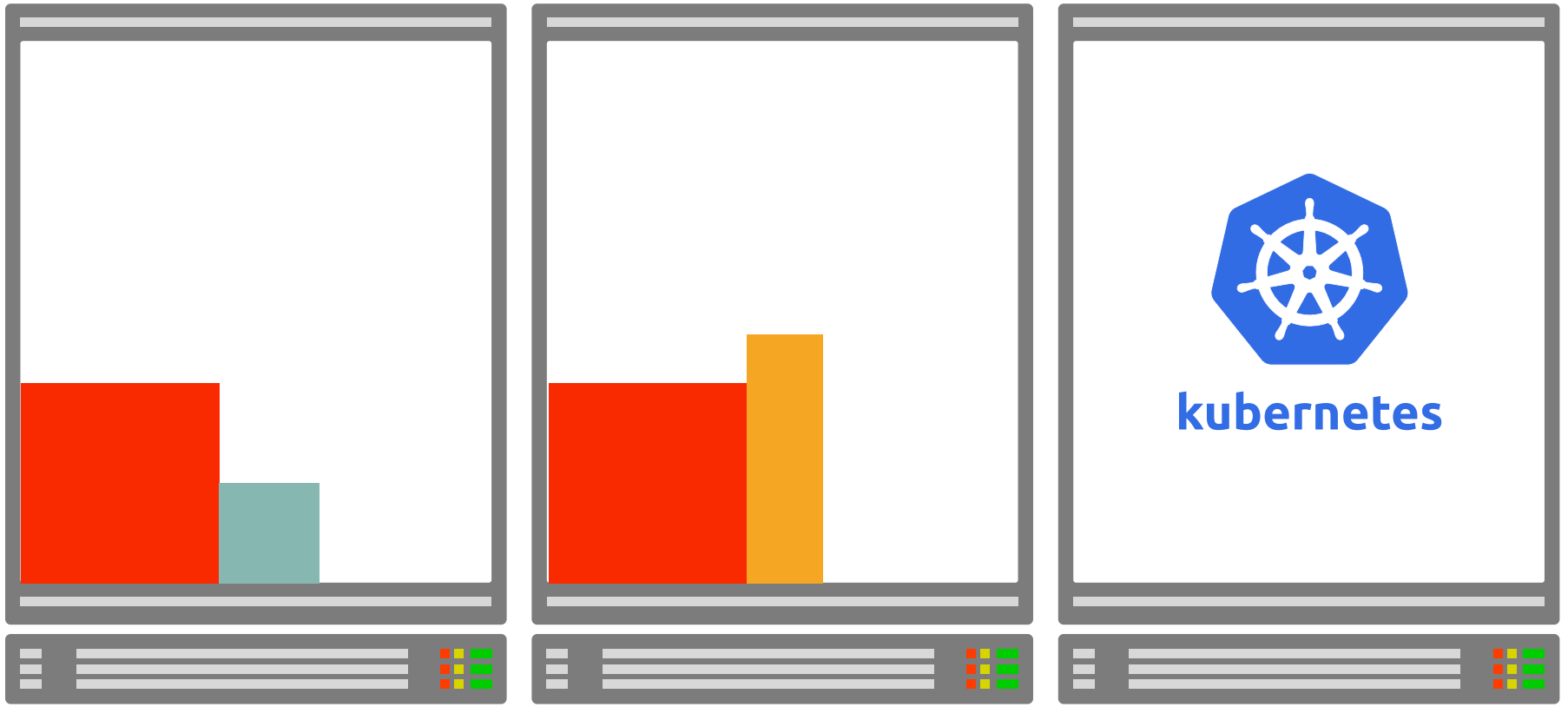

kubernetes

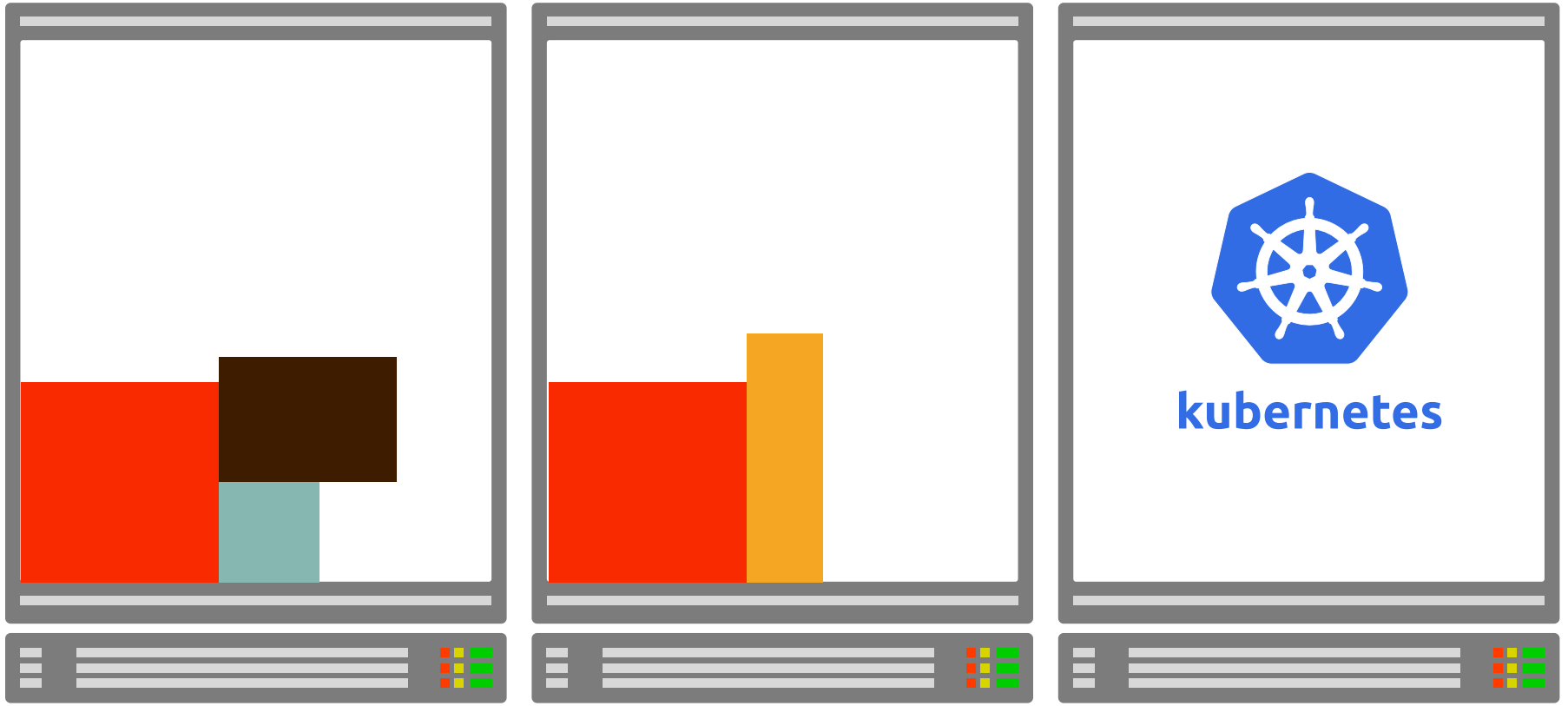
Next container

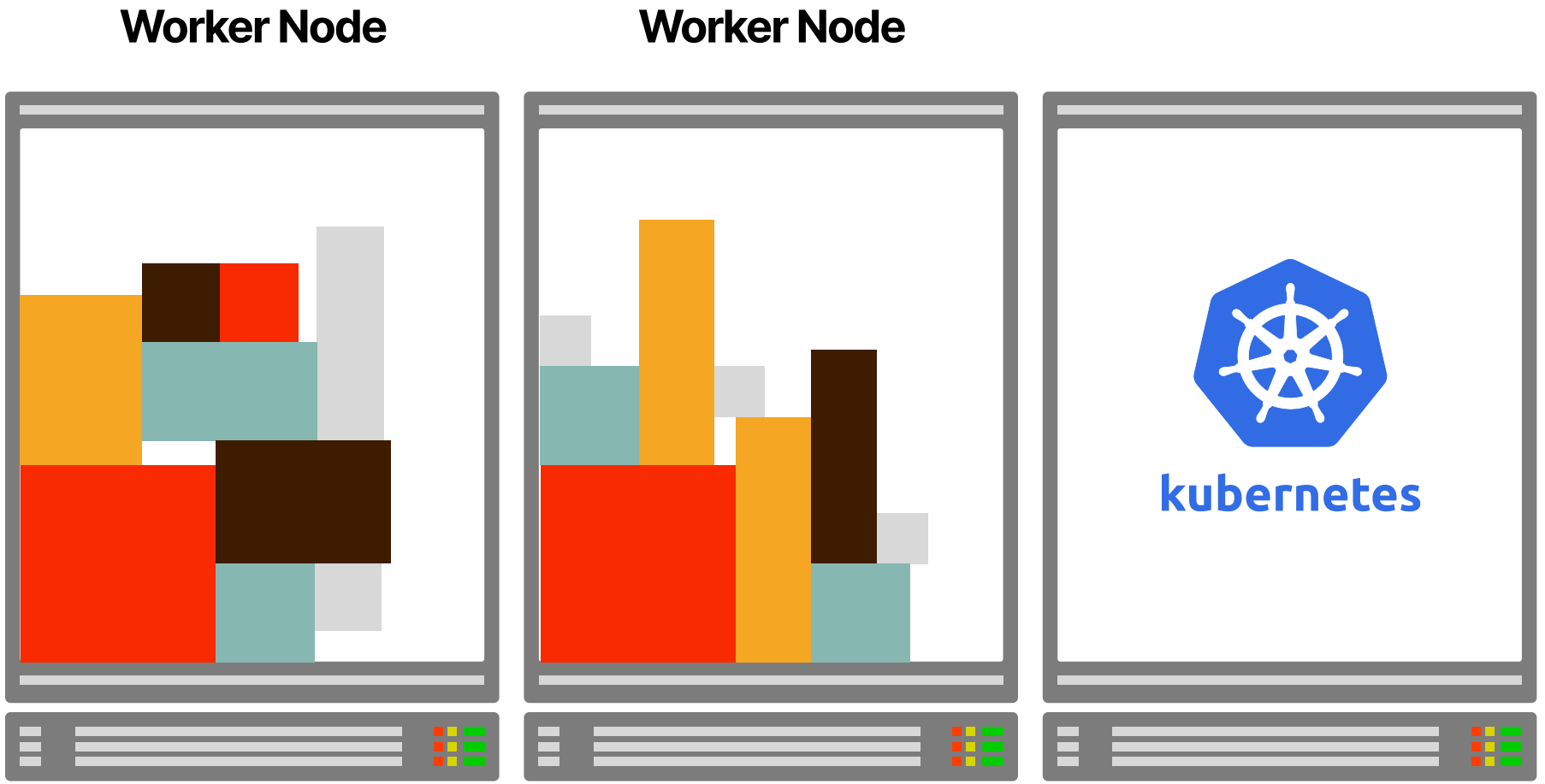



kubernetes

Next container




kubernetes


Worker Node
Worker Node
kubernetes

Infrastructure as an API

03


Kubernetes
AWS



Workloads

Kubernetes

AWS



Workloads

Kubernetes

Azure



Workloads

Kubernetes



On premise

Recap

Kubernetes, the orchestrator

Recap — Kubernetes, the orchestrator

1. An abstraction on the infrastructure
2. Combining resources
3. Smart scheduler
4. Managing infrastructure with an API
5. Not the only orchestrator

Recap — Kubernetes, the orchestrator

1. An abstraction on the infrastructure

2. Combining resources

3. Smart scheduler

4. Managing infrastructure with an API

5. Not the only orchestrator

Recap — Kubernetes, the orchestrator

1. An abstraction on the infrastructure
2. Combining resources
3. **Smart scheduler**
4. Managing infrastructure with an API
5. Not the only orchestrator

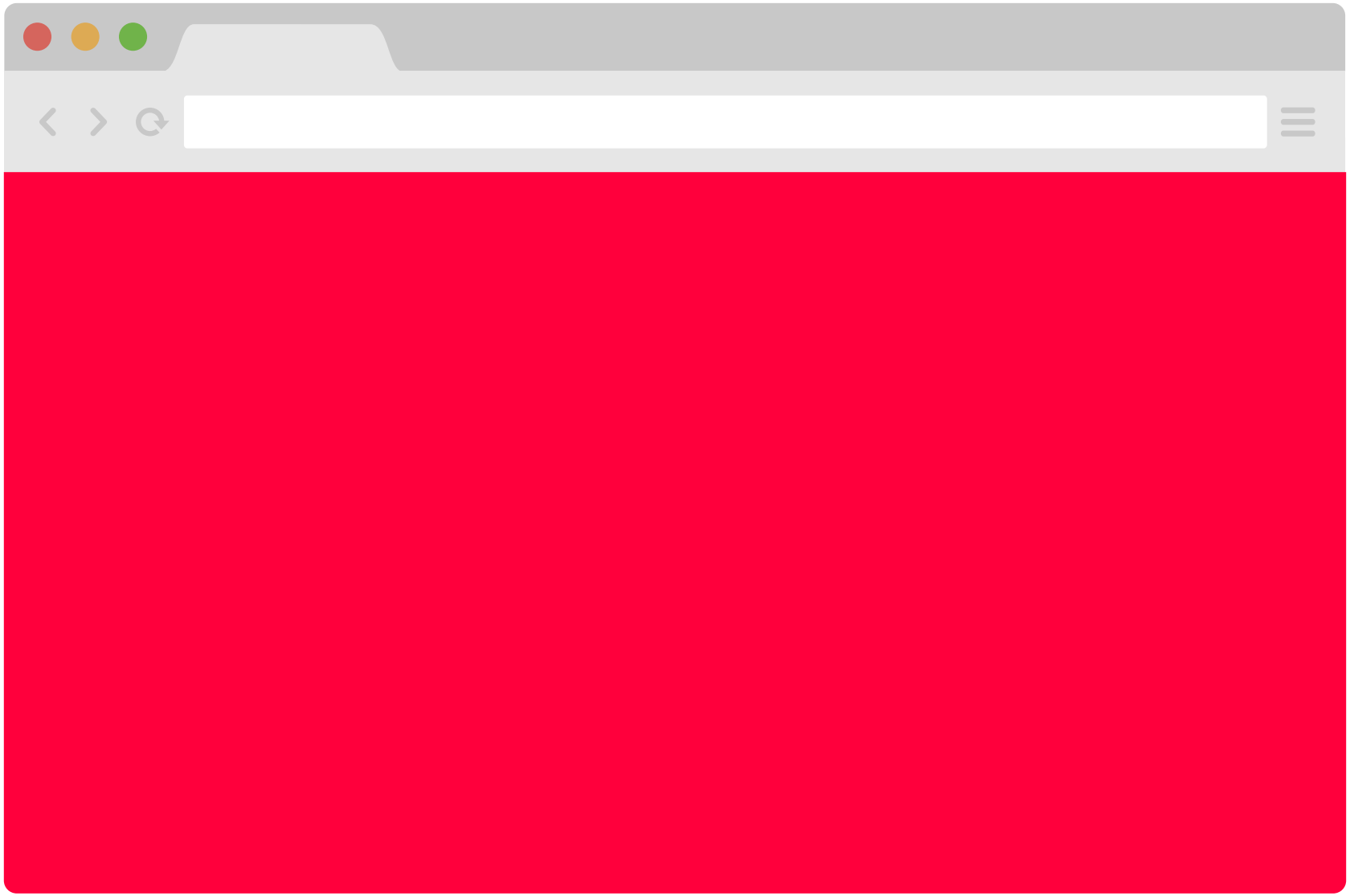
Recap — Kubernetes, the orchestrator

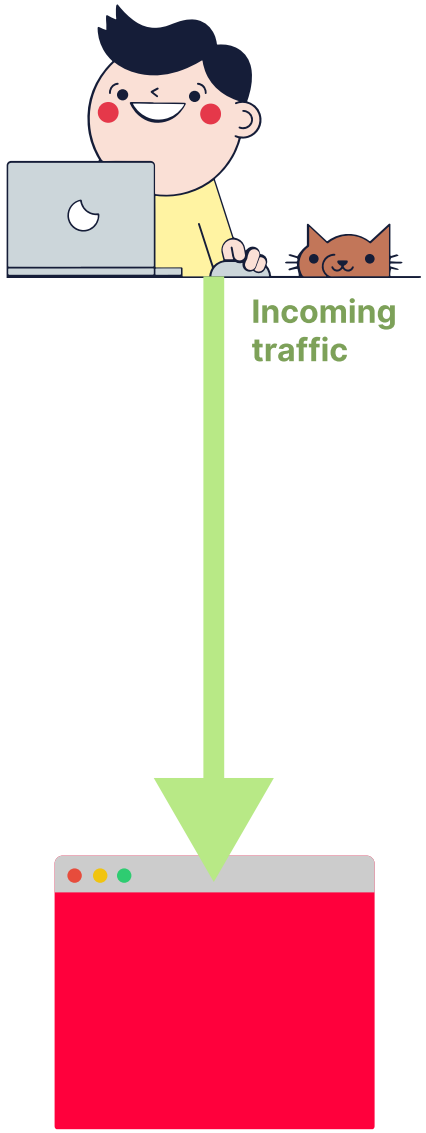
1. An abstraction on the infrastructure
2. Combining resources
3. Smart scheduler
4. **Managing infrastructure with an API**
5. Not the only orchestrator

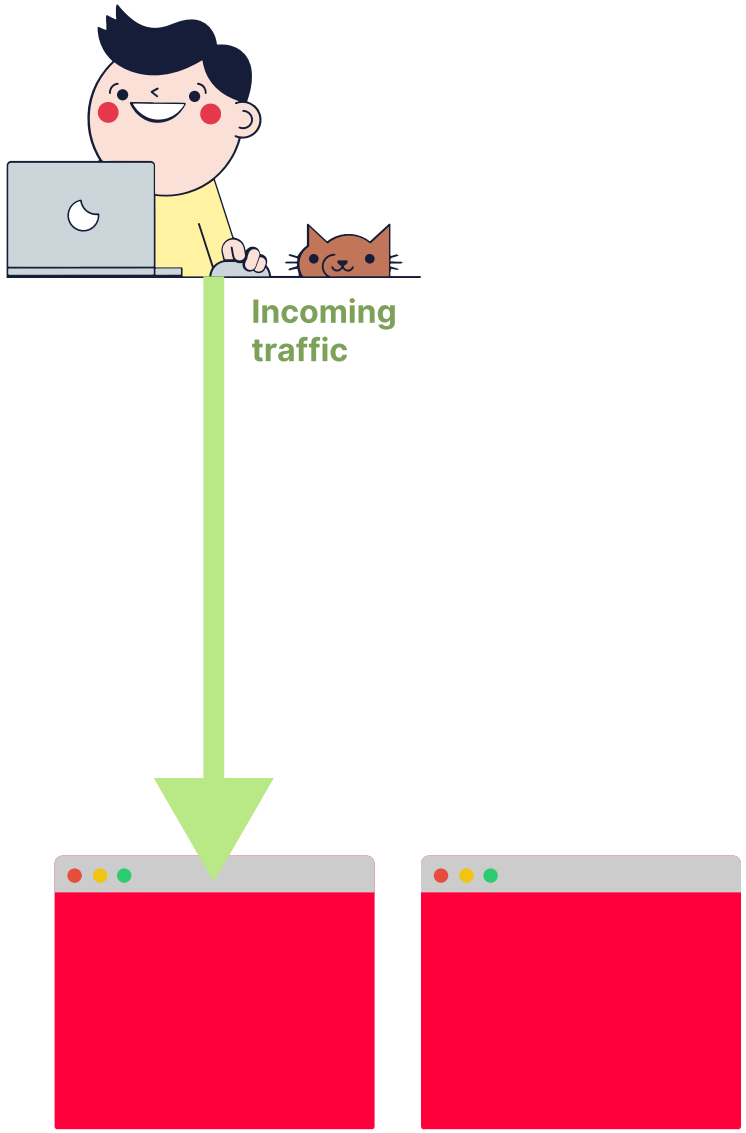
Recap — Kubernetes, the orchestrator

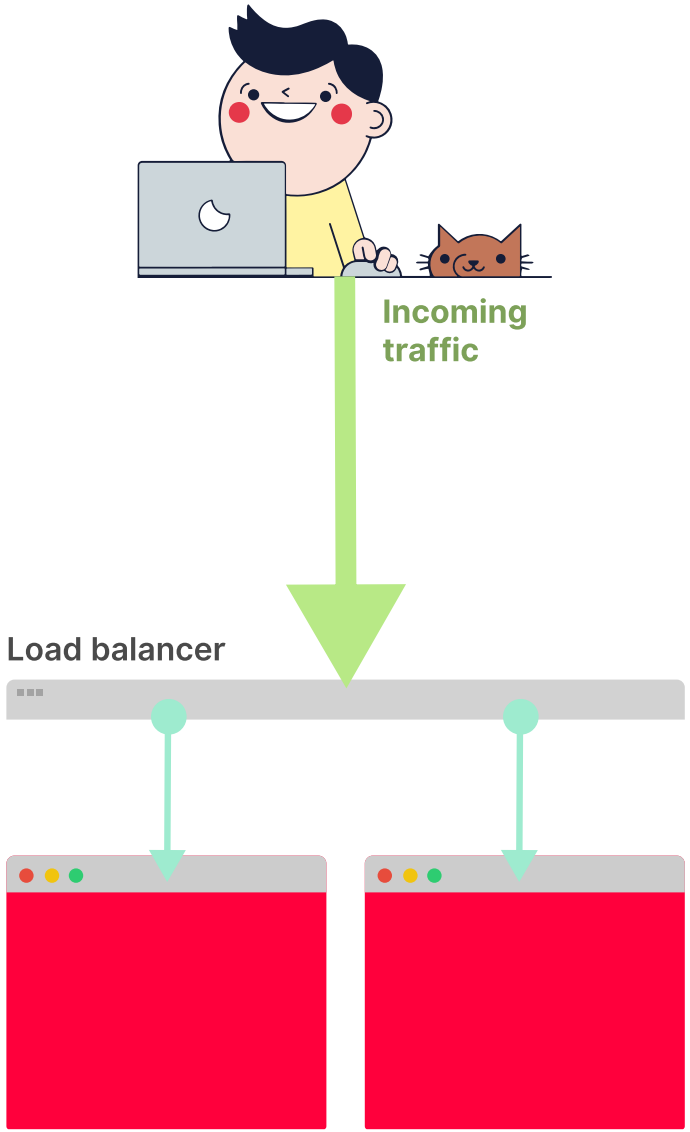
1. An abstraction on the infrastructure
2. Combining resources
3. Smart scheduler
4. Managing infrastructure with an API
5. Not the only orchestrator

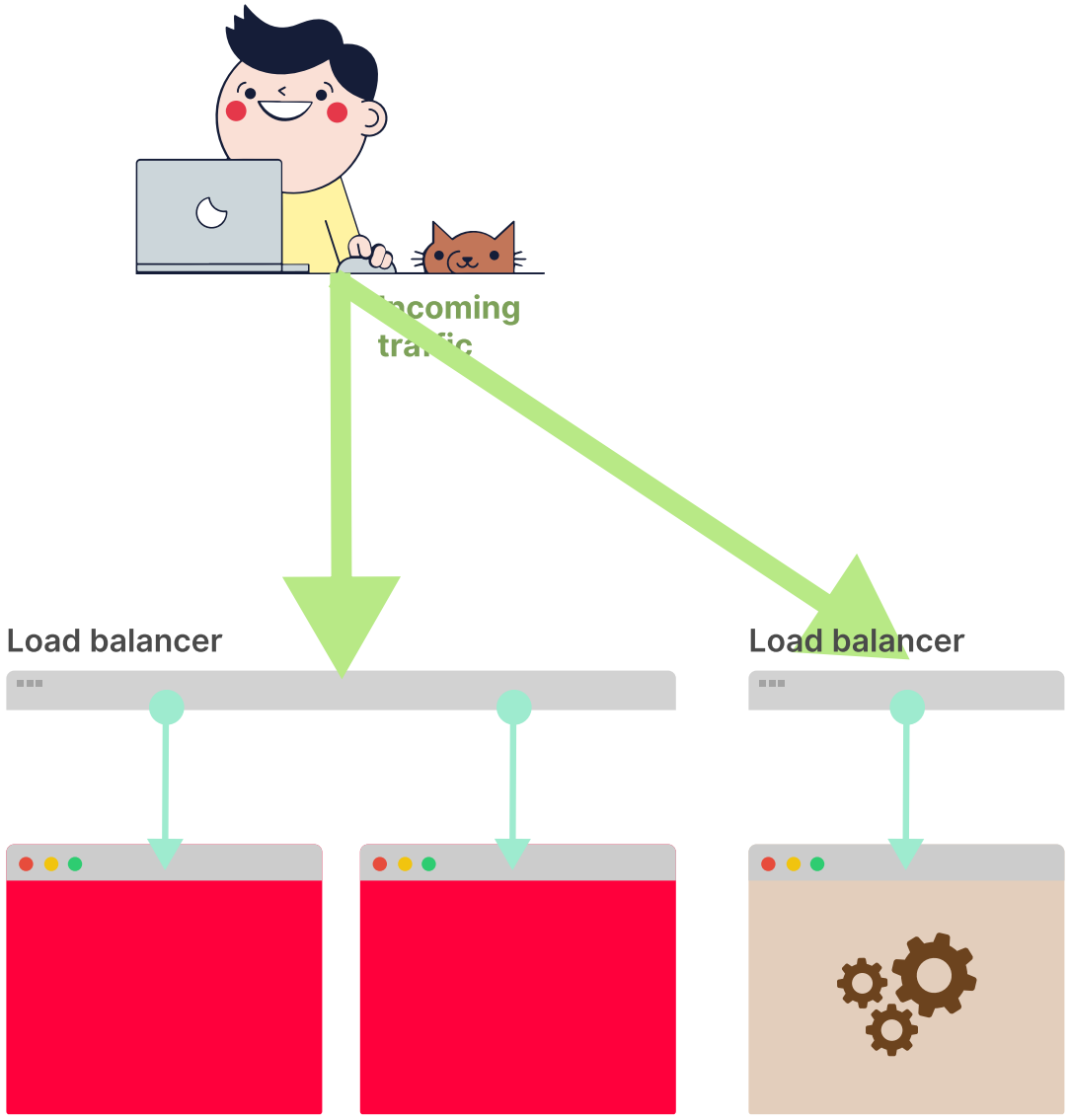
Deploying to Kubernetes

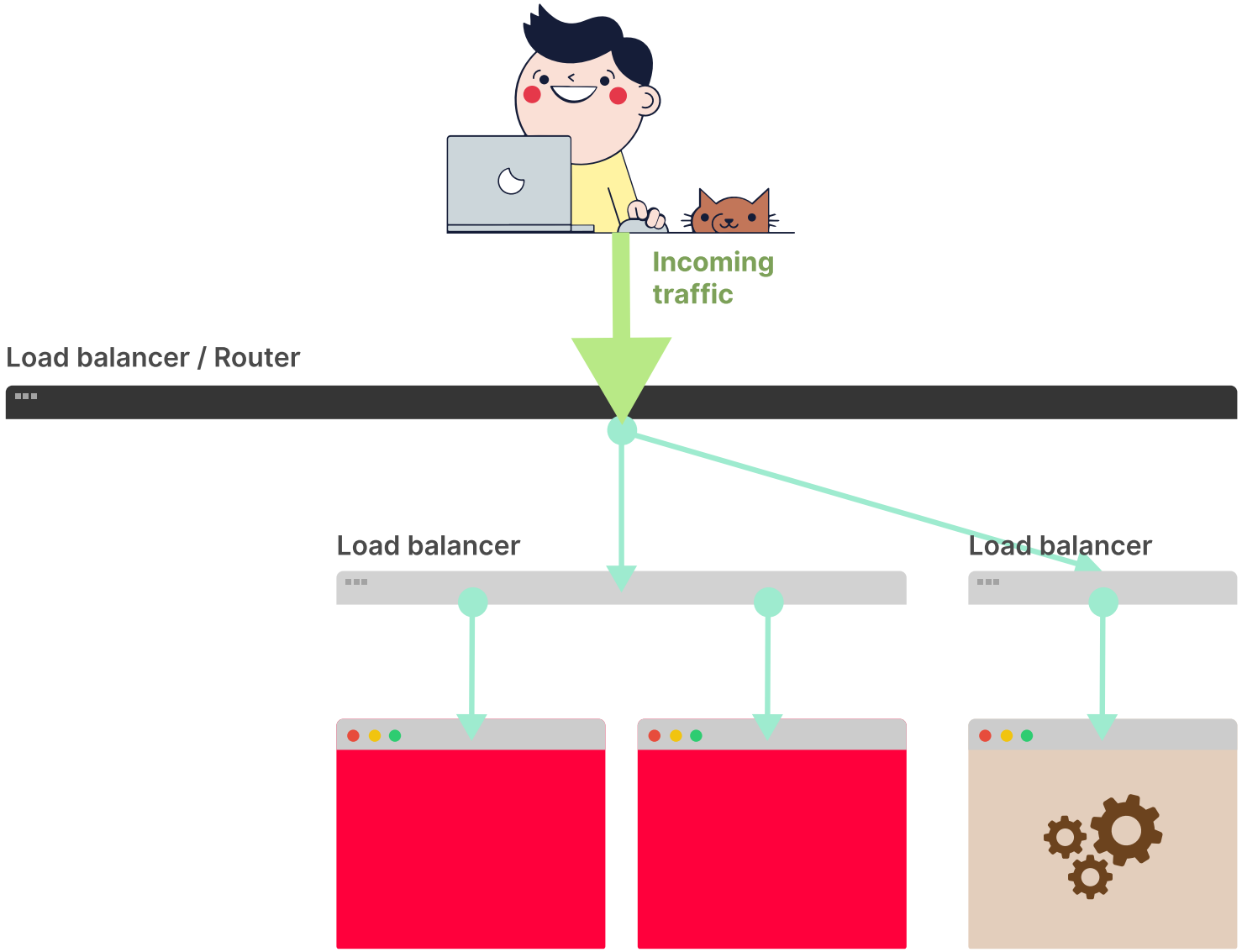


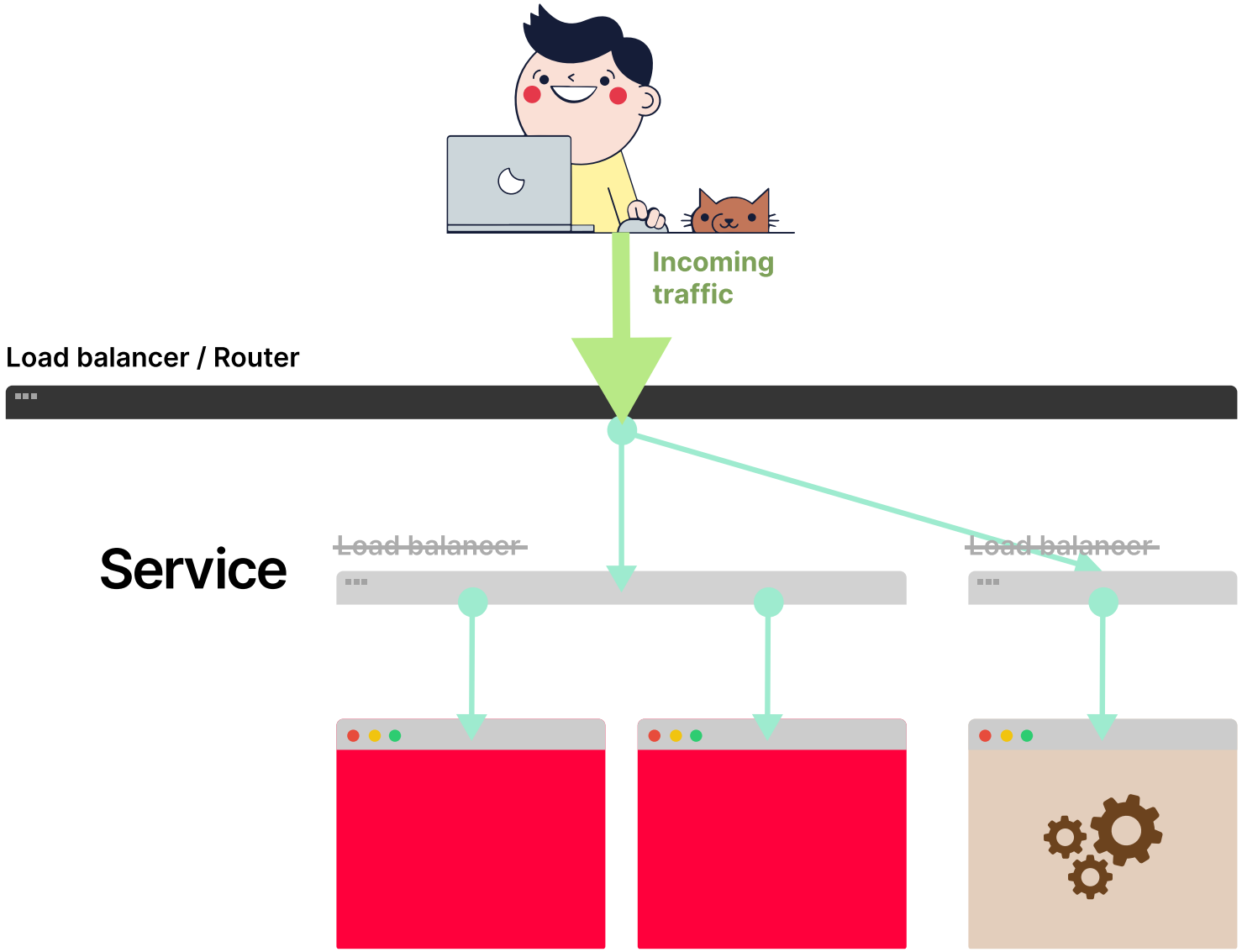

Incoming
traffic

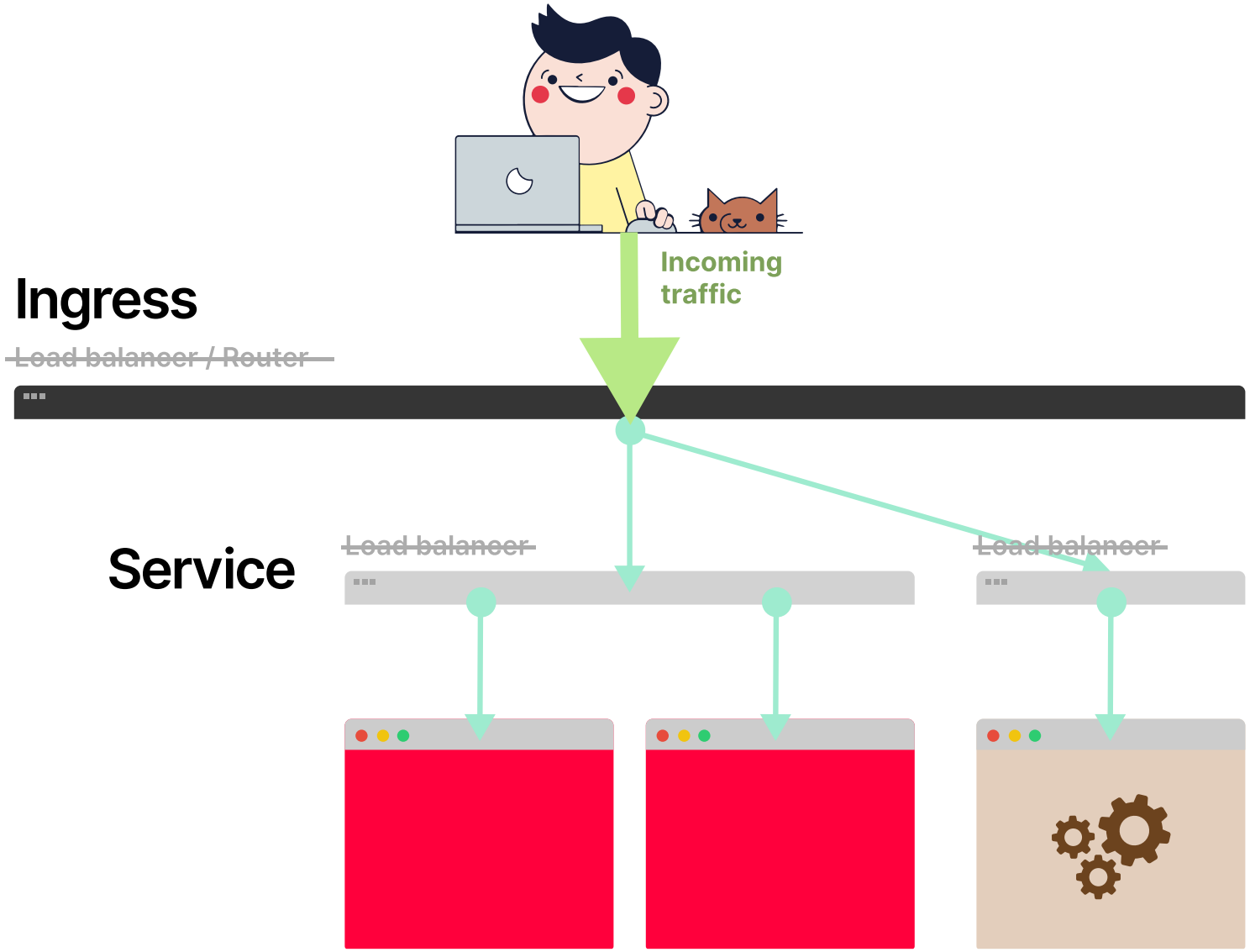

Incoming
traffic

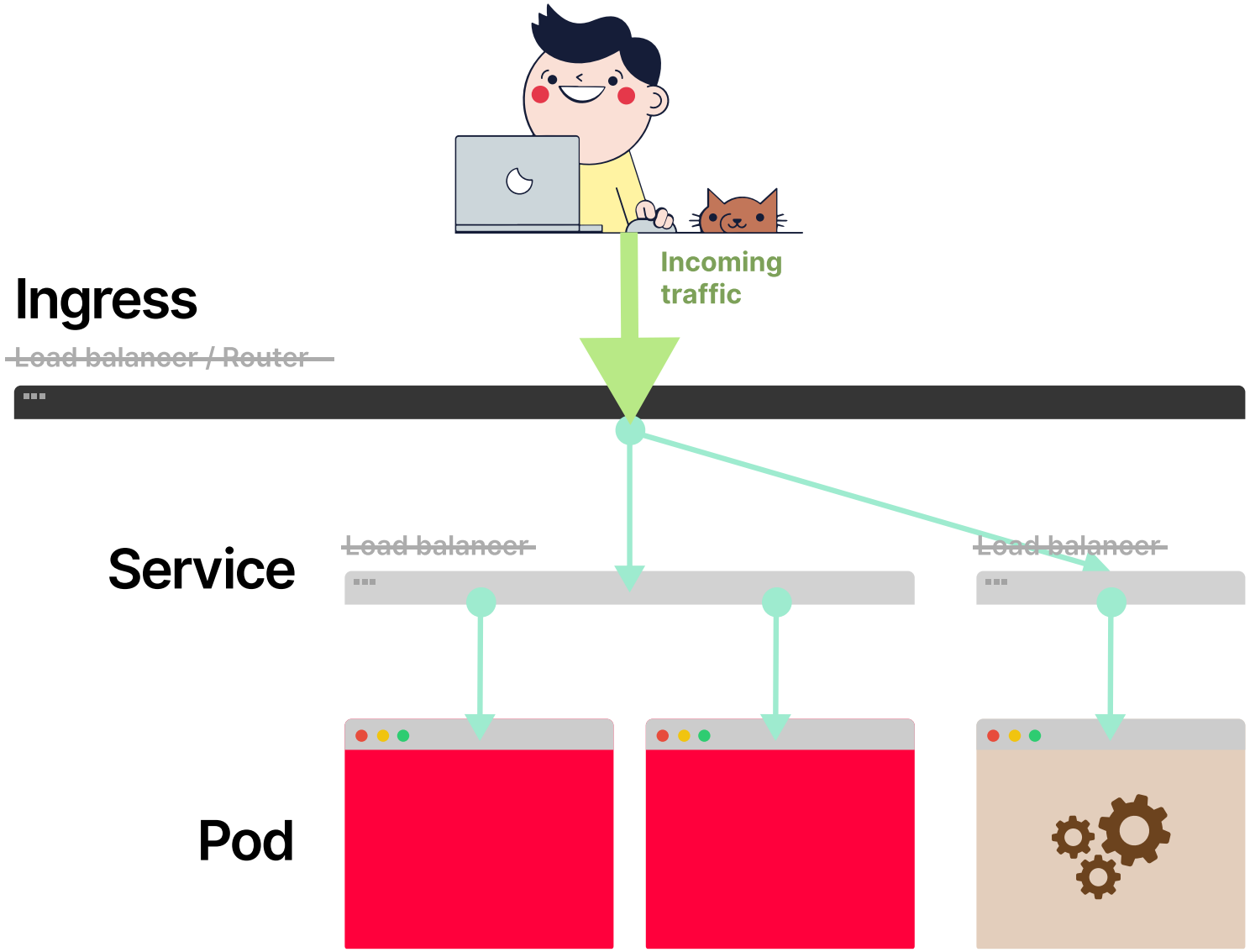

Incoming
traffic
Load balancer

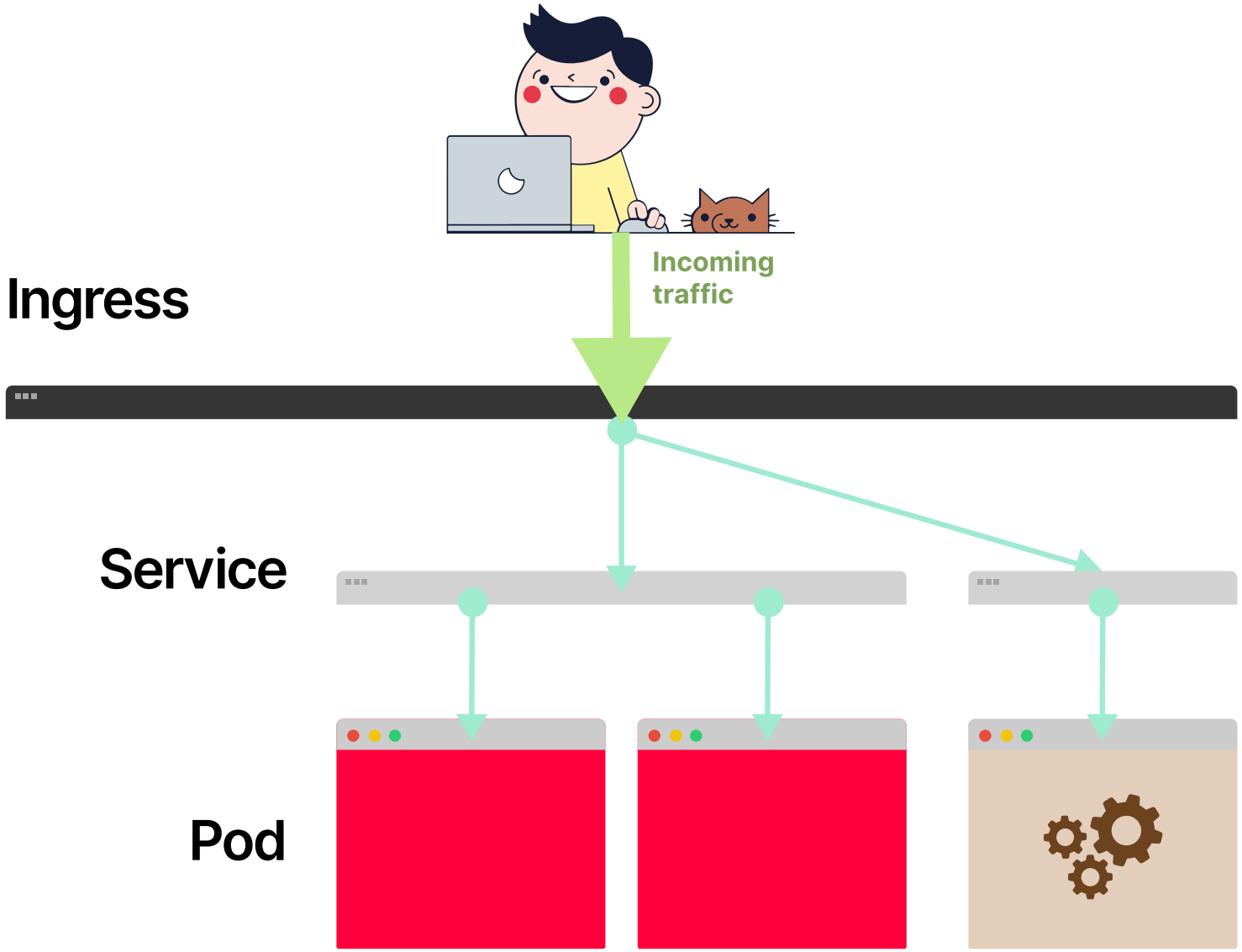

Load balancer
Load balancer


Incoming traffic
Load balancer / Router
Load balancer
Load balancer

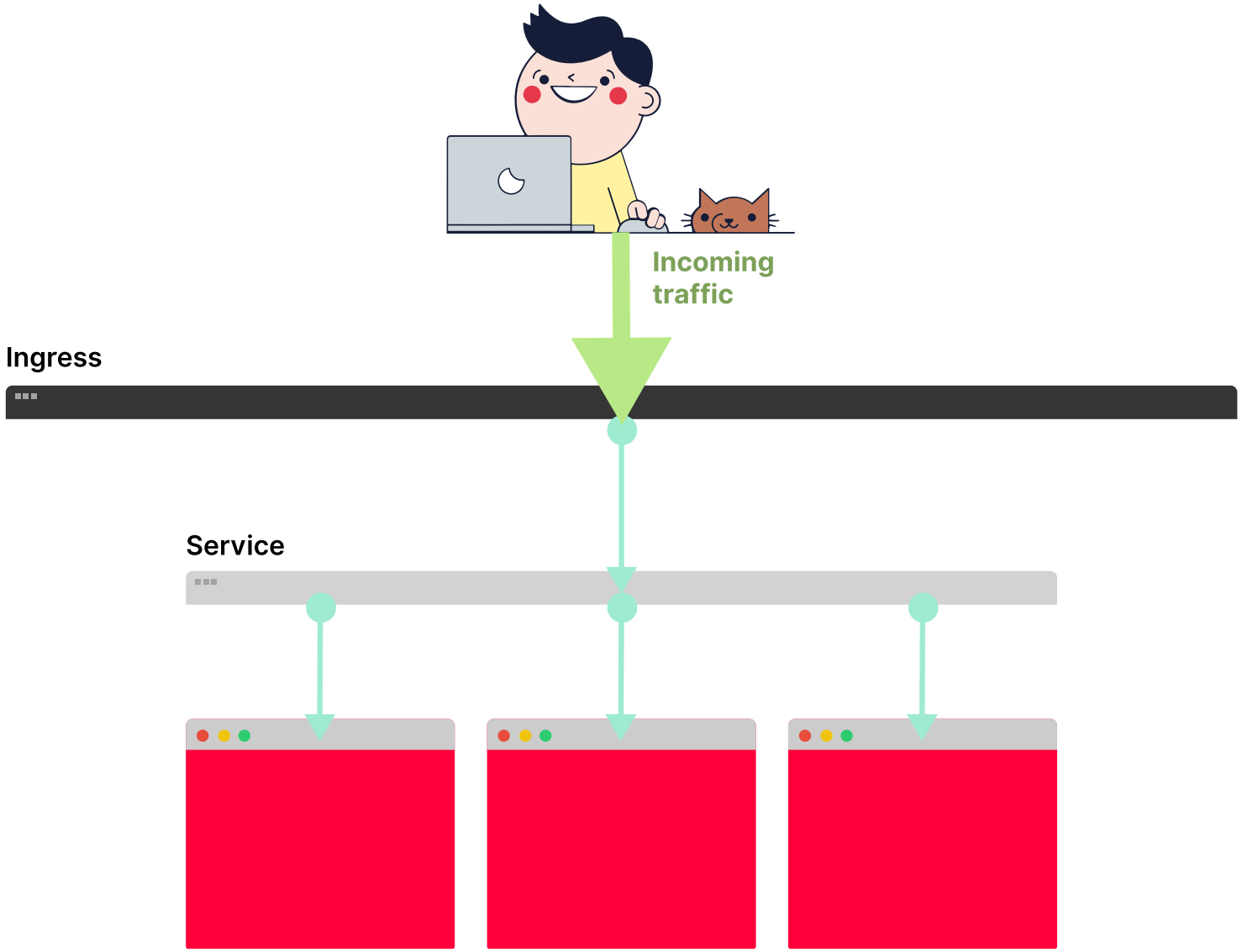

Incoming
traffic
Load balancer / Router
Service
Load balancer
Load balancer

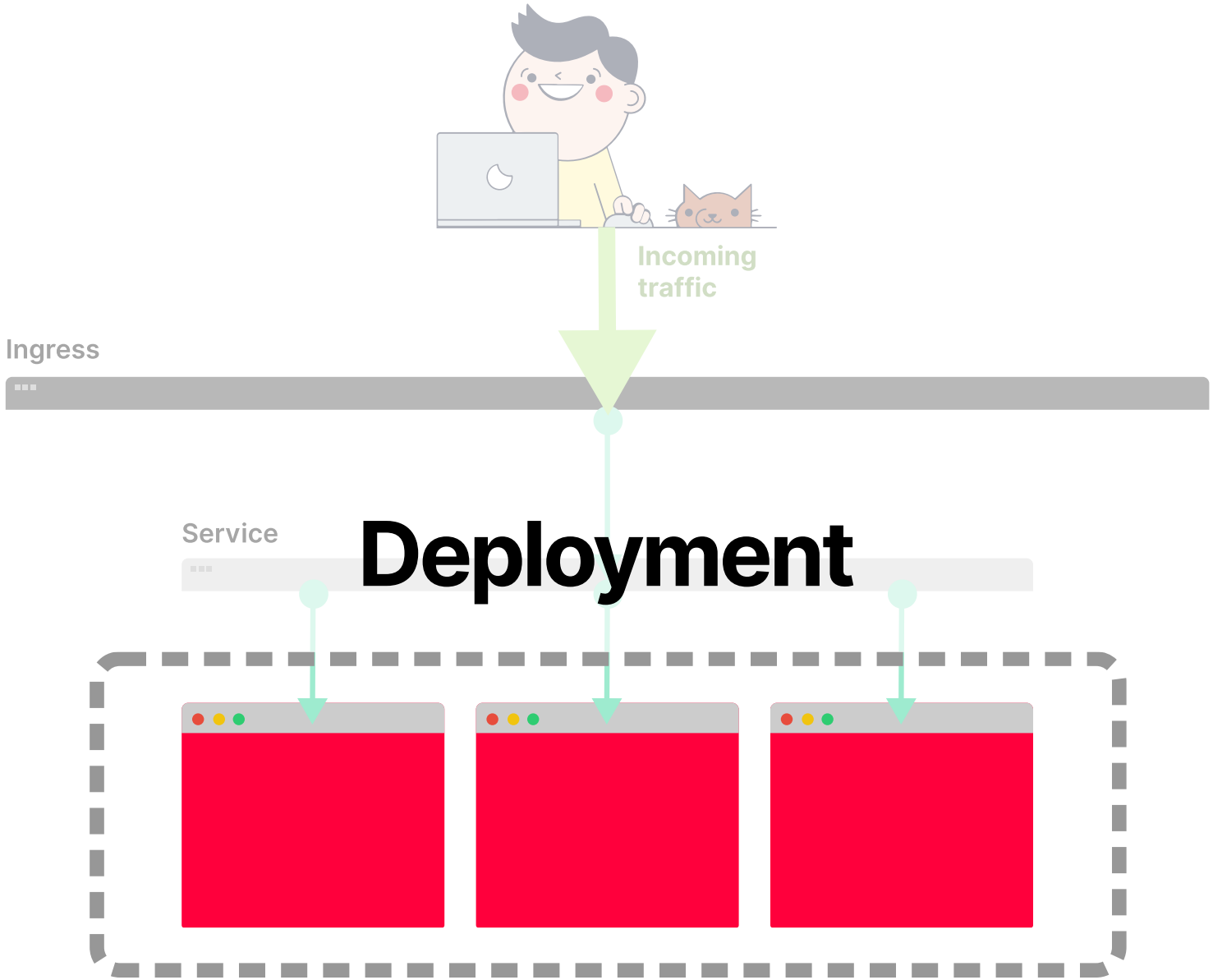

Incoming
traffic
Ingress
Load balancer / Router
Service
Load balancer
Load balancer

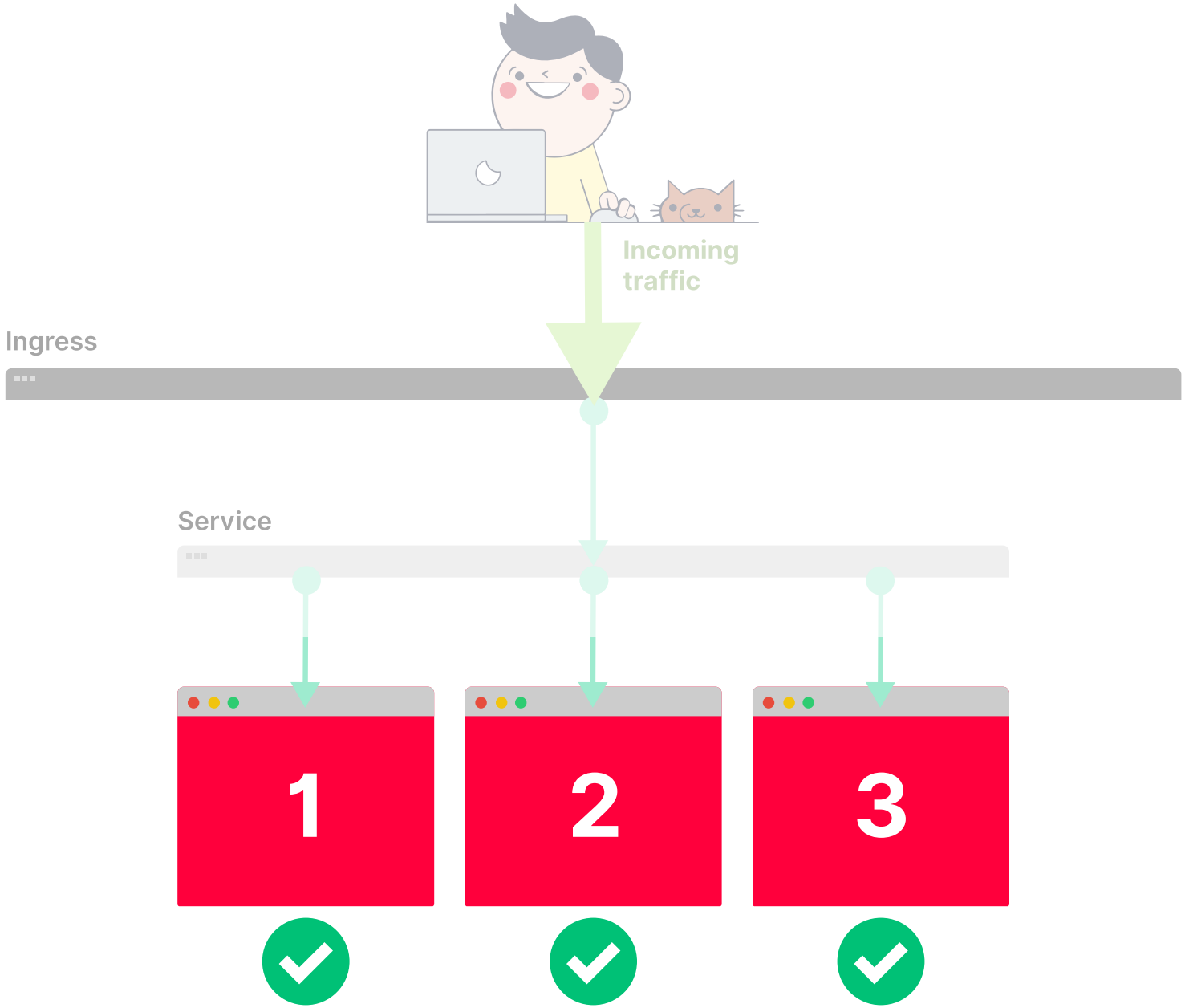

Incoming traffic
Ingress
Load balancer / Router
Service
Load balancer
Load balancer
Pod

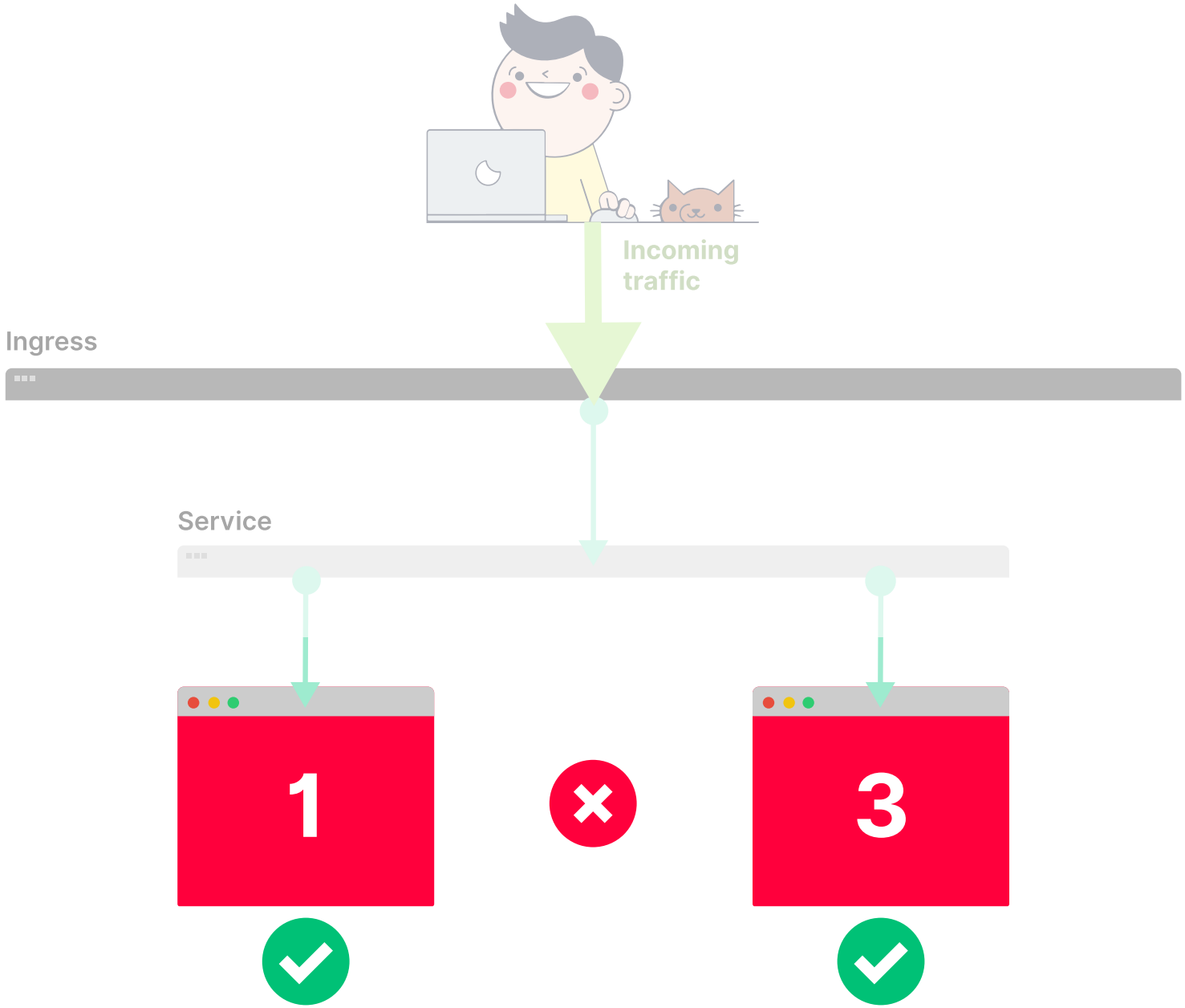

Ingress
Incoming traffic
Service
Pod

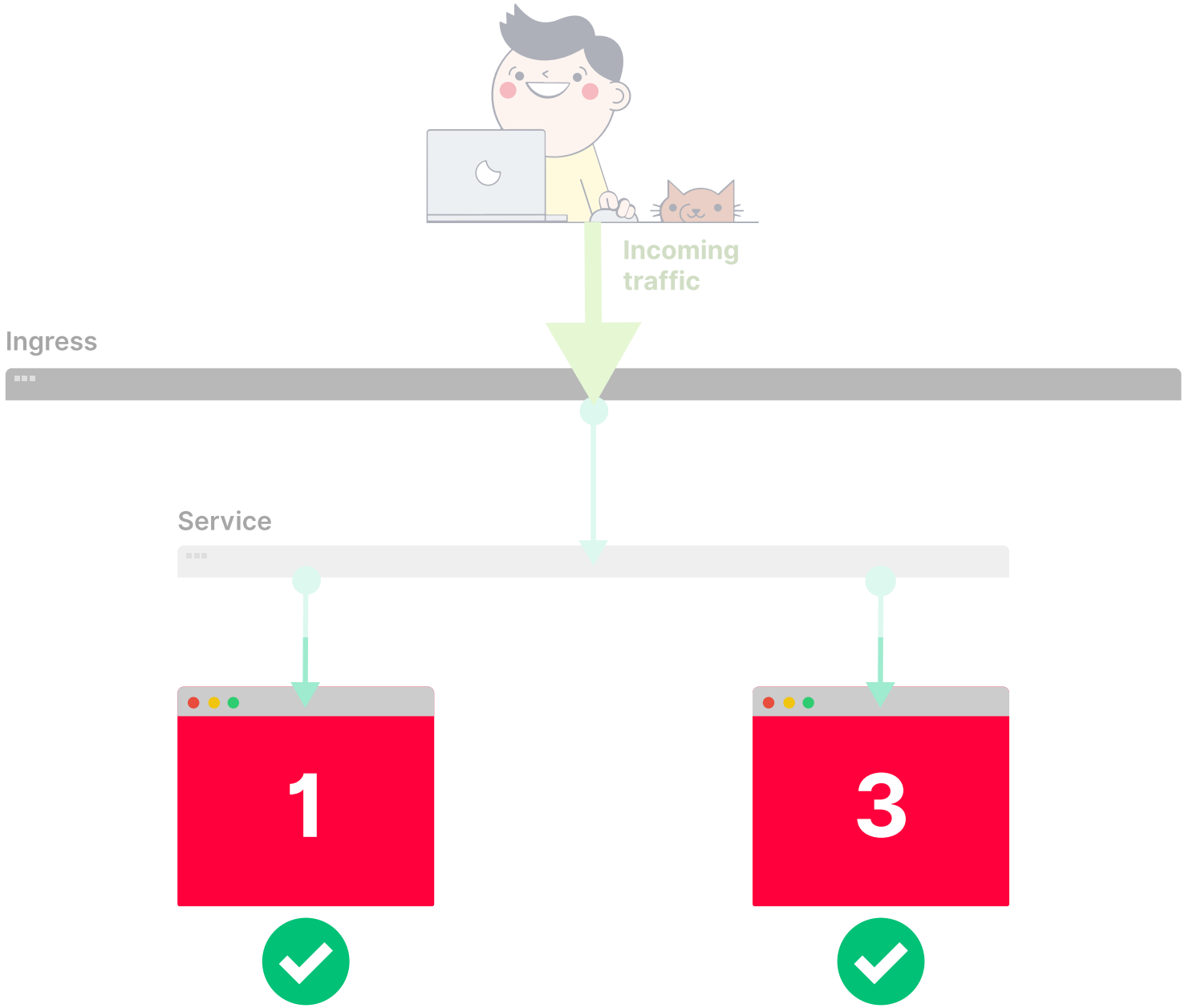
Deployment resource

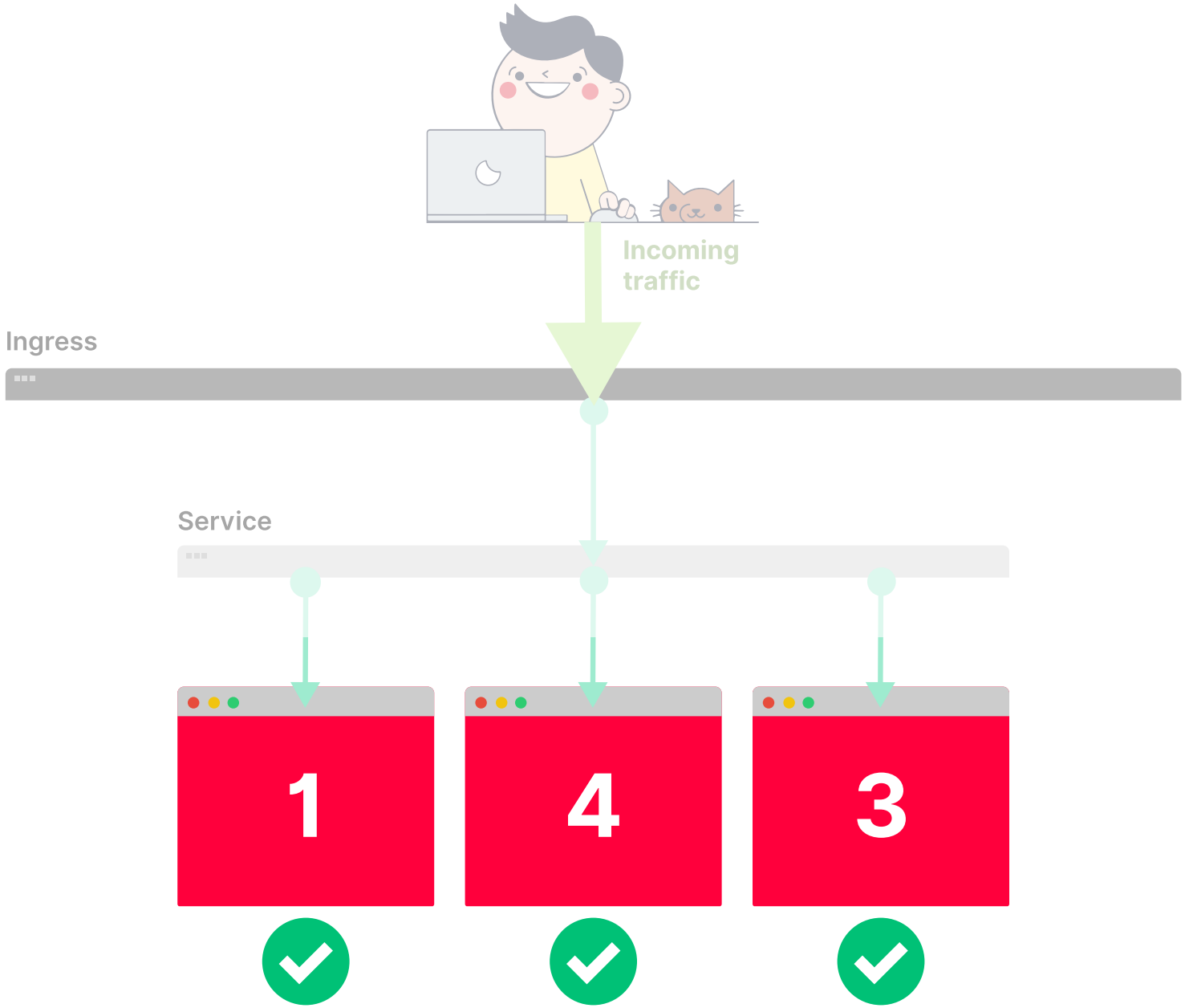

Incoming
traffic
Ingress
Service


Incoming
traffic
Ingress
Service
Deployment

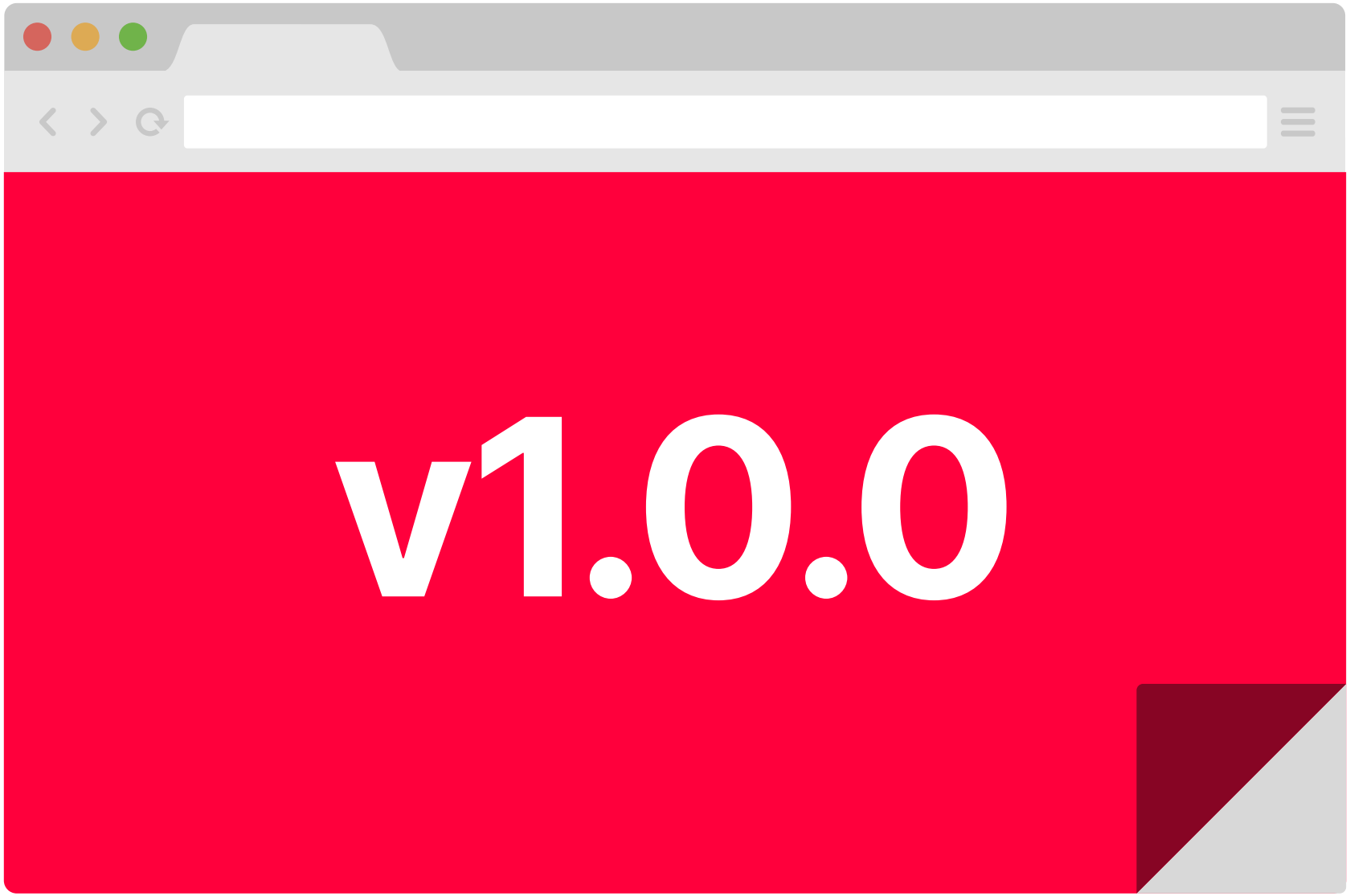

Incoming
traffic
Ingress
Service
1
2
3

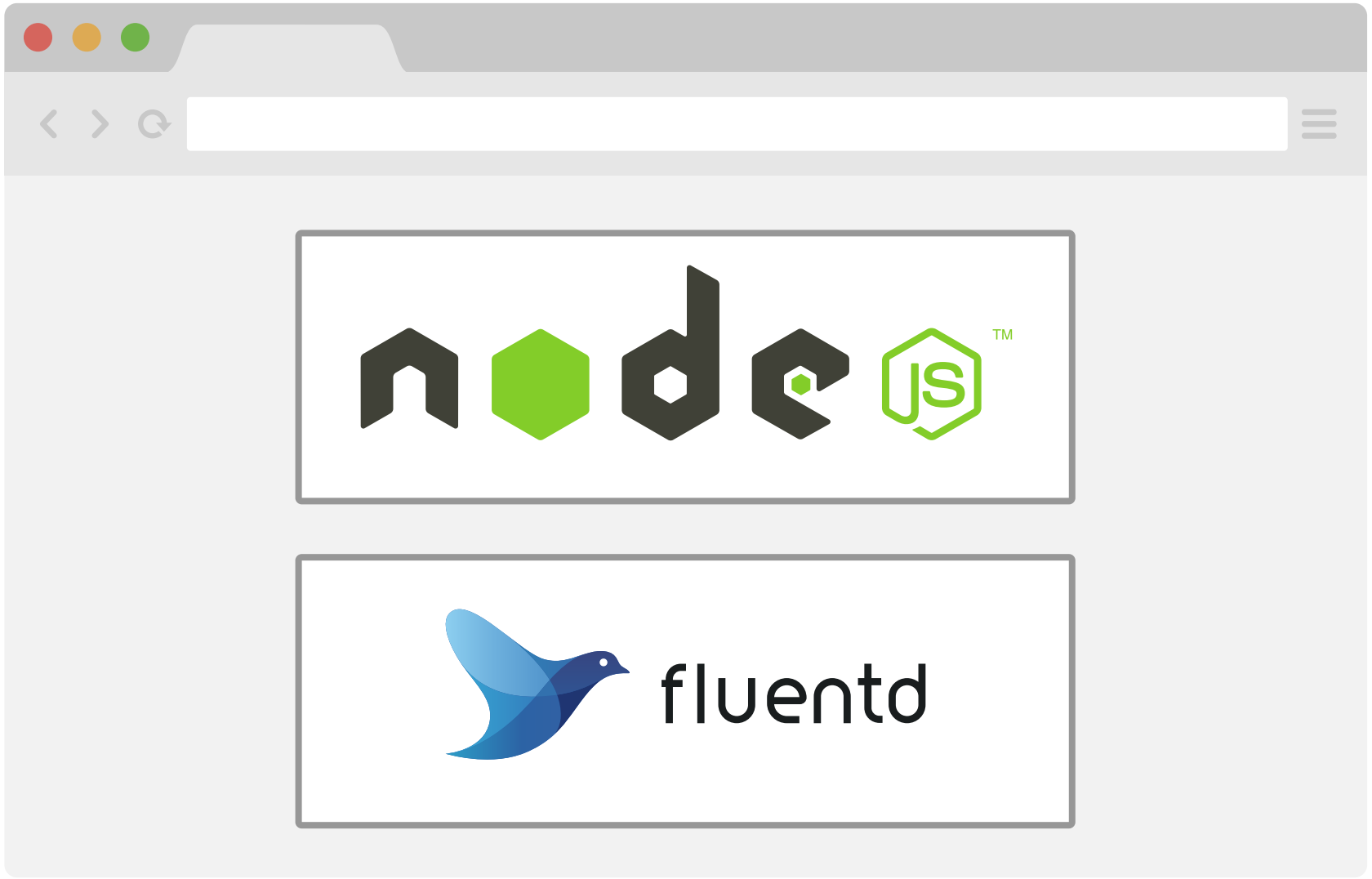

Incoming
traffic
Ingress
Service
1
3

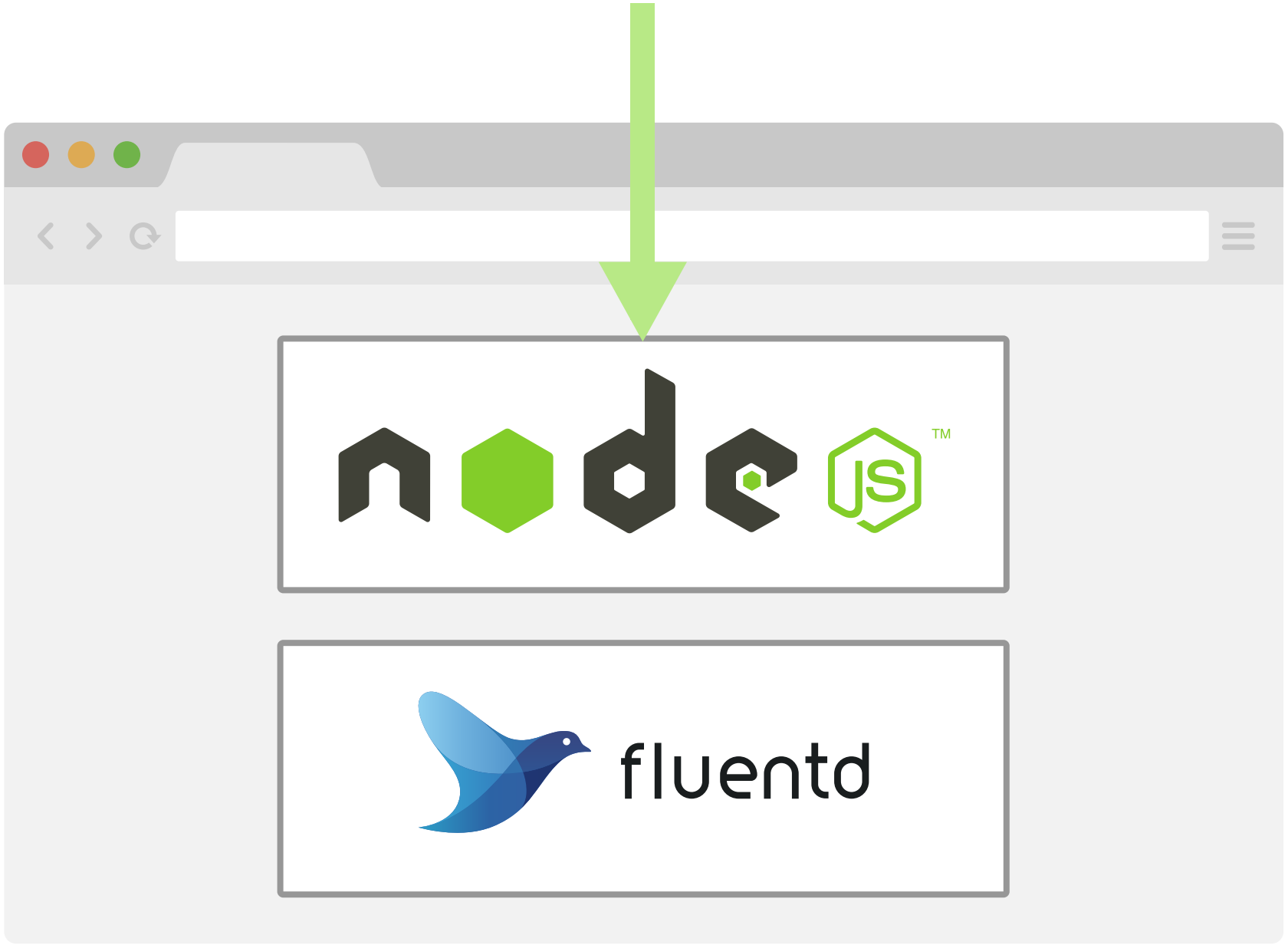

Incoming
traffic
Ingress
Service
1
3

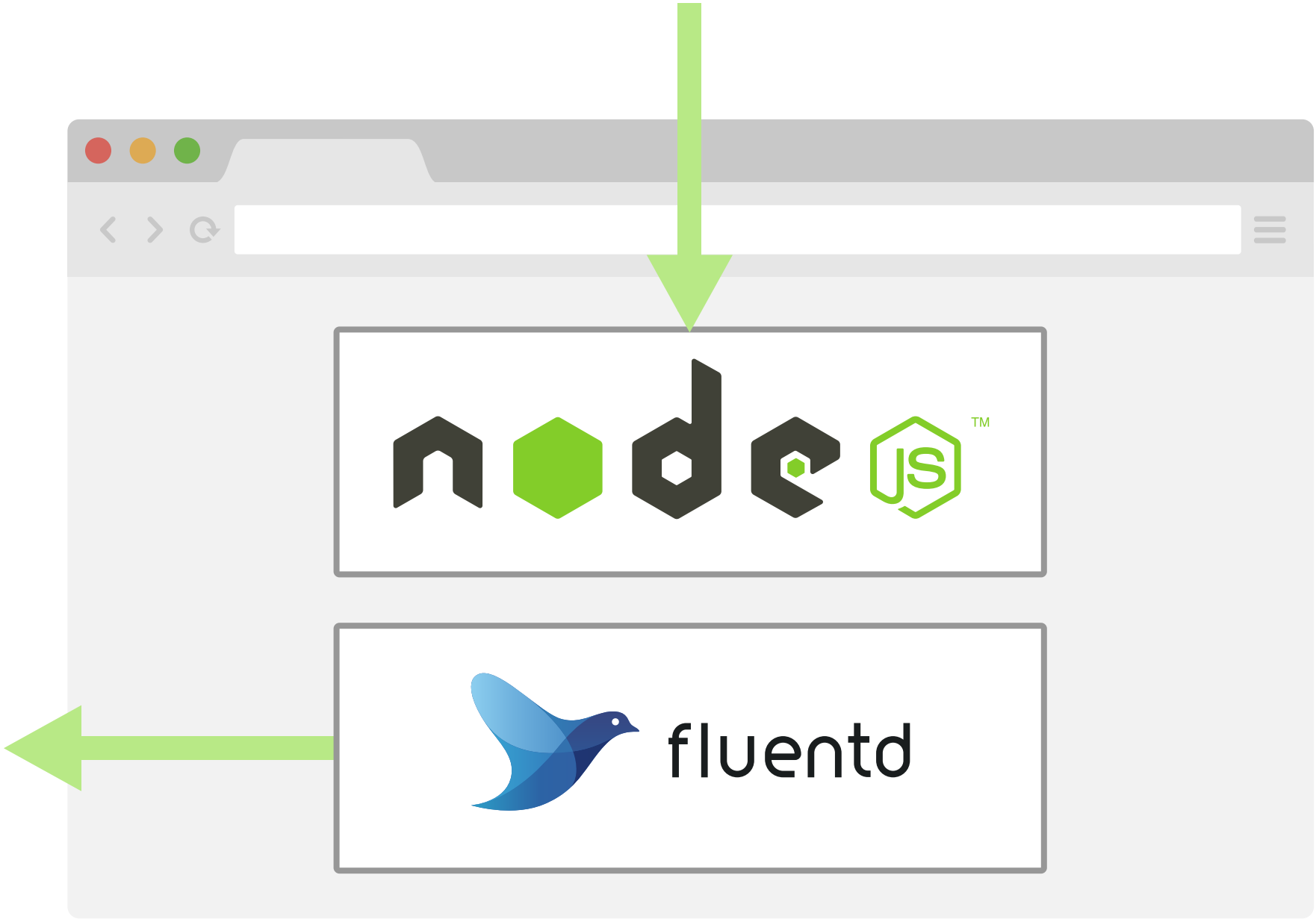

Incoming
traffic
Ingress
Service
1
4
3

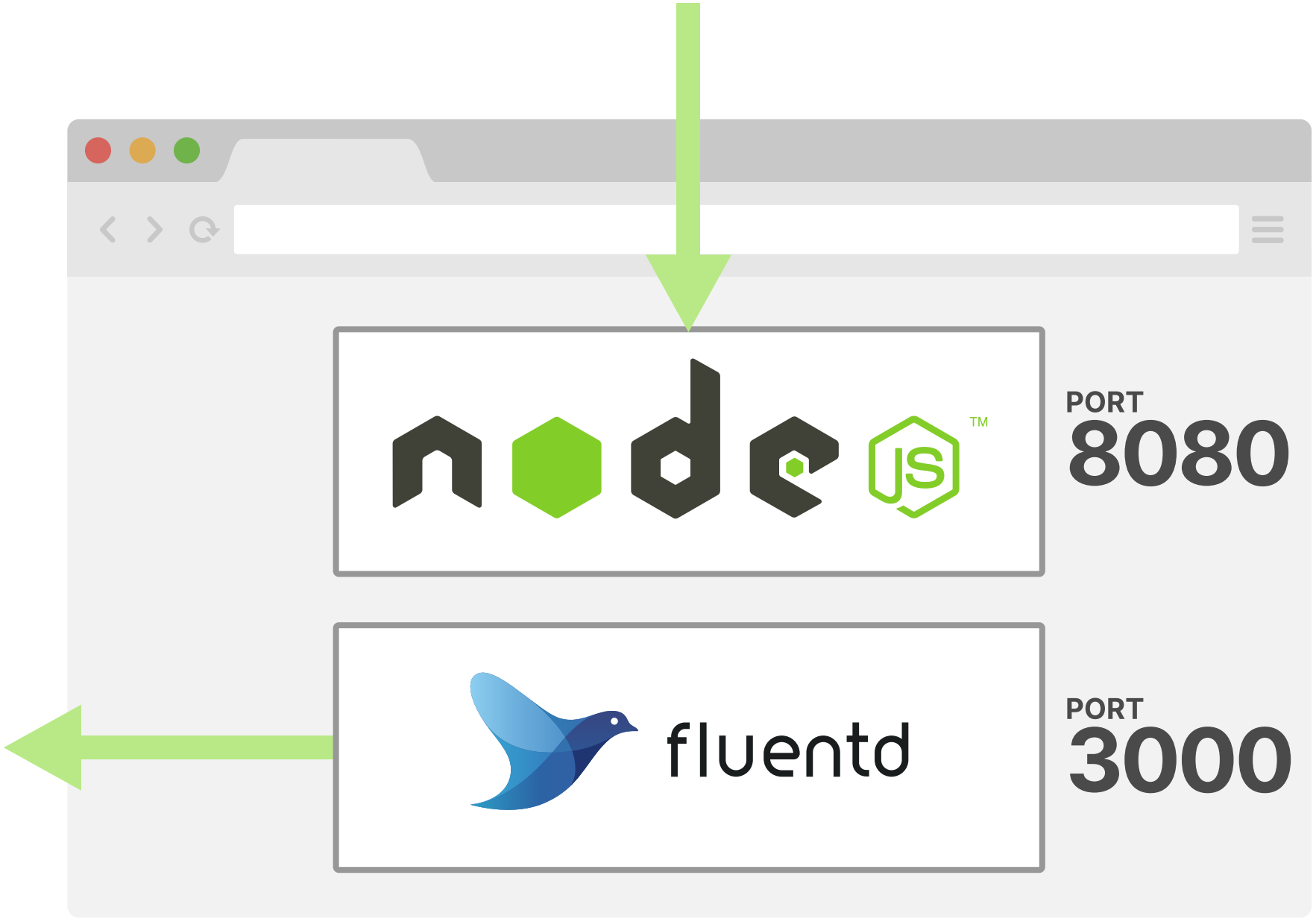
Why Pods?


v1.0.0

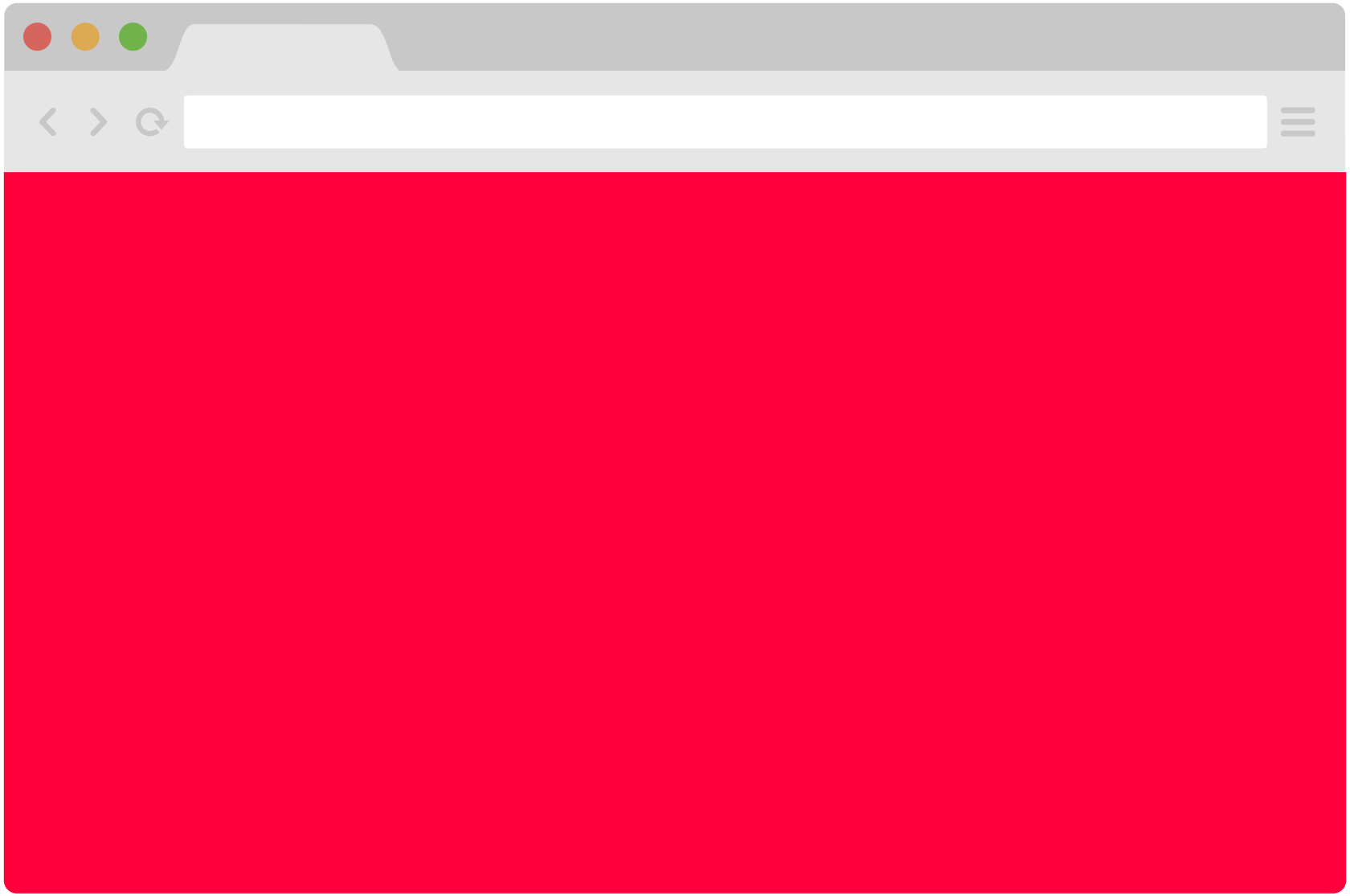

node
JS
TM
fluentd


node
JS
TM
fluentd

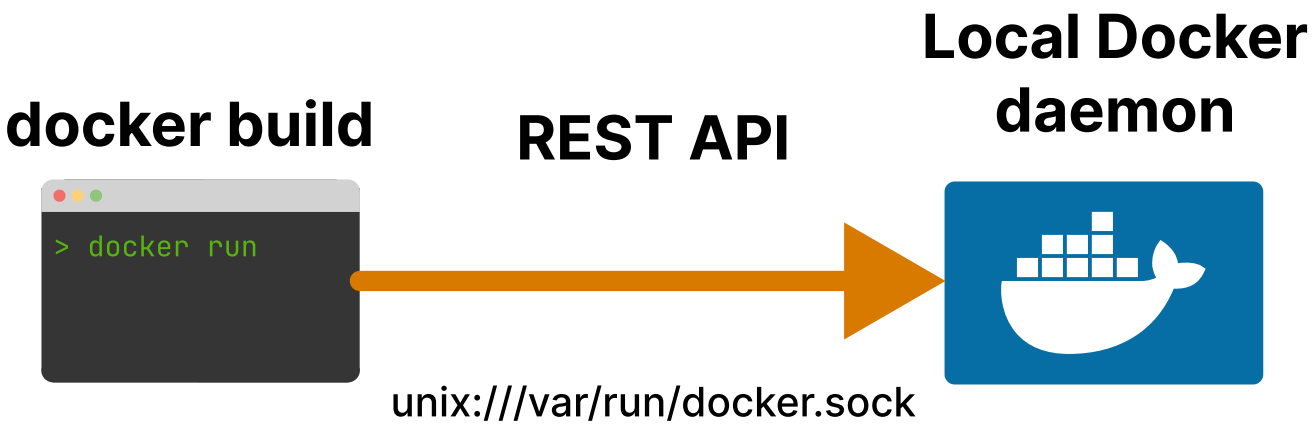

node
JS
TM
fluentd

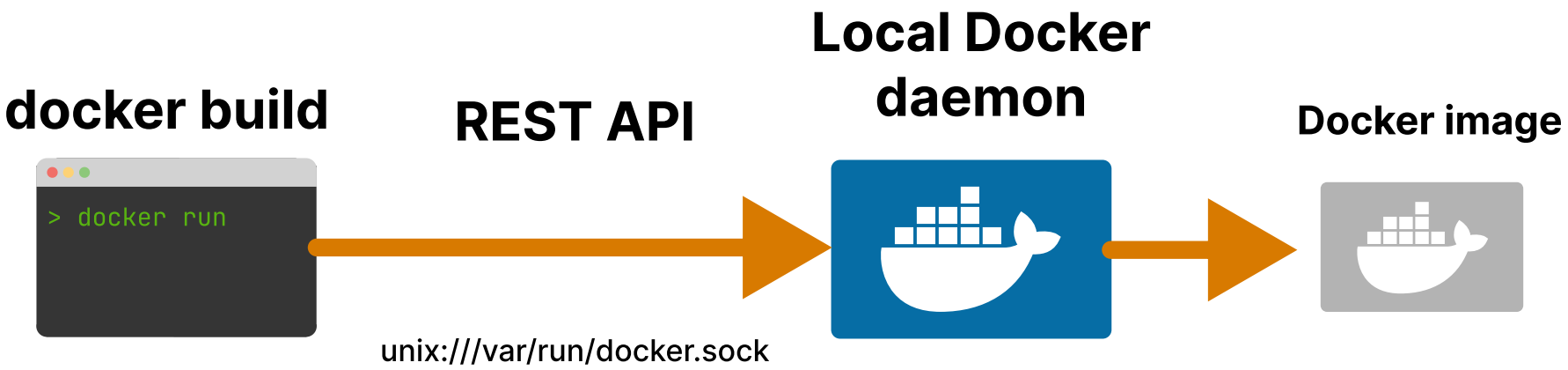

node
JS
TM
PORT
8080
fluentd
PORT
3000
LOCALHOST
127.0.0.1

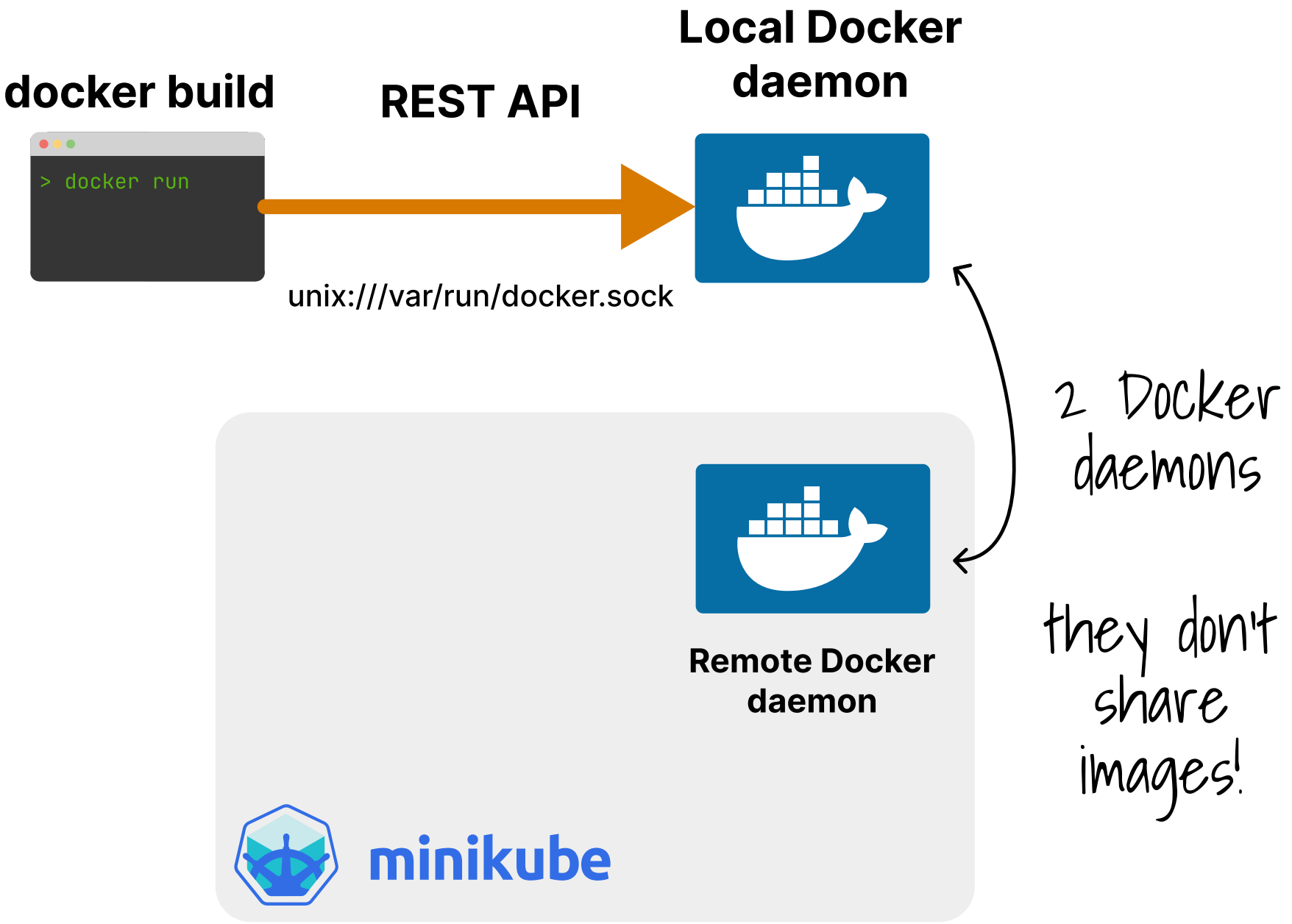
Lab

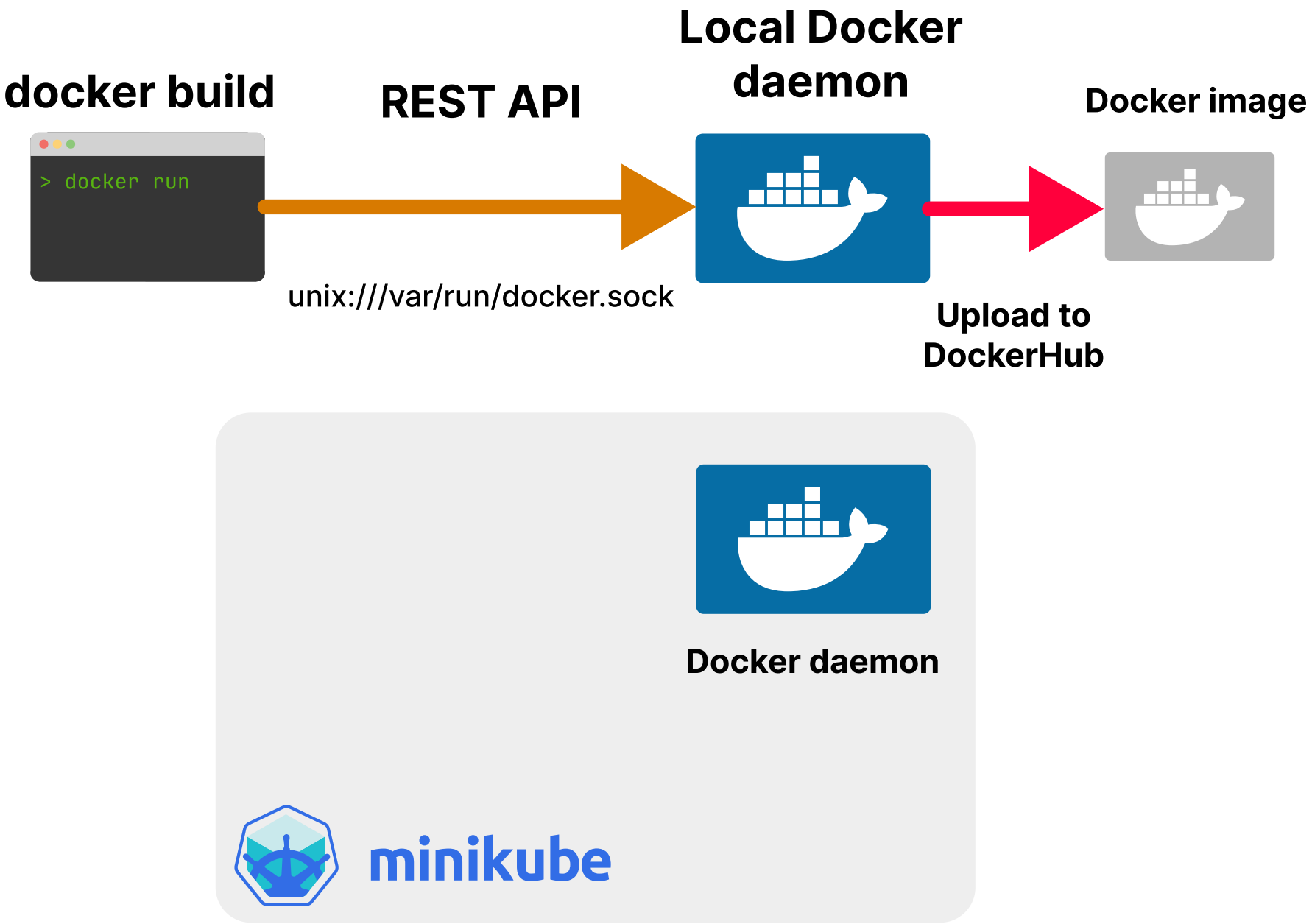


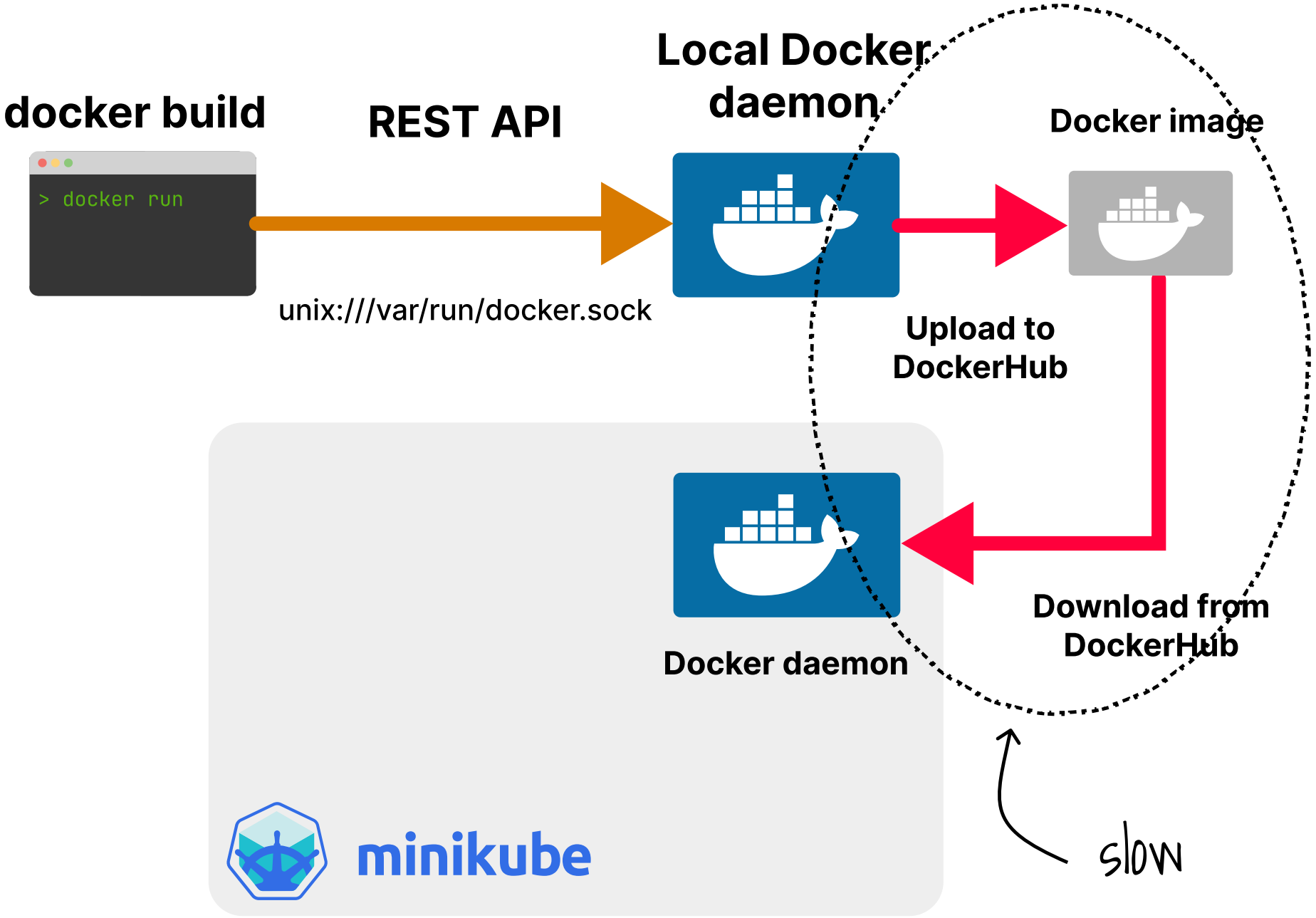
minikube

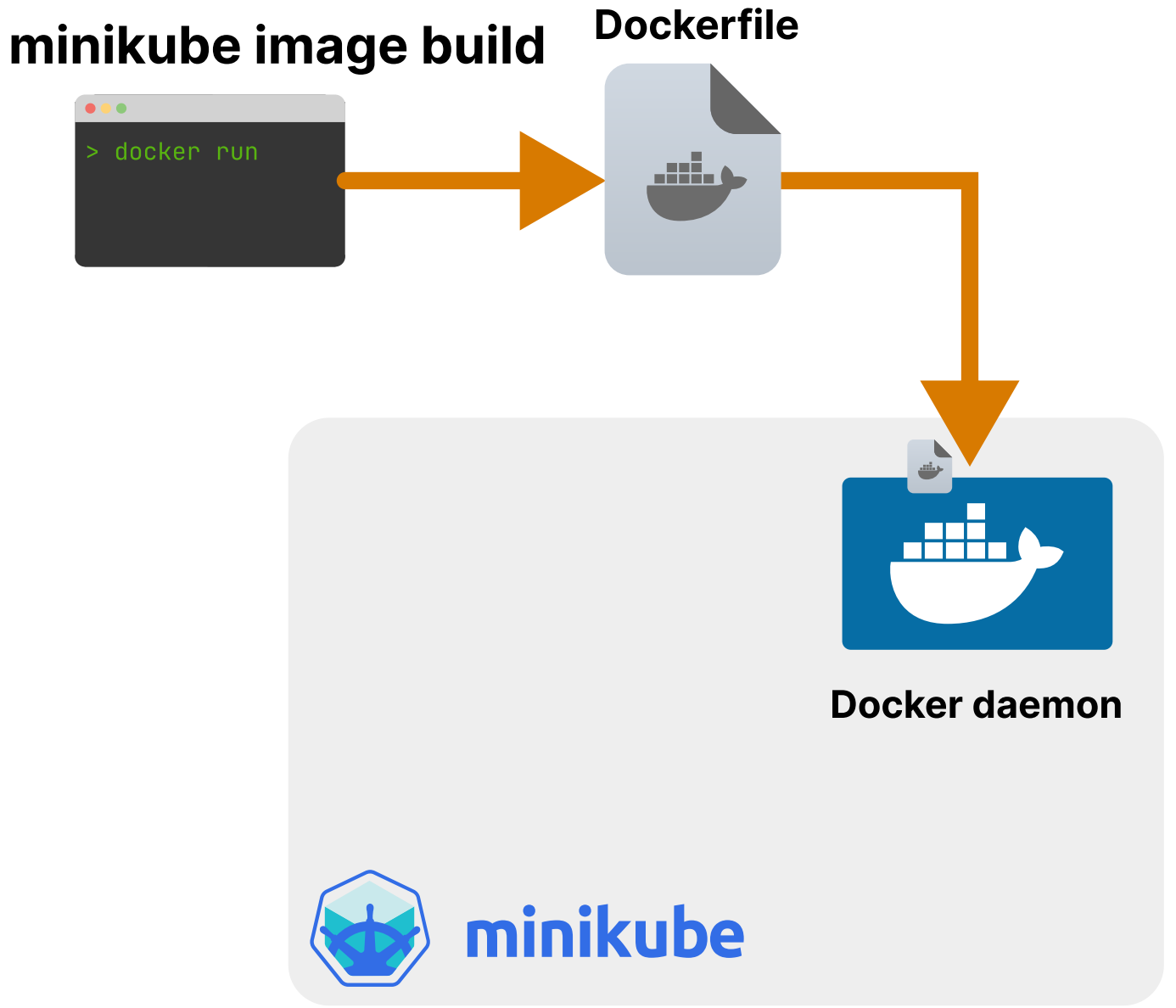

Local Docker daemon
docker build
REST API
> docker run
unix:///var/run/docker.sock

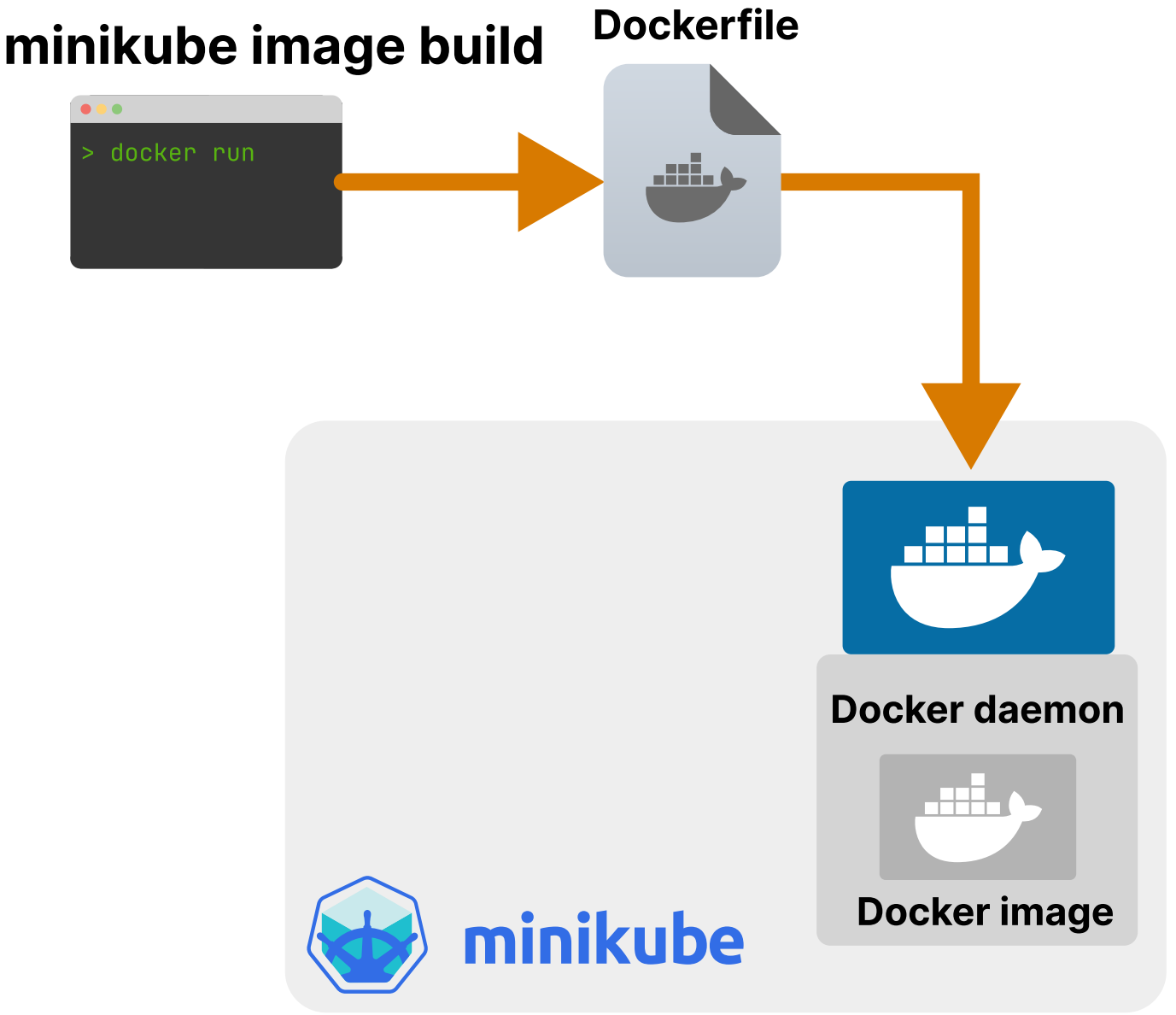

docker build
REST API
Local Docker daemon
Docker image
> docker run
unix:///var/run/docker.sock

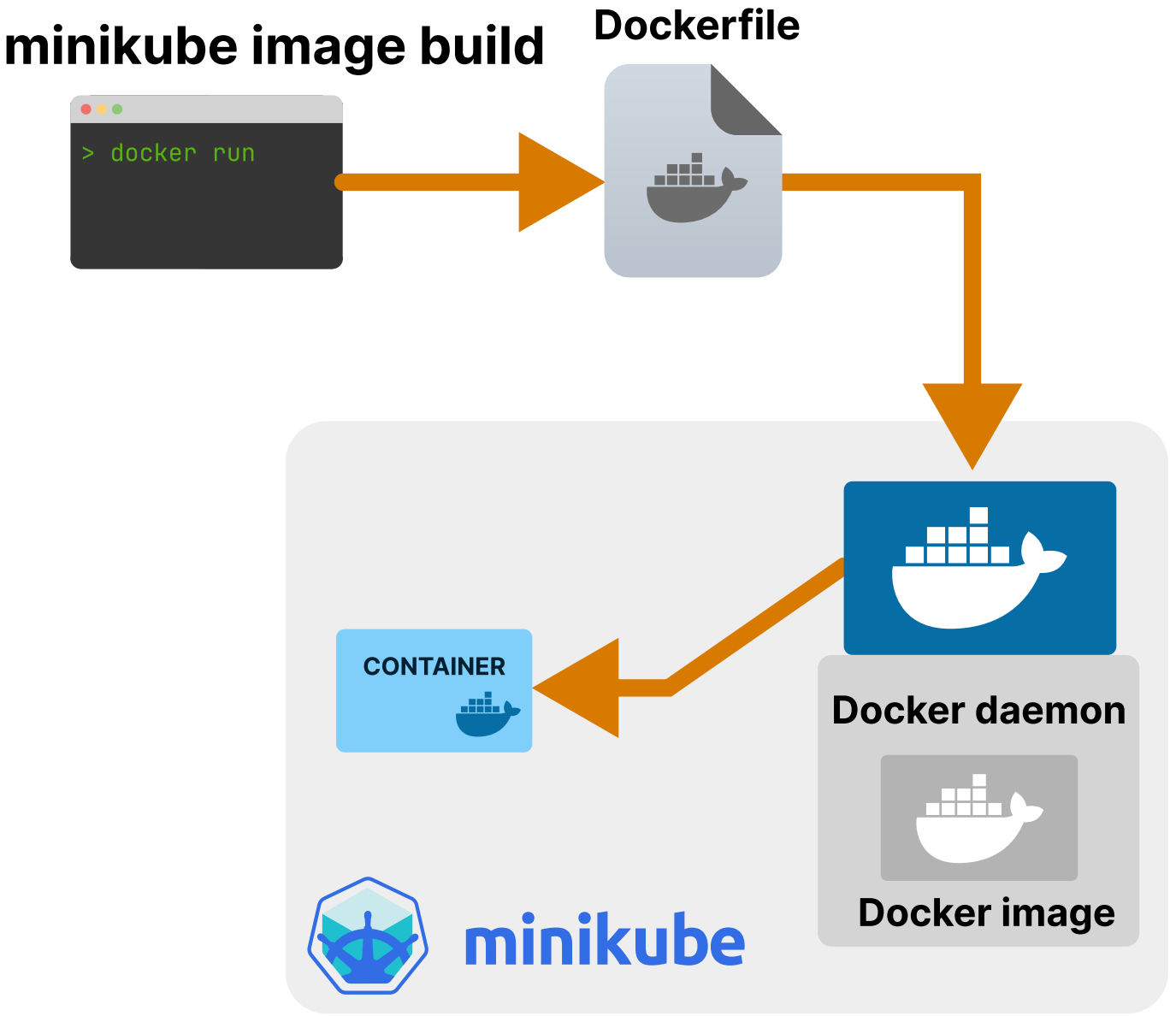

docker build
REST API
Local Docker daemon
> docker run
unix:///var/run/docker.sock
Remote Docker daemon
minikube
2 Docker daemons
they don't share images!


docker build
REST API
Local Docker daemon
Docker image
> docker run
unix:///var/run/docker.sock
Upload to DockerHub
Docker daemon
minikube


docker build
REST API
Local Docker daemon
Docker image
> docker run
unix:///var/run/docker.sock
Upload to DockerHub
Download from DockerHub
Docker daemon
minikube
slow


minikube image build
> docker run
Dockerfile
Docker daemon
minikube


minikube image build
> docker run
Dockerfile
Docker daemon
Docker image
minikube


minikube image build
> docker run
Dockerfile
CONTAINER
Docker daemon
Docker image
minikube

kubectl

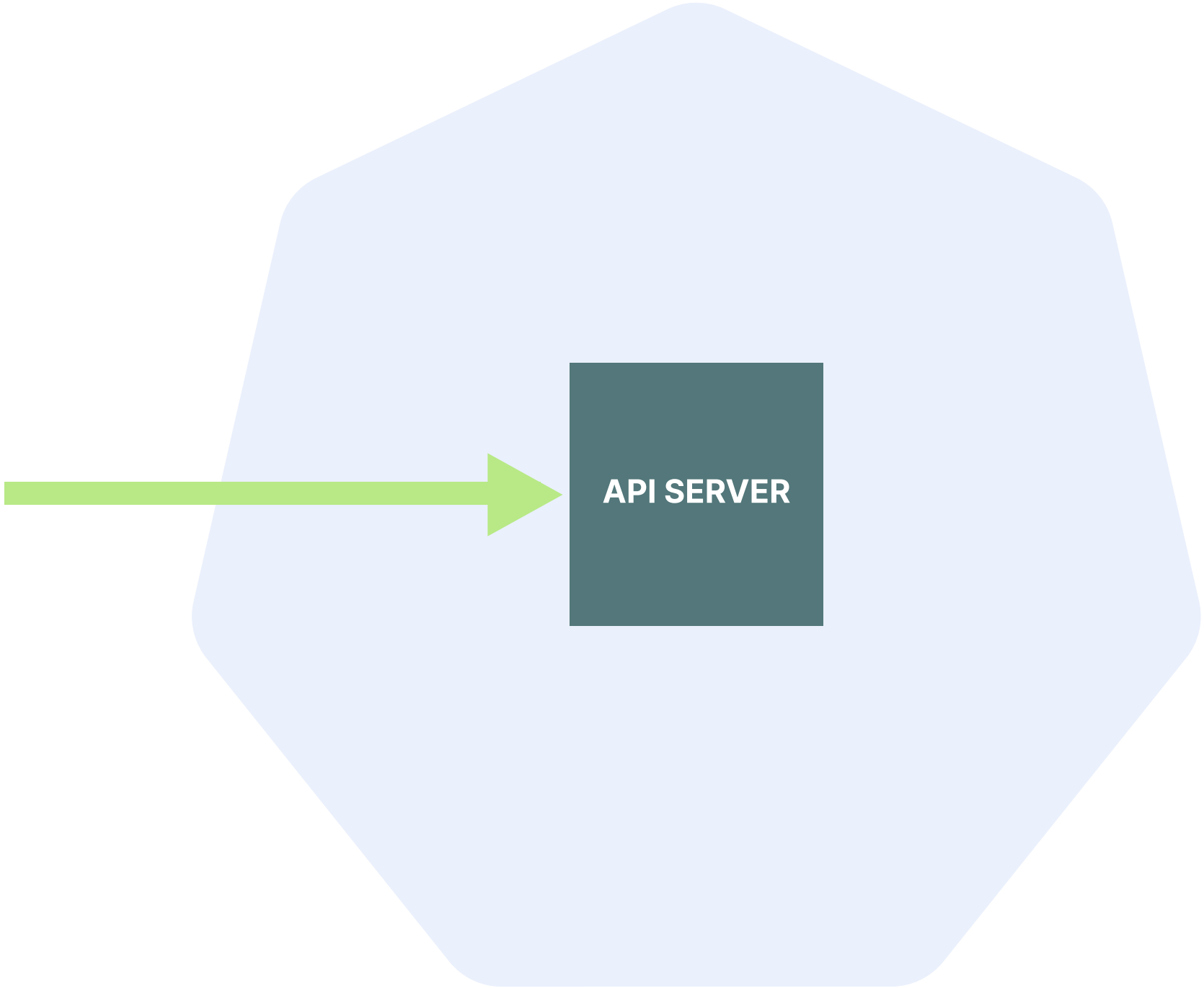




Deploy 3 replicas, please






API SERVER

YAML

```
~$ cat pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: example-pod
  labels:
    app: web
spec:
  containers:
  - name: app
    image: nginx

~$ _
```

```
~$ cat pod.json
{
  "apiVersion": "v1",
  "kind": "Pod",
  "metadata": {
    "name": "example-pod",
    "labels": {
      "app": "web"
    }
  },
  "spec": {
    "containers": [
      {
        "name": "app",
        "image": "nginx"
      }
    ]
  }
}

~$ _
```

Maps

YAML


~$ cat pod.yaml
apiVersion: v1
kind: Pod
~$ _

```
~$ cat pod.yaml

apiVersion: v1
kind: Pod

~$ _
```

```
~$ cat pod.json

{
  "apiVersion": "v1",
  "kind": "Pod"
}

~$ _
```

Nested maps

YAML


~$ cat pod.yaml
apiVersion: v1
kind: Pod
metadata:
name: example-pod
labels:
app: app
~$ _

```
~$ cat pod.yaml

apiVersion: v1
kind: Pod
metadata:
  name: example-pod
  labels:
    app: app

~$ _
```

```
~$ cat pod.json

{
  "apiVersion": "v1",
  "kind": "Pod",
  "metadata": {
    "name": "example-pod",
    "labels": {
      "app": "app"
    }
  }
}

~$ _
```

Lists

YAML


~$ cat args.yaml
args:
- sleep
- "1000"
~$ _

```
~$ cat args.yaml

args:
- sleep
- "1000"

~$ _
```

```
~$ cat args.json

{
  "args": [
    "sleep",
    "1000"
  ]
}

~$ _
```

Nested lists and maps

YAML

```
~$ cat pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: example-pod
  labels:
    app: web
spec:
  containers:
  - name: app
    image: nginx

~$ _
```

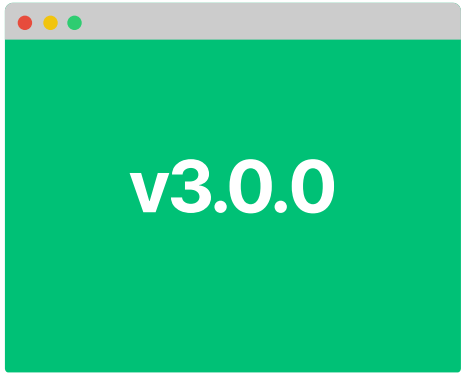
```
~$ cat pod.json
{
  "apiVersion": "v1",
  "kind": "Pod",
  "metadata": {
    "name": "example-pod",
    "labels": {
      "app": "web"
    }
  },
  "spec": {
    "containers": [
      {
        "name": "app",
        "image": "nginx"
      }
    ]
  }
}

~$ _
```

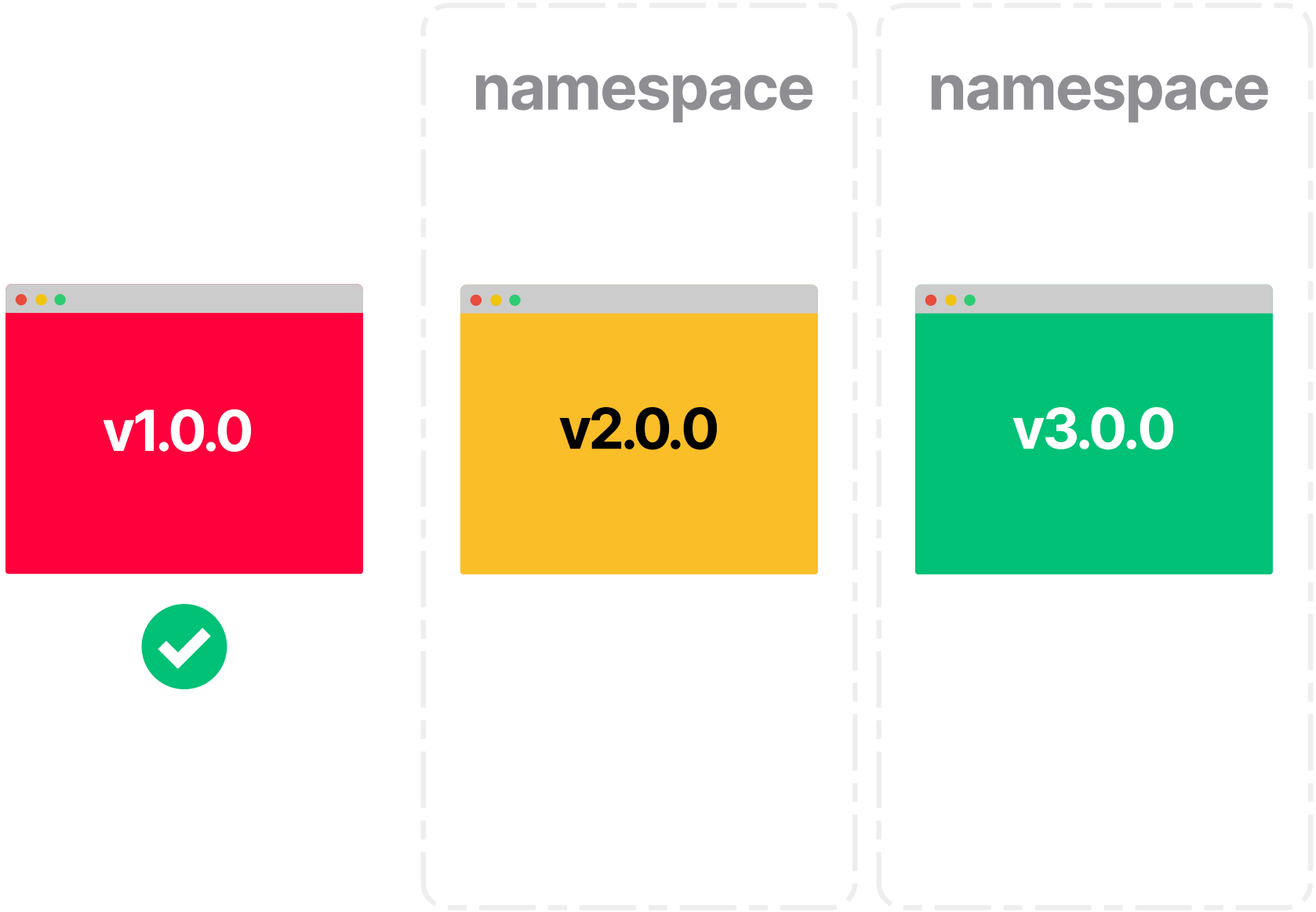
More labs

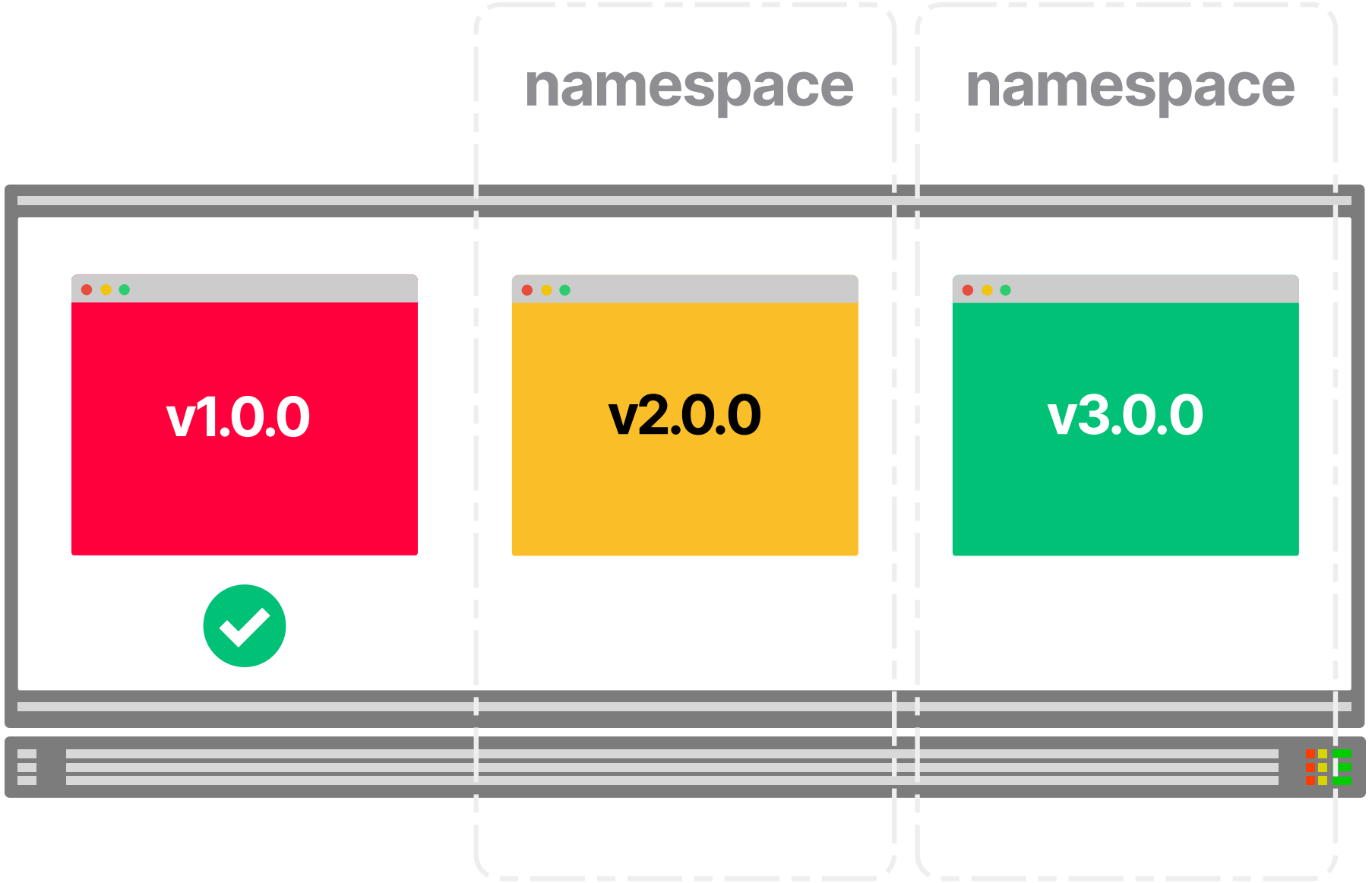

v1.0.0

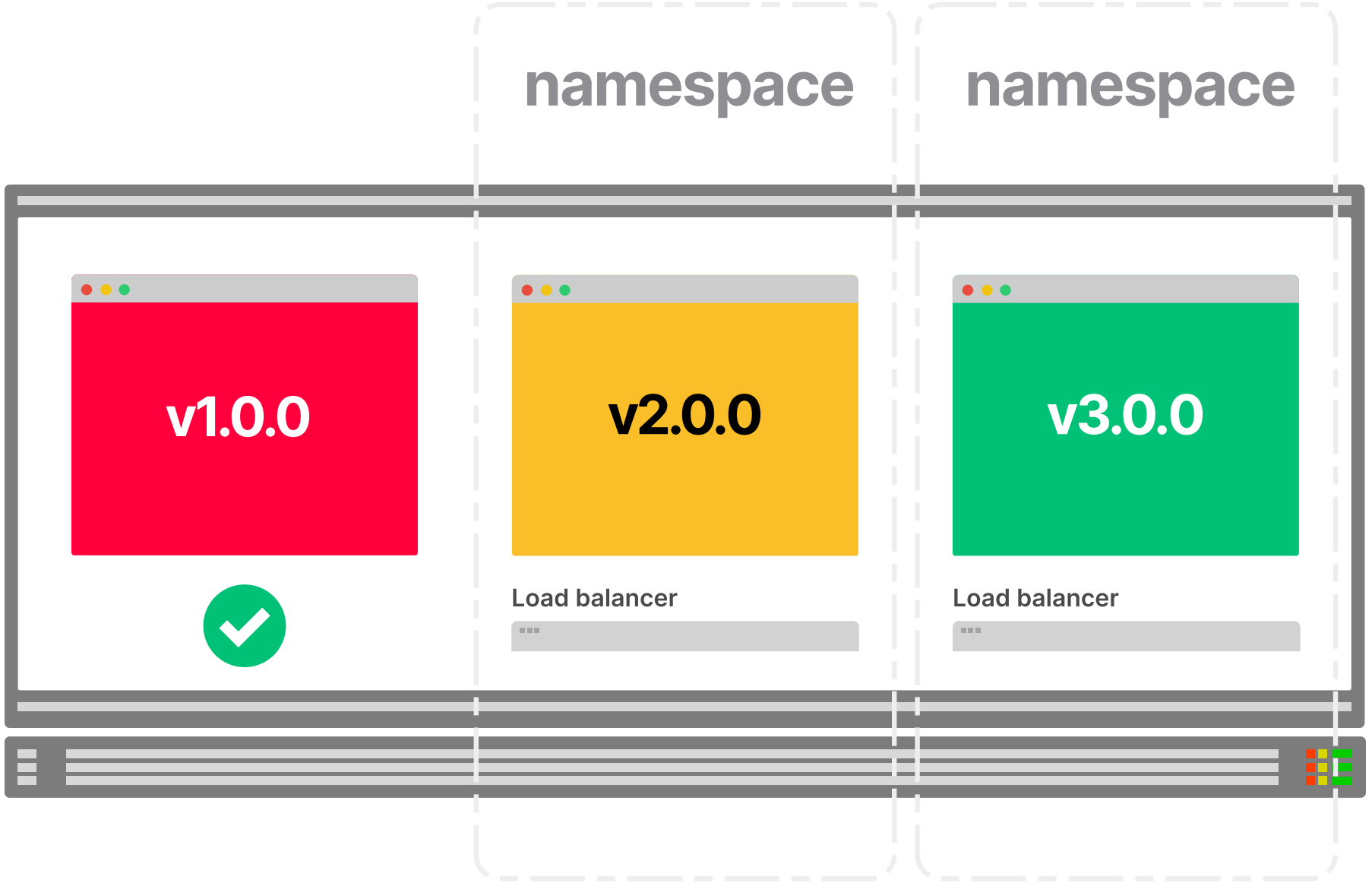

v2.0.0


v3.0.0




namespace
namespace
v1.0.0
v2.0.0
v3.0.0


namespace
namespace
v1.0.0
v2.0.0
v3.0.0


namespace
namespace
v1.0.0
v2.0.0
v3.0.0
Load balancer
Load balancer

Recap

Kubernetes fundamentals

Recap — Kubernetes fundamentals

1. Pod, Deployment, Service, Ingress
2. Pod can have multiple containers
3. Minikube architecture
4. Sending commands with kubectl
5. Namespaces

Recap — Kubernetes fundamentals

1. Pod, Deployment, Service, Ingress
2. **Pod can have multiple containers**
3. Minikube architecture
4. Sending commands with kubectl
5. Namespaces

Recap — Kubernetes fundamentals

1. Pod, Deployment, Service, Ingress
2. Pod can have multiple containers
3. **Minikube architecture**
4. Sending commands with kubectl
5. Namespaces

Recap — Kubernetes fundamentals

1. Pod, Deployment, Service, Ingress
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4. **Sending commands with kubectl**
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Recap — Kubernetes fundamentals

1. Pod, Deployment, Service, Ingress
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4. Sending commands with kubectl
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Thanks.